A DICHOTOMY FOR k-FOLD SUMSETS OF BOUNDED-GAP SEQUENCES (ERDŐS PROBLEM 1112)

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ABSTRACT. Let $k \geq 3$ and $1 \leq d_1 < d_2$ be integers. Erdős problem 1112 asks whether there is an integer r such that every lacunary sequence $B = \{b_1 < b_2 < \cdots\}$ of positive integers with $b_{i+1} \geq r\, b_i$ admits a sequence $A = \{a_1 < a_2 < \cdots\}$ of positive integers with all gaps in $[d_1, d_2]$ and $(kA) \cap B = \emptyset$, where kA is the k-fold sumset; write $r_k(d_1, d_2)$ for the least such r.

We prove that $r_k(d_1, d_2)$ exists if and only if $d_2 \geq k+1$ — so the answer to the recorded question is yes precisely when $d_2 \geq k+1$ — with the explicit bound $r_k(d_1, d_2) \leq 192\, d_2$ when it exists. Below the threshold, non-existence holds in a strong form: for every prescribed sequence of pointwise lower bounds on the successive ratios of B, however rapidly growing, a single B meets the k-fold sumset of every admissible A. The existence half uses a Beatty construction whose k-fold sumset threads the gaps a lacunary B must leave; the non-existence half reduces, through a word-combinatorial analysis of the gap sequence, to a bounded subset-sum covering lemma of possible independent interest. The proof is computer-assisted, with a finite certificate layer, and the full dichotomy is formally verified in Lean 4 in the problem's literal integer phrasing.

The proof is presented at length and from first principles: the background it needs is developed in the text, and the mechanism of each construction is separated from the estimates that bound its constants.

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Source, data, and the Lean 4 development: `https://github.com/beetree/math_erdos_1112`. §29 gives the release and reproduction instructions; §27 states how this work was produced.

Part I. The problem

1. The problem, from scratch

1.1. **On the form of this paper.** This paper settles a question posed by Paul Erdős and Ronald Graham in 1980 [ErGr80] and recorded as problem 1112 on `erdosproblems.com` [E1112], where the case $k \geq 3$ is listed as open. The result is the dichotomy stated as Theorem A in §1.8: the ratio $r_k(d_1, d_2)$ exists precisely when $d_2 \geq k+1$.

The proof is presented at greater length than is customary, and the reasons are practical. It combines several elementary but nonstandard reductions, draws on three separate bodies of background — remainder arithmetic, combinatorics on words, and a little real analysis — and rests at one point on a finite certificate layer. Few readers are fluent in all three areas at once, and a proof that is checkable only by someone who is has a very small pool of possible checkers.

We have therefore made three choices. The background the argument needs is developed in the text rather than cited, so that the paper can be verified without a reference shelf. The mechanism of each construction is stated separately from the estimates that bound its constants, so that a reader can see which lines carry the mathematical content and which perform bookkeeping. And routine expansions are placed in an appendix rather than in the reader's way. The result is roughly three times the length of a compressed presentation of the same theorems.

The cost is length; we think the trade is worth making for a proof of this shape, and §26 records what has been done to make the result checkable by other means as well. Nothing here is asserted without proof: the finite layer is machine-checked three independent ways, and the entire dichotomy is verified in Lean 4.

Some ground rules, so you know what you are reading.

- *Every mathematical subject is defined.* If a word like "sumset" or "lacunary" appears, it was defined earlier in this paper. Three self-contained interludes build the background the proof needs — remainders and divisibility (§7), infinite words and a little analysis (§11), and balanced words (§13). What is *not* redeveloped from zero is the general craft of reading proofs — contradiction, induction, "it suffices to show" — so Appendix A is a short guide to the conventions this paper uses. If you hit an undefined *mathematical term*, that is our mistake; if you hit an unfamiliar *proof move*, Appendix A should name it.
- *Every proof is complete, and one table travels separately.* When a step is genuinely hard, we slow down and build the tools — we do not say "it can be shown". Where a routine algebraic expansion would drown the idea it serves, the idea stays in the text and the expansion is written out, line by line, in Appendix C; nothing is left to "clearly". One exception, worth knowing from the start: the finite layer of Part IV rests on 350 certificate rows — six small integers each — and every one of them is printed here, in Appendix E. Each is checkable by hand in a few minutes; the method, and the three independent systems that have verified all of them, are in §§22–26.
- *Some passages are skippable.* A few stretches of pure verification are marked *(first pass: skippable)* — skip them, keep your momentum, and return when you want the details. The parts can also be read somewhat independently (a map, with three suggested routes, is in §1.10).
- *It is long.* A conventional paper compresses; this one decompresses. The length is the price of the experiment.

Two conventions of the genre are kept, because they exist for reasons that survive scrutiny. §1.9 states precisely what is new here and what is owed to others; and §§26–29 say exactly how the claims can be checked, by whom, and what remains trusted. Clarity is not an excuse for vagueness about credit or verifiability.

One notational convention up front: $\mathbb{N} = \{1, 2, 3, \dots\}$ denotes the positive integers (natural numbers), and when we say *number* without qualification we mean a positive integer.

1.2. **A note on prerequisites.** This paper builds all of its content from the ground up, but it does assume a little of the *craft* of reading proofs — what "it suffices to show means, how a chain of inequalities is to be read, when passing to a tail is legitimate. Appendix A sets out the whole of what is assumed, in two pages, and nothing else is. A reader for whom that is second nature should skip it; a reader who finds a move in a proof unfamiliar should look there first.

1.3. **Sequences and gaps.** The two main characters of this story are infinite sequences of positive integers. Let us fix the vocabulary.

Definition 1.1 (Sequence, gaps)**.** A *sequence* is an infinite list of positive integers

$$a_1 < a_2 < a_3 < \cdots ,$$

strictly increasing, going on forever. We write $A = \{a_1 < a_2 < \cdots\}$ and think of A interchangeably as a list or as an infinite set of positive integers. The *gaps* of A are the differences between consecutive terms:

$$g_n \;:=\; a_{n+1} - a_n \qquad (n = 1, 2, 3, \dots).$$

Example 1.2. The even numbers $A = \{2, 4, 6, 8, \dots\}$ have all gaps equal to 2. The squares $\{1, 4, 9, 16, 25, \dots\}$ have gaps $3, 5, 7, 9, \dots$, growing without bound. The powers of ten $\{10, 100, 1000, \dots\}$ have gaps $90, 900, 9000, \dots$, growing very fast indeed.

Our first character is a sequence whose gaps are *constrained to a window*: they may wobble, but only between two fixed bounds.

Definition 1.3 ((d_1, d_2)-sequence)**.** Let $d_1 \le d_2$ be positive integers. A (d_1, d_2)*-sequence* is a sequence A all of whose gaps lie between d_1 and d_2:

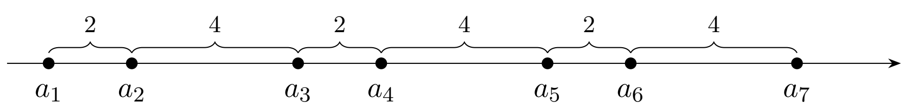
$$d_1 \;\le\; a_{n+1} - a_n \;\le\; d_2 \qquad \text{for every } n \ge 1.$$

Example 1.4. Take $(d_1, d_2) = (2, 4)$. Then

$$A = \{1,\ 3,\ 7,\ 9,\ 13,\ 15,\ 19,\ \dots\}$$

(gaps $2, 4, 2, 4, 2, 4, \dots$) is a $(2, 4)$-sequence, and so is $\{5, 8, 11, 14, \dots\}$ (all gaps 3), and so is any sequence you build by walking right along the number line taking steps of size 2, 3 or 4, in any order you like, forever.

That last phrasing is the right mental picture: *a (d_1, d_2)-sequence is a walk along the number line whose steps are all between d_1 and d_2.* You choose the starting point and you choose each step, subject only to the window.



1.4. **Sumsets.** Our second ingredient is addition: we will be interested not in A itself but in the set of all numbers you can make by adding up k members of A.

Definition 1.5 (k-fold sumset)**.** Let A be a sequence and $k \ge 1$ an integer. The *k-fold sumset* of A is

$$kA \;:=\; \{\, x_1 + x_2 + \cdots + x_k \;:\; x_1, \dots, x_k \in A \,\},$$

where the x_i need *not* be distinct — you may use the same member of A several times.

Two warnings about this (standard) notation. First, kA is *not* the set $\{k \cdot a : a \in A\}$ of multiples — though it contains all of those (take $x_1 = \cdots = x_k = a$). Second, repetitions are allowed: for $A = \{1, 3, 7, \dots\}$ and $k = 3$, the sums $1 + 1 + 1 = 3$ and $1 + 1 + 7 = 9$ and $3 + 7 + 7 = 17$ all belong to $3A$.

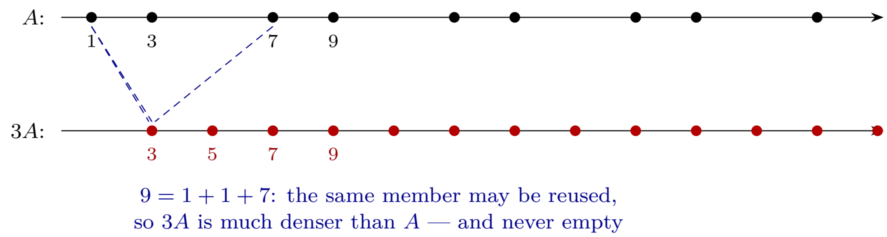


FIGURE 1. A sumset, seen. Above: the $(2,4)$-walk A of Example 1.4. Below: its 3-fold sumset $3A$, every element a sum of three members of A with repetitions allowed (dashed lines trace one such sum). The whole problem is whether the lower row can be steered to miss a given sparse sequence.

Example 1.6. Let $A = \{5, 8, 11, 14, \dots\}$, the numbers of the form $5+3j$ with $j \geq 0$. What is $3A$? Each element is a sum

$$(5+3j_1) + (5+3j_2) + (5+3j_3) \; = \; 15 + 3(j_1+j_2+j_3),$$

and since $j_1+j_2+j_3$ can be any integer ≥ 0, we get exactly $3A = \{15, 18, 21, 24, \dots\}$: again an arithmetic progression with the same common difference 3, starting at 15. This tidiness is special to arithmetic progressions — and, as we will see much later, it is precisely what makes them *bad* at dodging.

1.5. **Lacunary sequences.** The sequence to be dodged grows *multiplicatively*: each term is at least a fixed multiple of the one before.

Definition 1.7 (Lacunary sequence)**.** Let $r \geq 1$ be an integer. A sequence $B = \{b_1 < b_2 < \cdots\}$ is *lacunary with ratio r* if

$$b_{i+1} \; \geq \; r\, b_i \qquad \text{for every } i \geq 1.$$

Example 1.8. The powers $\{r, r^2, r^3, \dots\}$ are lacunary with ratio r — the minimal example, growing as slowly as the definition allows. But a lacunary sequence need not be this regular: $\{1, 7, 100, 1000, 10^{12}, \dots\}$ is lacunary with ratio 7, as long as every step multiplies by at least 7. The word *lacunary* comes from the Latin *lacuna*, a gap or hollow: on the number line, such a sequence is almost all gap.

The picture to hold: a lacunary sequence with a large ratio is *extremely sparse*. Below any number N it has at most about $\log_r N$ terms. Between consecutive terms b_i and $b_{i+1} \geq rb_i$ yawns an empty stretch r times longer than everything that came before. Sparseness is what should make B *easy* to dodge — that is the intuition the problem tests, and Figure 2 is what that intuition looks like.

1.6. **The question, as a game.** We can now state the problem. Here it is verbatim, as recorded on `erdosproblems.com` as problem 1112:

> Let $1 \leq d_1 < d_2$ and $k \geq 3$. Does there exist an integer r such that if $B = \{b_1 < \cdots\}$ is a lacunary sequence of positive integers with $b_{i+1} \geq rb_i$ then there exists a sequence of positive integers $A = \{a_1 < \cdots\}$ such that $d_1 \leq a_{i+1} - a_i \leq d_2$ for all $i \geq 1$ and $(kA) \cap B = \emptyset$, where kA is the k-fold sumset?

Every phrase in this is now something we know. But sentences with three quantifiers ("there exists r … such that if [every] B … then there exists A") are hard to hold in the

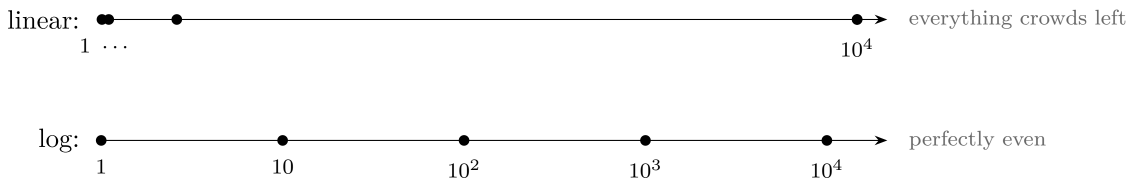
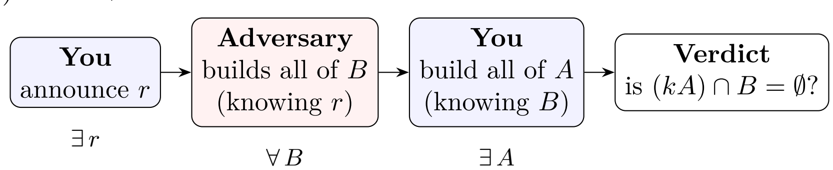


FIGURE 2. The powers of ten, drawn twice. On a linear axis they look like almost nothing at all — every term is dwarfed by the next. On a logarithmic axis they are perfectly evenly spaced. Read the picture carefully: the powers of ten are the *densest* sequence a ratio of 10 permits, and they mark the boundary. A general ratio-10 lacunary sequence is at least this widely separated on the log axis and may be far more irregular — lacunarity is a lower bound on separation, not a spacing law. What the picture does show is that no *scale* is special: sparseness of this kind offers the walk no foothold, which is why it will turn out to be no help below the threshold.

head, so let us restate it as a *game* between you and an Adversary. The numbers $k \geq 3$ and $1 \leq d_1 < d_2$ are fixed in advance and known to both players.

(1) **You** announce an integer r — once and for all.
(2) **The Adversary**, knowing r, builds a lacunary sequence B with ratio r: an infinite sequence with $b_{i+1} \geq rb_i$. The Adversary is infinitely clever and builds B specifically to hurt you.
(3) **You**, knowing B completely, build a (d_1, d_2)-sequence A: a walk with steps in $[d_1, d_2]$.

You win if kA avoids B entirely: no sum of k members of A ever equals a member of B. In symbols, $(kA) \cap B = \emptyset$.



The row of symbols underneath is how a logician would compress the timeline: *there exists* an r such that *for every* B *there exists* an A that wins. Whenever a quantifier statement threatens to blur, come back to this timeline: later moves see earlier moves, never the reverse.

The problem asks: *do you have a winning strategy?* That is, is there a single choice of r such that, whatever B the Adversary produces, you can answer with a winning A?

The order of moves is the soul of the statement, so let us dwell on it.

- You choose A *after* seeing all of B. You may tailor your walk to the particular sequence you must dodge. (If instead you had to choose A first, the Adversary would simply pick some element of kA — it is an infinite set — and start B there. You would lose always, and the problem would be boring.)
- But you choose r *before* seeing B. One fixed r must defeat every lacunary-ratio-r sequence at once. Making r enormous makes the Adversary's sequence extremely sparse — intuitively, easier to dodge. The question is whether *any* r, however large, is enough.

Why would this be hard? Your walk A is dense — steps at most d_2, so you have lots of members — and it is only kA, the k-fold sums, that must dodge B. The tension: A's gaps are *bounded*, so kA is forced to be rather evenly spread, and an evenly spread set has a hard time missing even a very sparse one, because "evenly spread" means it shows up everywhere at every scale. Whether the k-fold sums have enough slack to be steered around a sparse B — that is exactly what the problem is asking, and the answer will turn out to hinge on a precise comparison between k and d_2.

1.7. **A first fight: tiny r loses.** To get blood flowing, let us play the game and prove something: announcing $r = 1$ loses. This tiny result will not be needed later, but it makes the rules concrete, and it is our first complete proof.

Proposition 1.9. *Let $k \geq 3$ and $1 \leq d_1 < d_2$. If you announce $r = 1$, the Adversary wins: there is a lacunary sequence B with ratio 1 such that every (d_1, d_2)-sequence A has $(kA) \cap B \neq \emptyset$.*

Proof. With $r = 1$ the lacunarity condition reads $b_{i+1} \geq b_i$, which any increasing sequence satisfies. So the Adversary may play

$$B \;=\; \{1, 2, 3, 4, 5, \dots\},$$

all positive integers. Now let A be any (d_1, d_2)-sequence whatsoever. The sumset kA is not empty — it contains, for instance, ka_1 (add a_1 to itself k times), which is a positive integer, hence a member of B. So $(kA) \cap B \neq \emptyset$: the Adversary wins. □

So any winning r must be at least 2, and the real question starts there: the problem page asks about "an integer r", and we now see this means, in effect, an integer $r \geq 2$.

Remark 1.10 (Steady play loses, one walk at a time). It is worth seeing at once why a fixed arithmetic progression is not a strategy. Suppose the walker always plays $A = \{c, c + d, c + 2d, \dots\}$ for some c and some fixed $d \in [d_1, d_2]$. By Example 1.6, $kA = \{kc + sd : s \geq k\}$, so an Adversary who knew d and c could take $b_1 = kc + kd$ and each subsequent b_{i+1} far up the same progression — legal at any ratio, since an arithmetic progression contains arbitrarily large elements. What is *not* settled by this is whether one B, fixed in advance, can defeat every progression at once: different progressions own different congruence classes. That is the harder question, and §8 answers it.

1.8. **The answer.** Here is the theorem this paper proves. It answers the question completely, and the answer is a genuine dichotomy: a sharp line through parameter space, with a winning strategy on one side and a losing verdict on the other.

Theorem A (The dichotomy). Let $k \geq 3$ and $1 \leq d_1 < d_2$ be integers.

(1) (Existence.) If $d_2 \geq k + 1$, you have a winning strategy: announcing $r = 192\, d_2$ works — every lacunary B with that ratio can be dodged by some (d_1, d_2)-sequence A with $(kA) \cap B = \emptyset$. The same holds for every $r \geq 192\, d_2$: a sequence lacunary with ratio r is a fortiori lacunary with ratio $192\, d_2$, so raising r only shrinks the admissible class of B.

(2) (Non-existence, in a strong form.) If $d_2 \leq k$, no announcement wins. Worse: even if the fixed ratio r is replaced by any prescribed sequence $r_1, r_2, r_3, \dots$ of lower bounds — the Adversary now being required to satisfy $b_{i+1} \geq r_i\, b_i$ at every step, with the r_i growing as fast as you like — the Adversary can still build a *single* sequence B that defeats *every* (d_1, d_2)-sequence A at once.

So the answer to Erdős problem 1112 is: *yes precisely when $d_2 \geq k + 1$.*

Pause on how lopsided the two conclusions are. Above the threshold, one fixed ratio $192\,d_2$ tames every lacunary B. Below it, the collapse is total: you may demand $b_{i+1} \geq r_i b_i$ with $r_i = 10^{10^i}$ — forcing B to be unimaginably sparse — and still a single B meets the k-fold sums of *every* admissible walk. Sparseness, which looked like your best friend in §1.6, turns out to be completely irrelevant below the threshold: no amount of it saves you.

Notice also what the threshold does *not* depend on: d_1. Only the largest allowed step d_2 matters, measured against the number of summands k. Why the combination "d_2 versus $k+1$"? That is the subject of the next section, where one picture explains the whole dichotomy — and the rest of the document is the long, happy labor of making that picture rigorous.

1.9. **What was known, and what is new here.** *The problem's provenance.* The question above is recorded as problem 1112 on `erdosproblems.com` [E1112]. As that page itself notes, the formulation is a generous interpretation of an ambiguous remark of Erdős and Graham [ErGr80, p. 18]; it may be more accurate to call it a problem *inspired by* them. What we solve is the formulation as recorded there, quoted verbatim in §1.6.

What was known. The quantity $r_k(d_1, d_2)$ was introduced by Bollobás, Hegyvári and Jin [BHJ97]. For two summands the picture was nearly complete: Erdős–Graham [ErGr80] and [BHJ97] give $r_2(2,3) = 2$, and Chen [Ch00] shows $r_2(a,b) \leq 2$ for integers $a < b$ with $b \neq 2a$, and separately that $r_2(a, 2a) \geq 2$ — so on the excluded diagonal $b = 2a$ only a lower bound was available, and existence was not established. For three or more summands, [BHJ97] showed that $r_3(2,3)$ does not exist, and did so in the strong form we adopt here: no prescribed sequence of lower bounds on the successive ratios of B, however fast-growing, rescues the walker. That formulation is theirs; what is new below is its range — every $k \geq 3$ and every $d_2 \leq k$, rather than the single triple $(3,2,3)$, which Theorem A(2) recovers as a special case. Tang and Yang [TaYa21] proved, for $k \geq 3$, that if the gap sequence of A is restricted to a structured class (*block type*) then some B meets kA — in the spirit of our eventually-periodic case (§10). The general existence question for $k \geq 3$ was listed as open; as of July 2026 the problem page records no further progress on it, and we know of none elsewhere.

Notice where the difficulty sat. The structured gap sequences were understood; what resisted was precisely the *unstructured* ones — gap sequences that never settle into a period yet remain too regular to be exploited. Those are the Sturmian walks of §14, and they are the reason this proof needs a theory of infinite words rather than a counting argument.

A word on the excluded diagonal $b = 2a$, since it is easy to overstate what is open there. Chen's exclusion is not a gap in knowledge: the diagonal follows from the off-diagonal case by monotonicity. The parameter d_2 is an *upper* bound on the gaps, so enlarging it enlarges the class of admissible A and can only make avoidance easier:

$$d_2 \leq d_2' \quad \Longrightarrow \quad r_k(d_1, d_2') \;\leq\; r_k(d_1, d_2),$$

because a walk with gaps in $[d_1, d_2]$ is a fortiori one with gaps in $[d_1, d_2']$. Taking $d_2 = a+1$ and $d_2' = 2a$ (legitimate for $a \geq 1$) and applying [Ch00] off the diagonal — $a + 1 \neq 2a$ once $a \geq 2$ — gives $r_2(a, 2a) \leq r_2(a, a+1) \leq 2$, and with Chen's lower bound $r_2(a, 2a) \geq 2$,

$$r_2(a, 2a) \;=\; 2 \qquad (a \geq 2).$$

Our Theorem 6.1 also yields existence here, but only with the bound $192\,d_2 = 384\,a$, which the above supersedes completely. We record this because an earlier draft of this paper presented the $384\,a$ bound as filling a gap Chen left open; it does not, and we are grateful to the referee who pointed out the one-line monotonicity argument. Nothing elsewhere in this paper depends on the claim.

What is new here. Three things. First, the dichotomy itself (Theorem A): existence for every $d_2 \geq k+1$, with the explicit bound $r_k \leq 192\, d_2$, and non-existence for every $d_2 \leq k$ — in the strong form, where no prescribed sequence of pointwise lower bounds on the ratios of B can rescue the walker. Second, the combinatorial lemma the non-existence half consumes at its binding corner, which we isolate as Theorem 16.1 in Part IV and discuss immediately below. Third, a complete machine-checked verification (§26).

The subset-sum lemma, against the literature. Theorem 16.1 says: every finite set G of at least three positive integers with $\gcd(G) = 1$ and $\max(G) = M$ admits a multiset of at most $M - 1$ of its elements whose subset sums contain M consecutive integers. We use it only as a tool, but it may be of independent interest, so we place it against three traditions.

Dense-set subset-sum theorems. The results producing long intervals or progressions among subset sums — Sárközy [Sa89, Sa94], Lev [Lev03], Szemerédi–Vu [SV06], Conlon–Fox–Pham [CFP21], Alon–Freiman [AF88], and constructively Chen–Mao–Zhang [CMZ25] — all work with a *dense* set ($|A| \gtrsim \sqrt{n}$) and yield an arithmetic progression whose common difference need not be 1. The closest antecedent is Lev's consecutive-subset-sum theorem [Lev98, Thm. 1], which does give a genuine interval of consecutive integers — but again only for a dense set ($|A| \gtrsim \frac{2}{3}M$), which is vacuous when $|G| = 3$. Our lemma is the *density-free, multiset* analogue: it trades density for repetition, covering an interval of length exactly M at uniform cost $|S| \leq M - 1$, from an arbitrary gcd-one set, down to three elements.

Higher-sumset structure. The structure theorems for m-fold sumsets (Nathanson [Na72], Granville–Walker [GW21], Lev [Lev22]) are density-free but describe mG for large m — a total-degree simplex of coefficients, of which the subset sums of a bounded multiset form a per-generator box inside — so their conclusion is a weaker statement about a larger set.

Completeness and Frobenius. The completeness, Frobenius and postage-stamp traditions [Gr64, ErGr80, RA05, GS20] concern infinite sequences, or use unboundedly many summands, or optimise the generating set; none bounds a multiset drawn from an arbitrary given set.

What is new is the constant, not the phenomenon. It would be easy to read the above as claiming that no bounded multiset of any size was previously known to cover a run. That would be false, and we say so plainly: a short greedy argument gives $m(G) \leq a + M - 1 \leq 2M - 3$ with no hypothesis beyond $\gcd(G) = 1$ and $|G| \geq 3$ (Lemma 15.4). So the $O(M)$ phenomenon is elementary and we claim no novelty for it. What Theorem 16.1 adds is the sharp constant $M - 1$ — roughly a factor of two better, attained with equality at every maximum (Propositions 24.1 and 25.1), and small enough to be consumed by the slot lemma at the corner $M = d_2 = k$, where $2M - 3$ would be useless. Everything difficult in Part IV exists to buy that factor, and the novelty claim below is about that factor only.

Table 1 states, for each nearest result, exactly which hypothesis separates it from Theorem 16.1. A reader who wants to overturn the novelty claim should attack a row of this table.

Our claim is therefore narrow, and we state it in the narrow form: after a documented search[1] we found no result implying the sharp bound $m(G) \leq M - 1$. We do *not* claim that the $O(M)$ phenomenon is new — Lemma 15.4 shows it is elementary — only that the constant is. As with any negative claim in a mature area this warrants a specialist's confirmation, and we have not obtained one; the sharpest place to look is the neighbourhood of [Lev97, Lev98], and, in language we do not speak fluently, the Apéry-set and factorization-length literature for numerical semigroups.

[1]Conducted July 2026 over MathSciNet, zbMATH, arXiv and Google Scholar citation chains around the works cited here; the search record is in the repository as `paper/novelty-search.md`.

1.10. **Map of the paper.** **Part I** (§1–2) states the problem and gives the one picture — windows of width k striding at pace γ — from which everything grows; §2.4 turns that picture into an honest ledger of four gaps, which the rest of the document closes. **Part II** (§3–6) proves the existence half: floor toolbox, Beatty walks, the safe/unsafe dial, the free-gap lemma, nested intervals, the ratio $192\,d_2$. **Part III** (§8–15) reduces the non-existence half to walks (the certificate trick), then proves every admissible walk tail-covering: easy cases first, then the two-letter campaign — band and sweep, the Morse–Hedlund classification, the Sturmian endgame — and finally the slot lemma, which prices the corner at exactly the subset-sum lemma. **Part IV** (§16–24) proves that lemma: machinery, reduction to the hard core, six branches. **Part V** (§25–26) assembles the theorem and explains the computer verification.

Three reading routes, with honest time estimates. The *spine*: §1, §2, then the figures, Big idea boxes and theorem statements of each part — an afternoon, and you will know what is true and why it is plausible. The *existence route*: Parts I–II are self-contained and fully rigorous, using no modular arithmetic at all — a complete, checkable proof of the positive half; expect a week of evenings, with §5 the one steep climb. The *whole climb*: everything in order — a project for a season, not a sitting; the three interludes (§7, §11, §13) are built to be read exactly when they arrive, the marked *(first pass: skippable)* passages can be deferred without loss of thread, and Part IV is the steepest sustained stretch. Routine expansions live in Appendix C, and Appendix D is the map of the notation.

Two parameter sets, carried throughout. Rather than meet a fresh example in every section, it helps to keep two concrete cases in your pocket and ask what each new result says about them:

$$\boxed{k = 3,\ (d_1, d_2) = (1, 4)} \quad \text{(above the threshold: you win)} \qquad \boxed{k = 3,\ (d_1, d_2) = (1, 3)} \quad \text{(at the corner: you los}$$

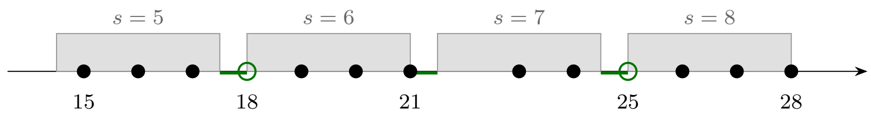
The winning case is the one §2.2 pictures, §5 computes with ($t = 10000$), and §6.4 plays to a finish (against $b_i = 800^i$, yielding $\gamma^* \approx 3.74999556$ and the walk $804, 808, 812, \ldots$). The losing case is where the two-letter analysis bites: $d_2 = 3 = k$ is the corner, so every admissible walk's gap word uses letters from $\{1, 2, 3\}$, the sweep needs width ≥ 2, and the Sturmian endgame handles what is left — and if the walk's tail alphabet has three letters, the slot lemma consumes SHARP at exactly $m(G) \leq M - 1$. Holding both in mind as you read, the sixty pages between the picture in §2 and its vindication in §25 will have an anchor.

The load-bearing results are machine-checked. Every result used in the proof of Theorem A corresponds, through §F, to a declaration in the Lean 4 development that states and proves it. Auxiliary results — worked examples, remarks, and the elementary facts assembled in the interludes — are proved here in prose and are not individually formalized; §F lists exactly what is covered.

2. The one picture that explains the threshold

Why should the dividing line fall at $d_2 = k + 1$? Before any rigor, this section shows you the picture from which the entire proof grows. Nothing here is officially part of the proof — everything will be re-done carefully later — but every later section is best read with this picture in mind. The section ends with an honest list of exactly what the picture does *not* establish; that list is the table of contents of the rest of the document.

2.1. **Steady walks, and rounding error.** Among all (d_1, d_2)-sequences, the simplest are the *steady* ones: walks that try to move at a constant average speed γ per step. If γ is an integer d, that is an arithmetic progression, $a_i = c + id$. But γ need not be an integer. To



windows $(3.5s - 3,\ 3.5s]$ for $k = 3$, $\gamma = 3.5$, $c = 0$; circled integers sit in daylight

FIGURE 3. The window picture for the walk of Example 2.1. Each 3-fold sum lands in a shaded window; the integers 18 and 25 fall in the daylight between windows, so they are *unreachable*: no three members of A sum to them. (The window $(18, 21]$ is open at its left end, which is why 18 escapes.)

walk at average speed $\gamma = 3.5$, alternate steps of 3 and 4; and in general, the canonical way to walk at any real speed γ is to place your i-th footfall at the integer just below $i\gamma$:

$$a_i \;=\; c + \lfloor i\gamma \rfloor \qquad (i = 1, 2, 3, \dots),$$

where $c \geq 0$ is an integer offset and $\lfloor x \rfloor$ denotes the *floor* of x, the greatest integer $\leq x$ (so $\lfloor 3.5 \rfloor = 3$, $\lfloor 7 \rfloor = 7$). We will call this the *Beatty walk* with slope γ and offset c.[2]

Example 2.1. Take $\gamma = 3.5$ and $c = 0$. Then $\lfloor 1 \cdot 3.5 \rfloor = 3$, $\lfloor 2 \cdot 3.5 \rfloor = 7$, $\lfloor 3 \cdot 3.5 \rfloor = 10$, and so on:

$$A \;=\; \{3,\ 7,\ 10,\ 14,\ 17,\ 21,\ 24,\ 28,\ 31,\ 35,\ \dots\},$$

with gaps $4, 3, 4, 3, 4, 3, \dots$ — a perfectly legal $(d_1, 4)$-sequence for any $d_1 \leq 3$.

Now the question the whole problem turns on: *where do the k-fold sums of a steady walk live?* Take k members of the walk, say with indices $i_1, \dots, i_k$, and write $s := i_1 + \cdots + i_k$ for the index total. If there were no floors, the sum would be exactly $kc + s\gamma$. Each floor rounds down by less than 1 — $\lfloor x \rfloor$ always lies in the half-open interval $(x - 1,\ x]$ — so k floors together round down by less than k:

$$a_{i_1} + \cdots + a_{i_k} \;=\; kc + \sum_{t=1}^{k} \lfloor i_t \gamma \rfloor \;\in\; \big(kc + s\gamma - k,\;\; kc + s\gamma\,\big].$$

Every k-fold sum therefore lands in a *window* of width k hanging just below one of the values $kc + s\gamma$. As s runs through the integers, these anchor values march along the number line in equal strides of γ:

Guiding heuristic. The k-fold sums of a steady walk of speed γ are confined to windows of width k, one hanging below each multiple of γ (shifted by kc). The windows stride along the line at pace γ. Everything — both halves of the dichotomy — follows from comparing the stride γ to the window width k.

2.2. **Above the threshold: daylight.** Suppose the stride exceeds the window width: $\gamma > k$. Then consecutive windows do not touch — between the end of one and the start of the next there is a stretch of length $\gamma - k$ that no k-fold sum of the walk can ever reach. Call these stretches *daylight*. They recur forever, once per stride, making up a fixed positive fraction of the number line at every scale.

[2] After Samuel Beatty, whose name is attached to sequences of the form $\lfloor i\gamma \rfloor$. Part II develops their basic properties from scratch; here we only need the rough picture.

Figure 3 shows this for Example 2.1, where $k = 3$ and $\gamma = 3.5$: the windows are $(3.5s - 3,\ 3.5s]$ and the daylight stretches have width 0.5. The integer 25 sits in daylight — so no three members of that walk sum to 25, a fact you can (and should) also confirm by hand.

Example 2.2 (Daylight, checked by hand)**.** For the walk of Example 2.1 ($\gamma = 3.5$, $c = 0$, $k = 3$), the integers 18 and 25 are not 3-fold sums. The search is finite: any sum of three terms involving a term exceeding 25 is itself larger than 25, so only $3, 7, 10, 14, 17, 21, 24$ can participate, and enumerating those triples produces neither value. The next unreachable integers are 32 and 39: daylight recurs, once per stride.

Now the strategic point. What strides can an admissible walk have? Steps are integers in $[d_1, d_2]$, so achievable average speeds fill the interval $[d_1, d_2]$: the integer speeds are the arithmetic progressions, and every speed in between is reached by mixing two adjacent step sizes. Daylight needs a stride exceeding k — but be careful about *which* strides will actually help. The integer speed $\gamma = d_2$ has daylight too, yet an arithmetic progression will turn out to lose *regardless* of the threshold (its sumset is a full congruence-class tail, Example 2.3, and Part III's single-B construction punishes exactly that). The usable resource is the *continuum* of non-integer speeds above k: the open interval $(d_2 - 1,\ d_2)$ of strides obtained by mixing the two largest steps. That interval lies above k exactly when $d_2 - 1 \geq k$, that is, when

$$d_2 \ \geq\ k + 1.$$

There is the threshold, appearing for the first time — not yet as a theorem, but as the exact condition under which an entire interval of usable dial positions exists. Keep this distinction in mind: *daylight is necessary but not sufficient*, and what the existence proof will actually exploit is the freedom to tune a real-valued dial (§6 returns to this point with the proof in hand).

Above the threshold, then, here is the plan for winning the game. The daylight stretches sit at predictable places, determined by the slope γ; *turning the dial γ slides the whole ladder of windows*, and with it the daylight. Given the Adversary's sequence B, we will tune γ (and the offset c) so that every single element of B falls into daylight. One dial must dodge infinitely many targets simultaneously — that sounds ambitious, but the targets b_i are lacunary, hence ever sparser, and it will turn out (Part II) that each successive target only forbids a tiny range of dial positions, leaving plenty for the next. The tool that makes "tiny" precise is the *free-gap lemma*, and the tool that finishes infinitely many rounds of tuning is the *nested interval theorem*. The mysterious ratio $192\, d_2$ of Theorem A is nothing deeper than the bookkeeping constant that keeps each round of tuning from exhausting the dial room the previous round left behind.

2.3. **Below the threshold: gridlock.** Now suppose $d_2 \leq k$, so every step is at most k and every stride is at most k: windows of width k marching at pace $\gamma \leq k$. Consecutive windows now overlap, or at best touch. There is no daylight anywhere. Figure 4 shows the whole dichotomy in one picture: the same windows at three strides.

The picture suggests that the k-fold sums of such a walk eventually swallow everything in their path — and for steady *integer* strides this is not merely a picture but a two-line calculation.

Example 2.3 (Steady walks are individually beatable)**.** Let A be the arithmetic progression $a_i = c + id$ with integer gap $d \leq k$. A k-fold sum is

$$(c + i_1 d) + \cdots + (c + i_k d) \ =\ kc + sd, \qquad s = i_1 + \cdots + i_k,$$

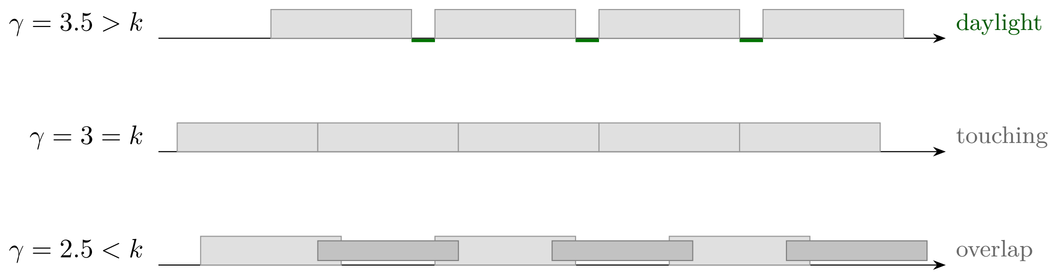


FIGURE 4. The dichotomy in one triptych ($k = 3$ throughout): windows of width 3 at stride γ. Above the threshold ($\gamma > k$) daylight recurs forever; at $\gamma = k$ the windows tile the line exactly; below ($\gamma < k$) they overlap. The theorem says the picture tells the truth: the game is winnable precisely when a stride $\gamma > k$ is available, i.e. when $d_2 \geq k + 1$. (One caution the proof will respect: daylight makes dodging *possible*, never automatic — integer strides above k still lose. See the closing remark of §6.)

and s ranges over *all* integers $\geq k$. So

$$kA \;=\; \{\, kc + sd \;:\; s \geq k \,\}:$$

from $kc + kd$ onwards, kA contains *every* number that leaves the same remainder as kc upon division by d — in the standard notation, every number $\equiv kc \pmod d$. (Read "$x \equiv y \pmod m$" as "x and y leave the same remainder when divided by m"; this *remainder arithmetic* is developed properly in §7, and until then the informal reading is all we use.) We say kA contains a *tail of a congruence class*. A set this fat cannot dodge: however sparse the Adversary is forced to be, the class contains arbitrarily large numbers, and the Adversary simply builds B out of them — choosing each b_{i+1} to be some member of the class beyond $r_i b_i$. Sparseness costs the Adversary nothing, which is exactly the brutal flavor of part (2) of Theorem A. (Mind the quantifiers, though: this beats each steady walk with a B *tailored to it*, and you move last. Beating every walk with one B, committed in advance, is a genuinely harder task — gap (2) of the ledger below.)

So a walk that wants to survive below the threshold must *wobble*: vary its steps, forever, in a way that never settles into an arithmetic progression. The entire difficulty of the non-existence half is the question: *can clever wobbling recreate daylight?* The window picture pushes toward "no" — wobbling changes where the windows sit but not their width, and there is no room between them — but the picture reasons about steady walks, and a wobbling walk has no single stride γ. That gap in the reasoning is not small print. Closing it is most of this paper.

2.4. **What the picture does not prove.** Here is the honest ledger — four gaps between the picture and the theorem, and where each one is closed.

(1) *Above the threshold, one dial setting must defeat all of B at once.* The daylight argument dodges one target at a time; to dodge every b_i simultaneously we tune γ through infinitely many rounds and must show the dial room never runs out. This is Part II: the free-gap lemma (§5) and the nested-interval construction (§6), yielding the explicit ratio $192\, d_2$.

(2) *Below the threshold, one B must defeat all walks at once.* Example 2.3 beats each steady walk with a B built from *its* congruence class — but different walks use different

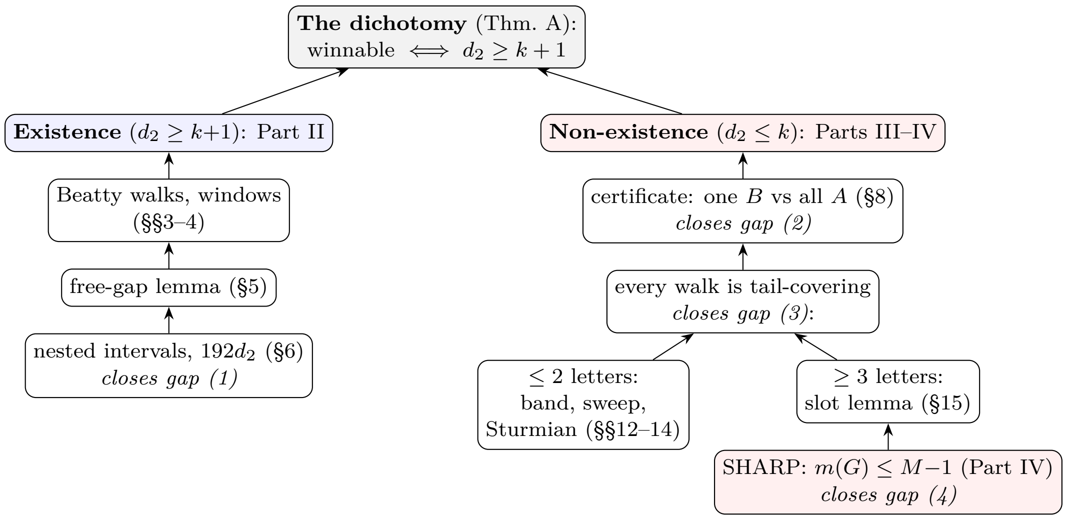


Figure 5. The proof as a map: arrows mean "feeds into". The four debts of the honest ledger annotate the boxes that pay them. Parts III and IV open with this map's relevant half redrawn in words (§9.3, §16.2).

classes, and the theorem demands a single B that beats every admissible walk. The device that accomplishes this is the *certificate trick* (§8), a diagonalization.

(3) *Below the threshold, wobbling walks must be shown hopeless too.* The plan: show every admissible walk's sumset contains a tail of a congruence class ("tail-covering"), by classifying how a walk can wobble — the analysis of *gap words* occupying §§9–15. The delicate walks are those that wobble between two step sizes as evenly and aperiodically as possible; these are the *Sturmian* walks, and taming them (§§13–14) is the deepest water in the document.

(4) *The corner $d_2 = k$ is razor-thin.* When the largest step equals k exactly, the windows touch without overlapping, and the walk has just barely too little room; converting "barely" into a proof costs a genuinely new piece of finite mathematics about subset sums, stated in §16 and proved across Part IV.

One more promissory note. In Example 2.3 the offset "kc" depended on the walk, and in gap (2) we waved at defeating all offsets and classes simultaneously; when we do this properly, the notion of *tail-covering* will be defined to absorb all of it. From here on, the document simply executes the ledger: Part II closes gap (1); Part III closes gaps (2) and (3), reducing the corner case to a finite lemma; Part IV proves that lemma, closing gap (4); Part V assembles the theorem and describes the computer verification. Figure 5 is the same ledger drawn as a map — worth a bookmark: every later section is one of its boxes.

Part II. Existence: winning when $d_2 \ge k+1$

3. Floors and Beatty walks

Part II proves the existence half of the dichotomy: if $d_2 \ge k+1$, announcing $r = 192\, d_2$ wins the game. The plan was sketched in §2.2; from here on we execute it rigorously. **Standing assumptions for all of Part II:** k, d_1, d_2 are integers with

$$k \ge 2, \qquad 1 \le d_1 < d_2, \qquad d_2 \ge k+1.$$

(The problem poses $k \geq 3$, but the existence proof works verbatim from $k = 2$, so we prove the slightly more general statement at no extra cost. Note $d_2 \geq 3$ follows.)

This section builds the two tools everything else stands on: the floor function, and the fact that a Beatty walk with slope between $d_2 - 1$ and d_2 is a legal (d_1, d_2)-sequence.

3.1. **The floor function.**

Definition 3.1 (Floor)**.** For a real number x, the *floor* $\lfloor x \rfloor$ is the unique integer m satisfying

$$m \;\leq\; x \;<\; m + 1.$$

So $\lfloor 3.5 \rfloor = 3$, $\lfloor 7 \rfloor = 7$, and (watch this one) $\lfloor -0.5 \rfloor = -1$: the floor is the integer just *below* x (or x itself), not "the integer part". That an integer $m \leq x < m + 1$ exists is visually clear — walk down the number line from above x until you first pass an integer — and we take it as known; but uniqueness deserves its one-line proof: if $m \leq x < m+1$ and $m' \leq x < m'+1$ with, say, $m < m'$, then $m + 1 \leq m'$ (they are integers), so $x < m + 1 \leq m' \leq x$, a contradiction.

Everything we ever use about floors is in the next lemma.

Lemma 3.2 (Floor toolbox)**.** *For all real x, y and every integer n:*

(1) $x - 1 \;<\; \lfloor x \rfloor \;\leq\; x$*;*
(2) $\lfloor x + n \rfloor = \lfloor x \rfloor + n$*;*
(3) *if $x \leq y$ then $\lfloor x \rfloor \leq \lfloor y \rfloor$;*
(4) *for reals $x_1, \ldots, x_k$:* $\;x_1 + \cdots + x_k - k \;<\; \lfloor x_1 \rfloor + \cdots + \lfloor x_k \rfloor \;\leq\; x_1 + \cdots + x_k$*;*
(5) *every real interval $[u,\; v]$ with $v - u \geq 1$ contains an integer.*

Proof. Write $m = \lfloor x \rfloor$, so $m \leq x < m + 1$ by definition.

(1) The right inequality is $m \leq x$ directly; the left is $x < m + 1$ rearranged.

(2) The integer $m + n$ satisfies $m + n \leq x + n < (m + n) + 1$, which is exactly the defining property of $\lfloor x + n \rfloor$; by uniqueness, $\lfloor x + n \rfloor = m + n$.

(3) Suppose $x \leq y$. Then $\lfloor x \rfloor \leq x \leq y$, so $\lfloor x \rfloor$ is an integer $\leq y$. But $\lfloor y \rfloor$ is the *greatest* integer $\leq y$: any integer $m' \leq y$ has $m' \leq \lfloor y \rfloor$, since otherwise $m' \geq \lfloor y \rfloor + 1 > y$. Hence $\lfloor x \rfloor \leq \lfloor y \rfloor$.

(4) Add the k instances of (1): summing $x_t - 1 < \lfloor x_t \rfloor \leq x_t$ over $t = 1, \ldots, k$ gives the claim.

(5) The integer $m := \lfloor u \rfloor + 1$ satisfies $u < m \leq u + 1 \leq v$. □

3.2. **Beatty walks, officially.**

Definition 3.3 (Beatty walk)**.** For a real number $\gamma > 1$ and an integer $c \geq 0$, the *Beatty walk* with slope γ and offset c is the sequence

$$A_{\gamma,c} \;:=\; \{\, c + \lfloor i\gamma \rfloor \;:\; i = 1, 2, 3, \ldots \}, \qquad \text{i.e.} \quad a_i = c + \lfloor i\gamma \rfloor.$$

We met $A_{3.5,\,0} = \{3, 7, 10, 14, 17, 21, \ldots\}$ in Example 2.1; Figure 6 shows the geometry: the walk is the line $y = \gamma\, i$ digitized downward to the integer grid, and the gaps are the vertical jumps of the resulting staircase.

One more, with a slope that is not a half-integer: for $\gamma = \frac{10}{3}$ and $c = 0$,

$$\lfloor \tfrac{10}{3} \rfloor = 3, \quad \lfloor \tfrac{20}{3} \rfloor = 6, \quad \lfloor \tfrac{30}{3} \rfloor = 10, \quad \lfloor \tfrac{40}{3} \rfloor = 13, \;\ldots$$

so $A_{10/3,\,0} = \{3, 6, 10, 13, 16, 20, 23, 26, 30, \ldots\}$, with gap pattern $3, 4, 3, 3, 4, 3, 3, 4, \ldots$: the gap 4 arrives once every three steps, keeping the average speed at exactly $\frac{10}{3}$.

The single structural fact we need is that a Beatty walk's gaps take at most two values, and we can say which ones.

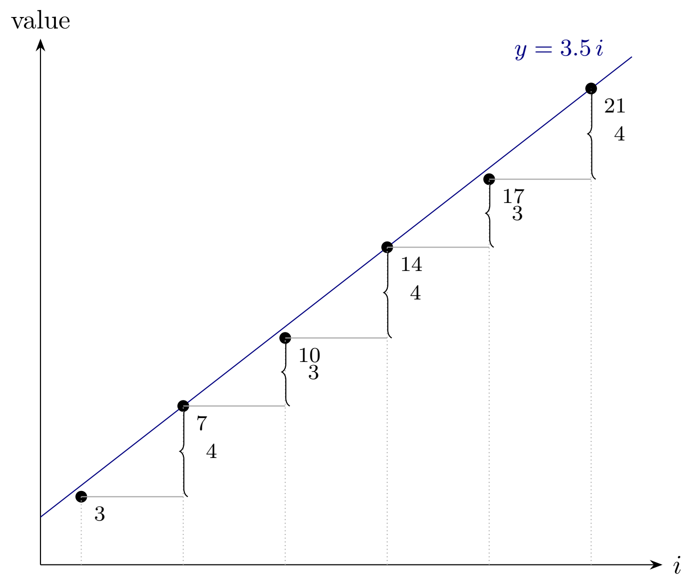


FIGURE 6. The Beatty walk $a_i = \lfloor 3.5\,i \rfloor$ as a digitized line: each dot sits at the integer just below the line $y = 3.5i$, and consecutive dots differ by $\lfloor\gamma\rfloor = 3$ or $\lfloor\gamma\rfloor + 1 = 4$ — the two-value gap phenomenon of Lemma 3.4, seen rather than computed.

Lemma 3.4 (Two-value gaps)**.** *Let $\gamma > 1$ be real. Every gap of the Beatty walk $A_{\gamma,c}$ lies in $\{\lfloor\gamma\rfloor,\ \lfloor\gamma\rfloor + 1\}$:*

$$\lfloor (i+1)\gamma \rfloor - \lfloor i\gamma \rfloor \ \in\ \{\lfloor\gamma\rfloor,\ \lfloor\gamma\rfloor + 1\} \qquad \textit{for every } i \geq 1.$$

Proof. The offset c cancels from every gap, so consider $g := \lfloor (i+1)\gamma \rfloor - \lfloor i\gamma \rfloor$. Write $x := i\gamma$ and $y := \gamma$, so that $g = \lfloor x + y \rfloor - \lfloor x \rfloor$; the claim is the following standard fact, which we record for reference:

$$\lfloor x + y \rfloor - \lfloor x \rfloor \ \in\ \{\,\lfloor y \rfloor,\ \lfloor y \rfloor + 1\,\} \qquad \text{for all real } x, y.$$

Proof of the display. Split each number into integer and fractional parts, $x = \lfloor x \rfloor + \{x\}$ and $y = \lfloor y \rfloor + \{y\}$, with $\{x\}, \{y\} \in [0, 1)$. Then

$$x + y \ =\ \big(\lfloor x \rfloor + \lfloor y \rfloor\big) + \big(\{x\} + \{y\}\big),$$

where the first bracket is an integer and the second lies in $[0, 2)$. If $\{x\} + \{y\} < 1$ the floor of the whole is $\lfloor x \rfloor + \lfloor y \rfloor$; if $\{x\} + \{y\} \geq 1$ it is $\lfloor x \rfloor + \lfloor y \rfloor + 1$ (Lemma 3.2(2), moving the integer bracket through the floor). Subtracting $\lfloor x \rfloor$ gives the display, and substituting $y = \gamma$ gives the lemma. $\square$

(If γ is an integer, only the value $\lfloor\gamma\rfloor = \gamma$ actually occurs, and the walk is an arithmetic progression. For non-integer γ, both values occur infinitely often, though we will not need that.)

3.3. **Beatty walks are admissible.** Now aim the slope at the top of our permitted window.

Proposition 3.5 (Admissibility)**.** *Let γ be any real number with $d_2 - 1 < \gamma < d_2$, and $c \geq 0$ any integer. Then $A_{\gamma,c}$ is a (d_1, d_2)-sequence.*

Proof. Three things to check: the terms are positive, the sequence is strictly increasing, and the gaps lie in $[d_1, d_2]$.

Since $d_2 - 1 < \gamma < d_2$ and $d_2 - 1, d_2$ are integers, the defining property of the floor gives $\lfloor \gamma \rfloor = d_2 - 1$. By Lemma 3.4, every gap of $A_{\gamma,c}$ is $d_2 - 1$ or d_2. Both lie in $[d_1, d_2]$: indeed $d_1 \le d_2 - 1$ because $d_1 < d_2$ are integers. As $d_2 \ge k + 1 \ge 3$, every gap is at least $2 > 0$, so the sequence is strictly increasing; and its first term is $a_1 = c + \lfloor \gamma \rfloor \ge 0 + (d_2 - 1) \ge 2 > 0$. All three checks pass. □

Two remarks worth pausing on. First, *the proofs of Part II never ask whether the slope is rational*: the final winning slope will simply be some real number produced by the nested-interval theorem, and no argument here tests its rationality. (This is not the same as irrationality being *irrelevant*: a closing remark in §6 shows that against the strongest adversary the produced slope is *forced* to be irrational — the proof just never needs to check.) Second, notice how little of the freedom of (d_1, d_2)-sequences we are using: only steps $d_2 - 1$ and d_2, the two largest. The lower bound d_1 plays no role beyond $d_1 \le d_2 - 1$, which is automatic. This is the promised explanation of why d_1 is absent from the threshold: the existence strategy never has any reason to take a small step.

4. Where the k-fold sums live

This section turns the window picture of §2 into theorems, and then performs the change of viewpoint on which the whole existence proof runs: instead of asking *which numbers* the walk's sums can reach, we ask *which dial positions* γ are safe for a given target. By the end of the section, dodging a single element of B will be a clean geometric condition on γ alone.

4.1. **The window containment.**

Proposition 4.1 (Windows)**.** *Let $\gamma > 1$ be real, $c \ge 0$ an integer, and $k \ge 1$. Then*

$$kA_{\gamma,c} \;\subseteq\; \bigcup_{s \ge k} \big(kc + s\gamma - k, \;\; kc + s\gamma \,\big],$$

the union over integers $s \ge k$.

Proof. A member of $kA_{\gamma,c}$ is a sum of k terms of the walk, say with indices $i_1, \dots, i_k \ge 1$ (repetitions allowed):

$$\sum_{t=1}^{k} \big(c + \lfloor i_t \gamma \rfloor\big) \;=\; kc + \sum_{t=1}^{k} \lfloor i_t \gamma \rfloor.$$

Set $s := i_1 + \cdots + i_k$; since each $i_t \ge 1$, we have $s \ge k$. By the sum-of-floors bound (Lemma 3.2(4)) applied to the numbers $x_t = i_t \gamma$, whose sum is $s\gamma$,

$$s\gamma - k \;<\; \sum_{t=1}^{k} \lfloor i_t \gamma \rfloor \;\le\; s\gamma,$$

so the member lies in $(kc + s\gamma - k, \; kc + s\gamma]$. □

Alongside the windows we record where the sums *start*: nothing in $kA_{\gamma,c}$ is small.

Lemma 4.2 (Sums start high)**.** *If $d_2 - 1 < \gamma < d_2$, the smallest element of $kA_{\gamma,c}$ is $k(c + d_2 - 1)$. In particular every integer $b \le kc$ satisfies $b \notin kA_{\gamma,c}$: targets at or below kc are dodged automatically.*

Proof. Every term of the walk is at least its first term $a_1 = c + \lfloor \gamma \rfloor = c + d_2 - 1$, so every k-fold sum is at least $ka_1 = k(c + d_2 - 1)$ — and this value is attained, by summing a_1 with itself k times. Finally $k(c + d_2 - 1) > kc$, since $d_2 \geq 2$. □

4.2. **The avoidance criterion.** Fix a target $b > kc$ that we wish to keep out of $kA_{\gamma,c}$, and write

$$t \;:=\; b - kc \;\geq\; 1$$

for the *shifted target*. Subtracting kc throughout, Proposition 4.1 says: if $b \in kA_{\gamma,c}$, then $t \in (s\gamma - k,\; s\gamma]$ for some integer $s \geq k$ — equivalently, $s\gamma \in [\,t,\; t + k)$ for some $s \geq k$. Turning this around gives the criterion we will use forever after.

Definition 4.3 (Certified-safe slope)**.** Let $t \geq 1$ be an integer. A real $\gamma > 0$ is *certified safe for* t (*safe* for short) if

$$s\gamma \;\notin\; [\,t,\; t + k\,] \qquad \text{for every integer } s \geq 1.$$

The word *certified* is doing honest work: the condition is a checkable guarantee, deliberately *stronger* than the mere fact of avoidance — a slope can dodge b without earning the certificate, and we will see that happen. First, the guarantee itself:

Proposition 4.4 (Avoidance criterion)**.** *Let $d_2 - 1 < \gamma < d_2$, let $c \geq 0$, and let $b > kc$ be an integer. If γ is certified safe for $t = b - kc$, then $b \notin kA_{\gamma,c}$.*

Proof. Contrapositive. If $b \in kA_{\gamma,c}$, the discussion above produces an integer $s \geq k \geq 1$ with $s\gamma \in [t, t + k) \subseteq [t, t + k]$, so γ is not certified safe for t. □

And see it succeed once, before any fine print. Take $k = 3$, $d_2 = 4$, and the target $t = 46$: the slope $\gamma = 3.8$ is certified safe, because the products $s\gamma$ march upward in steps of 3.8 and the two nearest the window are

$$12 \cdot 3.8 = 45.6 \;<\; 46 \qquad \text{and} \qquad 13 \cdot 3.8 = 49.4 \;>\; 49,$$

with every smaller s falling shorter and every larger s overshooting further. So for any offset c, the number $b = 46 + 3c$ is *provably* absent from $3A_{3.8,\,c}$: the criterion converts one two-line check into an exclusion from an infinite sumset.

Now look closely: Definition 4.3 demands *more* than the proof of Proposition 4.4 needs, in two ways, and both are deliberate.

- It forbids $s\gamma \in [t, t + k]$ for *all* $s \geq 1$, though only $s \geq k$ can actually occur in a window. Ruling out extra values of s is a stronger demand on γ — harmless for the conclusion, and it frees every later argument from tracking which s are realizable.
- It uses the *closed* interval $[t, t + k]$, though the windows are half-open. Again this only makes "safe" harder to achieve, never easier — and closed intervals are more comfortable to work with (their endpoints belong to them, so "the least s with…" arguments have no edge cases).

The cost of this generosity is real but affordable: some slopes that would in fact dodge b are not certified safe. You will meet a concrete instance in Remark 4.5 — the walk of Example 2.1 really does miss 25, yet the criterion declines to vouch for it. We simply will not care: the free-gap lemma of §5 finds slopes that pass even the stricter test.

Reduction to slope avoidance. For a Beatty walk, "the sum of k members never equals b" has become a statement about the dial position alone: no multiple $s\gamma$ may land in the closed window $[t, t+k]$, where $t = b - kc$. The game of dodging an infinite sequence B is now a game of placing one real number γ clear of infinitely many forbidden zones.

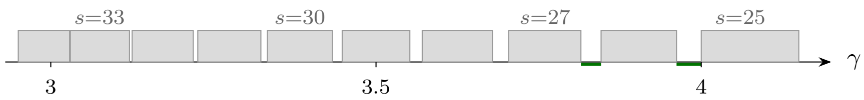


FIGURE 7. The unsafe set U_{100} for $k = 3$, on the stretch $2.95 \le \gamma \le 4.18$ of the dial: the interval $[\frac{100}{s}, \frac{103}{s}]$ for $s = 25, 26, \ldots, 34$. Safe gaps (two are marked) open up between consecutive unsafe intervals, wider toward the right; toward the left the intervals crowd together and, below $\gamma = 3 = k$, merge into a solid unsafe block (Proposition 4.6).

Remark 4.5 (The criterion is sufficient, not necessary). Definition 4.3 is deliberately stronger than avoidance, and the gap is visible in the running example. At $\gamma = 3.5$ the target $t = 25$ is not certified: $s = 8$ gives $3.5 \cdot 8 = 28 \in [25, 28]$. Yet $25 \notin 3A$, as Example 2.2 records — membership would require $s\gamma \in [25, 28)$, and the only candidate, 28, is excluded by the half-open window. The closed certification interval contains it; the true window does not.

At this slope the failure is systematic rather than accidental: multiples of 3.5 are integers or half-integers, so for any integer t the least multiple $u \ge t$ satisfies $u \le t + 3$, and every closed window $[t, t+3]$ is hit. Thus $\gamma = 3.5$ certifies nothing at all, while dodging infinitely many targets. The slopes selected in §5 are chosen to avoid this, and §6 shows that against the strongest adversary they must be irrational.

4.3. **The dial, and its unsafe zones.** Fix the shifted target t and look at the condition of Definition 4.3 from the dial's point of view. For each single $s \ge 1$,

$$s\gamma \in [t,\ t+k] \quad\Longleftrightarrow\quad \gamma \in \Big[\frac{t}{s},\ \frac{t+k}{s}\Big],$$

so the slopes ruled out by index s form a closed interval, and γ is safe for t exactly when it avoids the *unsafe set*

$$U_t \ :=\ \bigcup_{s\ge 1} \Big[\frac{t}{s},\ \frac{t+k}{s}\Big].$$

As s grows, these unsafe intervals march down the dial toward 0, shrinking as they go (both endpoints are proportional to $1/s$). Figure 7 shows the portion of U_{100} near the dial region we shop in, for $k = 3$.

Two features of this picture are the pivot of the whole story. First, the safe gaps — the white space between consecutive unsafe intervals — *exist* in the region $\gamma > k$, and we will learn to find them (§5). Second, below $\gamma = k$ there is nothing to find:

Proposition 4.6 (No safety below k)**.** *Let $t \ge 1$ be an integer. Every real γ with $0 < \gamma \le k$ is unsafe for t.*

Proof. We must find an integer $s \ge 1$ with $s\gamma \in [t, t+k]$, i.e. with

$$s \ \in\ \Big[\frac{t}{\gamma},\ \frac{t+k}{\gamma}\Big].$$

This is a real interval of length $k/\gamma \ge 1$, so it contains an integer s by Lemma 3.2(5). That integer is positive, since $s \ge t/\gamma > 0$. ∎

This is the threshold, seen from the dial. A walk whose stride is at most k cannot be certified to dodge *anything*: every single target's unsafe set swallows the entire dial region $(0, k]$. Our standing assumption $d_2 \ge k+1$ is exactly what places the shopping district

$(d_2 - 1,\ d_2)$ of Proposition 3.5 inside the living zone $\gamma > k$. (That the picture "no safety below k" hardens into the actual non-existence theorem below the threshold — not merely a failure of this particular criterion — is the business of Parts III and IV.)

What remains for the existence half is quantitative: given the target t, *how wide* are the safe gaps near the top of the dial, and how little of the dial does each new target destroy? That is the free-gap lemma, to which we now turn.

5. The free-gap lemma

We now build the engine of the existence proof. The task left over from §4.3: given a target t, find a safe stretch of dial — and find it *quantitatively*, because we will have to do this infinitely often, once per element of B, each time inside whatever room the previous rounds left us. The lemma of this section says: inside any dial interval that is not too narrow (compared to $1/t$), there is a safe subinterval that is not too much narrower. The two "not too"s are what the constant 192 will be made of in §6.

One more tool for the toolbox first.

Definition 5.1 (Ceiling)**.** For a real x, the *ceiling* $\lceil x \rceil$ is the unique integer m with $m-1 < x \le m$ — the least integer $\ge x$. (Uniqueness: as for the floor. Note $\lceil x \rceil = \lfloor x \rfloor$ when x is an integer, and $\lfloor x \rfloor + 1$ otherwise.)

5.1. The statement.

Lemma 5.2 (Free gap)**.** *Let $t \ge 32\, d_2^2$ be an integer, and let $I = [lo,\ hi]$ be a closed real interval with*

$$I \ \subseteq\ \left[\, d_2 - \tfrac{1}{2},\ d_2 \,\right] \qquad \textit{and} \qquad |I| \ =\ hi - lo \ \ge\ \frac{4d_2^2}{t}.$$

Then I contains a closed subinterval I' with

$$|I'| \ \ge\ \frac{d_2}{48\, t}$$

such that every *$\gamma \in I'$ is safe for t (Definition 4.3): $s\gamma \notin [t,\ t+k]$ for every integer $s \ge 1$.*

Read the shape of this before the proof. The input interval must have width at least $4d_2^2/t$; the output is guaranteed width at least $d_2/(48t)$. Both scale like $1/t$: a bigger target is dodged with proportionally finer dial adjustments. The ratio between them,

$$\frac{4d_2^2/t}{d_2/(48t)} \ =\ 192\, d_2,$$

is the number of times the output of one round fits into the input the next round demands — hold that thought until §6, where it becomes the theorem's constant.

A word on the constants before they arrive, because none of them is magic. The proof needs exactly one quantitative fact: the gap J it will name has width $t - k\, s_1$ over roughly s_1^2, and since $s_1 \approx t/d_2$ while $k \le d_2 - 1$, the numerator should be about s_1-sized — there is a whole "spare 1" in $d_2 - k$. Every fraction with a 32 in it below ($\frac{15}{32}$, $\frac{17}{32}$, and the hypothesis $t \ge 32d_2^2$) is just a concrete way of banking part of that spare 1 so that the chain of inequalities closes with room at each step; any similar choices would work and would only change the final 192. *Throughout the proof, a bracketed line* [like this] *tracks one running instance:* $d_2 = 4$, $k = 3$, $t = 10000$, $I = [3.5,\ 4]$ — so every symbolic step has a number attached.

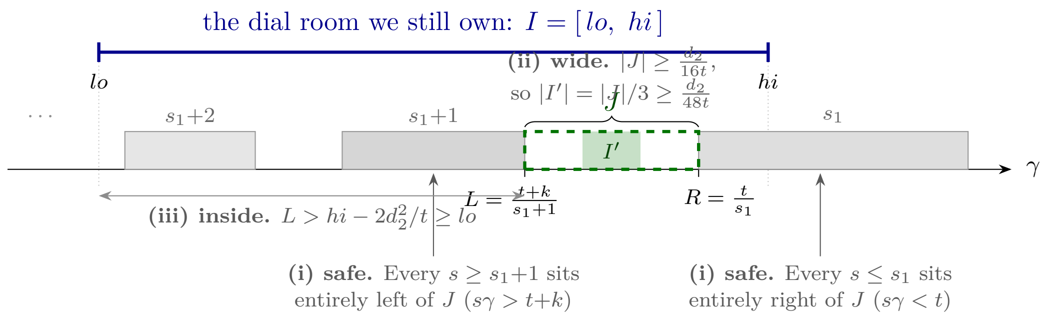


FIGURE 8. Anatomy of the free gap. The unsafe intervals $[\,t/s,\ (t{+}k)/s\,]$ march leftward and shrink as s grows, so naming the single gap J between indices s_1 and $s_1{+}1$ dodges *all* of them at once — no enumeration. The three checks of the proof are the three annotations: J is safe, J is wide enough that its closed middle third I' still has width $\geq d_2/(48t)$, and J lies inside the dial room I we still own. Numerically, for $d_2 = 4$, $k = 3$, $t = 10000$, $I = [3.5, 4]$: $s_1 = 2500$, $R = 4$, $L = 10003/2501 \approx 3.99960$, so $|J| = 1/2501 \approx 4.0 \times 10^{-4}$ — some 16 times the promised $d_2/(16t) = 2.5 \times 10^{-5}$.

5.2. **The idea: name one gap, count nothing.** The unsafe set $U_t = \bigcup_{s\geq 1}[t/s,\ (t+k)/s]$ is an infinite union, and a first instinct is to estimate how many of its intervals meet I and how much room they eat. That accounting can be made to work, but there is something better: we can point at *one specific gap* between two *specific* consecutive unsafe intervals, and prove that this single gap already avoids every unsafe interval at once, has the width we need, and sits inside I. No counting, no union bounds — three clean checks on one explicitly named object.

Which gap? Look at Figure 7 again: near the top of the dial the unsafe intervals are the sparsest and the gaps the widest, so we aim as high in I as possible. Let

$$s_1 \ := \ \left\lceil \frac{t}{hi} \right\rceil \qquad \text{(the least integer } s \text{ with } t/s \leq hi\text{)},$$

so that the unsafe interval of index s_1 is the highest-placed one whose left endpoint has not yet escaped past the top of I. Our gap is the space just below that interval and just above the next one:

$$L \ := \ \frac{t+k}{s_1+1}, \qquad R \ := \ \frac{t}{s_1}, \qquad J \ := \ (L,\ R)$$

— the open interval between the right end of unsafe interval $s_1 + 1$ and the left end of unsafe interval s_1.

[Instance: $s_1 = \lceil 10000/4 \rceil = 2500$, $R = 10000/2500 = 4$, $L = 10003/2501 \approx 3.99960$; so $J \approx (3.99960,\ 4)$.]

5.3. **The mechanism, alone.** Before any estimate, here is the entire idea, as a lemma that mentions no constants at all.

Lemma 5.3 (Geometric free gap)**.** *Let $t \geq 1$ and let $s_1 \geq 1$ be any integer. Then every γ in the open interval*

$$J \ = \ (L,\ R), \qquad L := \frac{t+k}{s_1+1}, \qquad R := \frac{t}{s_1}$$

is certified safe for t — provided J is nonempty.

Proof. Let $\gamma \in J$ and let $s \geq 1$ be any integer; we show $s\gamma \notin [t,\ t+k]$, splitting on the size of s.

- If $s \leq s_1$: then $t/s \geq t/s_1 = R > \gamma$, so $s\gamma < t$. The product undershoots the window.
- If $s \geq s_1 + 1$: then $(t+k)/s \leq (t+k)/(s_1+1) = L < \gamma$, so $s\gamma > t+k$. The product overshoots it.

□

That is the whole invention, and it is worth pausing over: the unsafe intervals $[t/s, (t+k)/s]$ march monotonically leftward as s grows, so the single gap between indices s_1 and $s_1{+}1$ is automatically clear of *all* of them — those with small s lie entirely to its right, those with large s entirely to its left. **One comparison per side replaces an infinite enumeration.**

[Instance: at $\gamma = 3.9998 \in J$, index $s = 2500$ gives $s\gamma = 9999.5 < 10000$, and $s = 2501$ gives $10003.5 > 10003$; every smaller s undershoots further, every larger s overshoots further.]

Everything that follows is bookkeeping: choosing s_1 so that J is nonempty, lands inside I, and is wide enough to keep the construction alive. If you are reading for the idea, you now have it, and may rejoin at §6.

5.4. **The audit: size and placement.** *(First pass: skippable.) This subsection is where every constant in the lemma comes from — and where the* 192 *of the main theorem is ultimately made. It proves nothing new about the mechanism: Lemma 5.3 already has that.*

Throughout, remember the standing facts: $2 \leq k \leq d_2 - 1$ (from $d_2 \geq k+1$), $d_2 \geq 3$, $t \geq 32d_2^2$, and $d_2 - \frac{1}{2} \leq lo \leq hi \leq d_2$. We take

$$s_1 \ :=\ \left\lceil \frac{t}{hi} \right\rceil$$

as in §5.2, so that Lemma 5.3 supplies safety the moment we know $J \neq \emptyset$ — which check (ii) below establishes by giving J a positive width.

We first pin down the size of s_1. Its defining property gives the two-sided estimate

$$\frac{t}{hi}\ \leq\ s_1\ <\ \frac{t}{hi} + 1. \tag{1}$$

From the left half and $hi \leq d_2$:

$$s_1\ \geq\ \frac{t}{d_2}\ \geq\ \frac{32d_2^2}{d_2}\ =\ 32\,d_2. \tag{2}$$

From the right half and $hi \geq d_2 - \frac{1}{2} \geq \frac{d_2}{2}$ (using $d_2 \geq 1$):

$$s_1 + 1\ <\ \frac{t}{hi} + 2\ \leq\ \frac{2t}{d_2} + 2\ \leq\ \frac{2t}{d_2} + \frac{t}{d_2}\ =\ \frac{3t}{d_2}\ \leq\ \frac{5t}{d_2}, \tag{3}$$

where $2 \leq t/d_2$ holds because $t/d_2 \geq 32d_2 \geq 96$.

Check (i): every point of J is safe for t. This is Lemma 5.3, already proved — it holds for any choice of s_1 whatever, and needs no hypothesis on t, I or the constants.

Check (ii): J is wide enough: $|J| = R - L \geq \dfrac{d_2}{16\,t}$. Put the difference over a common denominator:

$$R - L\ =\ \frac{t}{s_1} - \frac{t+k}{s_1+1}\ =\ \frac{t(s_1+1) - (t+k)s_1}{s_1(s_1+1)}\ =\ \frac{t - k\,s_1}{s_1(s_1+1)}.$$

So the width question is: how much does t exceed $k\,s_1$? *The goal:* a bound of the form $t - ks_1 \geq c\,s_1$ for some definite constant $c > 0$. *Why it should hold:* $s_1 \approx t/hi \approx t/d_2$ while

$k \le d_2 - 1$, so $ks_1 \approx t\,(d_2-1)/d_2$, leaving a margin of about $t/d_2 \approx s_1$ — a whole s_1, of which we only need a fixed fraction. *The choice:* we take $c = \frac{15}{32}$, picked so that the chain below closes against the hypothesis $t \ge 32\,d_2^2$ with a little room to spare. Now the chain. Using $s_1 < t/hi + 1 \le t/(d_2 - \frac{1}{2}) + 1$ and $k \le d_2 - 1$:

$$\Big(k + \tfrac{15}{32}\Big)\, s_1 \;\le\; \Big(d_2 - \tfrac{17}{32}\Big)\Big(\frac{t}{d_2 - \frac{1}{2}} + 1\Big) \;=\; t\,\frac{d_2 - \frac{17}{32}}{d_2 - \frac{1}{2}} + \Big(d_2 - \tfrac{17}{32}\Big).$$

The fraction equals $1 - \frac{1/32}{d_2 - 1/2}$, so the right side is

$$t \;-\; \frac{t}{32\,(d_2 - \frac{1}{2})} \;+\; \Big(d_2 - \tfrac{17}{32}\Big) \;\le\; t,$$

provided the middle term dominates the last, i.e. provided $\frac{t}{32(d_2 - 1/2)} \ge d_2 - \frac{17}{32}$. And it does: from $t \ge 32d_2^2$,

$$\frac{t}{32(d_2 - \frac{1}{2})} \;\ge\; \frac{d_2^2}{d_2 - \frac{1}{2}} \;\ge\; d_2 + \tfrac{1}{2} \;>\; d_2 - \tfrac{17}{32},$$

where the middle inequality is $d_2^2 \ge (d_2 + \frac{1}{2})(d_2 - \frac{1}{2}) = d_2^2 - \frac{1}{4}$. Altogether $(k + \frac{15}{32})s_1 \le t$, i.e.

$$t - k\,s_1 \;\ge\; \tfrac{15}{32}\,s_1.$$

Feeding this into the width formula, then using (3):

$$R - L \;\ge\; \frac{\frac{15}{32}\,s_1}{s_1(s_1+1)} \;=\; \frac{15}{32}\cdot\frac{1}{s_1+1} \;\ge\; \frac{15}{32}\cdot\frac{d_2}{5t} \;=\; \frac{3}{32}\cdot\frac{d_2}{t} \;\ge\; \frac{d_2}{16\,t}.$$

[Instance: $R - L = 4 - 10003/2501 = 1/2501 \cdot (4 \cdot 2501 - 10003) = 1/2501 \approx 4.0 \times 10^{-4}$, comfortably above $d_2/(16t) = 4/160000 = 2.5 \times 10^{-5}$.]

Check (iii): J sits inside I. The top is easy: $R = t/s_1 \le hi$ by the defining property of s_1. For the bottom we need $L \ge lo$, and we get it by showing J hangs only a little below hi. First, R is close below hi: from $s_1 < t/hi + 1$,

$$R \;=\; \frac{t}{s_1} \;>\; \frac{t}{t/hi + 1} \;=\; \frac{hi\,t}{t + hi} \;=\; hi\Big(1 - \frac{hi}{t + hi}\Big) \;>\; hi - \frac{hi^2}{t} \;\ge\; hi - \frac{d_2^2}{t}.$$

Second, J itself is short: using (1),

$$R - L \;=\; \frac{t - ks_1}{s_1(s_1+1)} \;\le\; \frac{t}{s_1^2} \;\le\; \frac{t}{(t/hi)^2} \;=\; \frac{hi^2}{t} \;\le\; \frac{d_2^2}{t}.$$

Adding: $L = R - (R - L) > hi - 2d_2^2/t$. Finally the hypothesis $hi - lo \ge 4d_2^2/t$ says exactly that $2d_2^2/t \le \frac{1}{2}(hi - lo)$, so

$$L \;>\; hi - \tfrac{1}{2}(hi - lo) \;\ge\; lo.$$

[Instance: $R = 4 = hi$, and $L \approx 3.99960 > 4 - 2\cdot 16/10000 = 3.9968 > 3.5 = lo$: the whole of J hangs just under the top of I.]

Conclusion. Check (ii) gives $|J| > 0$, so J is a nonempty open interval and Lemma 5.3 applies to it. Let I' be the closed middle third of J. Then $I' \subseteq J \subseteq I$ by check (iii); every $\gamma \in I'$ is certified safe for t by check (i); and $|I'| = |J|/3 \ge d_2/(48t)$ by check (ii). (This proves Lemma 5.2.) □

Why the middle third, and not J itself? Because the lemma promises a *closed* subinterval — the next round of the construction will want to apply the lemma again *inside* I', and closed intervals are what the nested-interval theorem of §6 consumes. The open gap J safely contains its closed middle third, and a third is the cheapest way to buy closedness with an explicit constant.

Remark 5.4 (Where the hypotheses are spent). The hypothesis $t \geq 32d_2^2$ is used three times, each a different demand. It gives $s_1 \geq 32d_2$ in (2); it makes $t/\big(32(d_2 - \frac{1}{2})\big) \geq d_2 + \frac{1}{2}$ in the width chain, which is what lets check (ii) close; and it bounds $2d_2^2/t$ against half the width of I in check (iii), which keeps J inside I. The third use fails first: for $d_2 = 4$, $k = 3$ and t of order 20, the named gap can protrude below lo.

6. Nested intervals, and the ratio $192\,d_2$

We can now assemble the existence half. Here is the theorem, stated for the record.

Theorem 6.1 (Existence)**.** *Let $k \geq 2$ and $1 \leq d_1 < d_2$ with $d_2 \geq k+1$. Then announcing $r = 192\,d_2$ wins the game: for every lacunary B with ratio $192\,d_2$ there is a (d_1, d_2)-sequence A with $(kA) \cap B = \emptyset$.*

The plan: fix the Adversary's B, choose a good offset c, convert each element of B into a shifted target, and then apply the free-gap lemma over and over — each application shrinking our interval of candidate slopes, each shrunken interval still wide enough for the next application *because B is lacunary.* A limit point of the shrinking intervals is a slope safe for every target at once.

6.1. **One more axiom: the nested interval theorem.** The construction produces an infinite chain of intervals, each inside the last, and needs a point lying in all of them. That such a point exists is a fact about the real number line itself.

> **Nested interval theorem.** If $I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots$ are nonempty closed bounded intervals of real numbers, then there is at least one real number γ^* belonging to every I_j.

Why believe it? Write $I_j = [lo_j,\ hi_j]$. The left endpoints $lo_0 \leq lo_1 \leq lo_2 \leq \cdots$ form a nondecreasing sequence, bounded above (by hi_0): it keeps rising yet can never pass hi_0, so it must pile up against some limiting value γ^* — and making "pile up against some value" precise is exactly what completeness asserts. This γ^* is $\geq$ every lo_j and $\leq$ every hi_j (if $\gamma^* > hi_{j_0}$ for some j_0, then some lo_j would eventually exceed $hi_{j_0} \geq hi_j$, impossible). The honest fine print: the existence of that limiting value is the *completeness* of the real numbers — the defining property that separates $\mathbb{R}$ from the rationals, where the theorem is false (nested intervals can squeeze down onto $\sqrt{2}$, which a rational line does not contain). We take completeness as an axiom of the real line, as every treatment must at some level; a rigorous construction of $\mathbb{R}$ is a beautiful topic that lives one course away from this paper. Note the theorem needs *closed* intervals: the open intervals $(0, 1/n)$ are nested with empty intersection — this is why the free-gap lemma delivered a closed I'.

6.2. **Clearing the small elements of B.** Fix $B = \{b_1 < b_2 < \cdots\}$, lacunary with ratio $r = 192\,d_2$. The dial criterion of §4 handles targets above kc; elements of B at or below kc are dodged automatically (Lemma 4.2). So we choose the offset to split B cleanly. Since B is infinite and increasing, some index j_0 has

$$b_{j_0} \;\geq\; 4\,d_2^2;$$

fix such a j_0 and set

$$c \;:=\; b_{j_0} + 1.$$

Every element b_i with $i \le j_0$ satisfies $b_i \le b_{j_0} < kc$, hence is dodged automatically, whatever slope we pick. For the remaining elements define the shifted targets

$$t_i \;:=\; b_i - kc \qquad (i > j_0).$$

Three facts about these targets, each a short computation with the lacunarity.

(1) The targets are positive — indeed the first one is already huge. Using $b_{j_0+1} \ge 192\, d_2\, b_{j_0}$ and $kc = k(b_{j_0}+1) \le d_2\,(b_{j_0}+1) \le 2d_2 b_{j_0}$ (as $b_{j_0} \ge 1$):

$$t_{j_0+1} \;=\; b_{j_0+1} - kc \;\ge\; 192\, d_2\, b_{j_0} - 2\, d_2\, b_{j_0} \;=\; 190\, d_2\, b_{j_0} \;\ge\; 190\, d_2 \cdot 4d_2^2 \;\ge\; 32\, d_2^2,$$

comfortably meeting the free-gap lemma's threshold $t \ge 32d_2^2$. Later targets are larger still, since B increases.

(2) Shifting preserves the ratio. For $x \ge y > \tau \ge 0$ one has

$$\frac{x-\tau}{y-\tau} \;\ge\; \frac{x}{y}.$$

Cross-multiplying by the positive quantities y and $y-\tau$, this is $y(x-\tau) \ge x(y-\tau)$, i.e. $xy - \tau y \ge xy - \tau x$, i.e. $\tau x \ge \tau y$ — true since $x \ge y$ and $\tau \ge 0$. Apply this with $x = b_{i+1}$, $y = b_i$, $\tau = kc$ (valid: for $i > j_0$, $b_i \ge b_{j_0+1} \ge 192 d_2 b_{j_0} > 2d_2 b_{j_0} \ge kc$, so $y > \tau$):

$$\frac{t_{i+1}}{t_i} \;\ge\; \frac{b_{i+1}}{b_i} \;\ge\; 192\, d_2 \qquad (i > j_0). \tag{4}$$

Subtracting the same amount from both terms of a ratio only helps the ratio — the targets inherit the lacunarity of B, undiluted.

(3) If every target is dodged, we win. Suppose $\gamma^* \in (d_2-1,\ d_2)$ is safe for every t_i, $i > j_0$. Then by the avoidance criterion (Proposition 4.4), no b_i with $i > j_0$ lies in $kA_{\gamma^*,c}$; and no b_i with $i \le j_0$ does either, by the automatic clearance above. So $(kA_{\gamma^*,c}) \cap B = \emptyset$, and $A_{\gamma^*,c}$ is a legal (d_1, d_2)-sequence by Proposition 3.5.

6.3. **The nest.** Everything is now in place. Start from the closed interval

$$I_0 \;:=\; \bigl[\, d_2 - \tfrac{1}{2},\;\; d_2 - \tfrac{1}{4} \,\bigr],$$

of width exactly $\frac{1}{4}$. (We stop at $d_2 - \frac{1}{4}$ rather than d_2 out of tidiness only; any closed subinterval of $[d_2 - \frac{1}{2}, d_2]$ of width $\frac{1}{4}$ would do. Note $I_0 \subseteq (d_2-1, d_2)$ since $d_2 \ge 3$.)

Round 1. Apply the free-gap lemma to the first target t_{j_0+1} and the interval I_0. Its hypotheses hold: $t_{j_0+1} \ge 32d_2^2$ by fact (1), and

$$|I_0| \;=\; \tfrac{1}{4} \;\ge\; \frac{4d_2^2}{t_{j_0+1}} \quad\Longleftrightarrow\quad t_{j_0+1} \ge 16\, d_2^2,$$

true again by fact (1). Out comes a closed $I_1 \subseteq I_0$, safe for t_{j_0+1}, with $|I_1| \ge d_2/(48\, t_{j_0+1})$.

Round $j+1$, given round j. Suppose inductively that the closed interval I_j is safe for the targets $t_{j_0+1}, \ldots, t_{j_0+j}$ and satisfies

$$|I_j| \;\ge\; \frac{d_2}{48\, t_{j_0+j}}.$$

We want to apply the free-gap lemma to the next target $t := t_{j_0+j+1}$ inside I_j. The only hypothesis in doubt is the width demand $|I_j| \ge 4d_2^2/t$, which in view of the inductive width bound follows from

$$\frac{d_2}{48\, t_{j_0+j}} \;\ge\; \frac{4\, d_2^2}{t_{j_0+j+1}} \quad\Longleftrightarrow\quad t_{j_0+j+1} \;\ge\; 192\, d_2 \cdot t_{j_0+j},$$

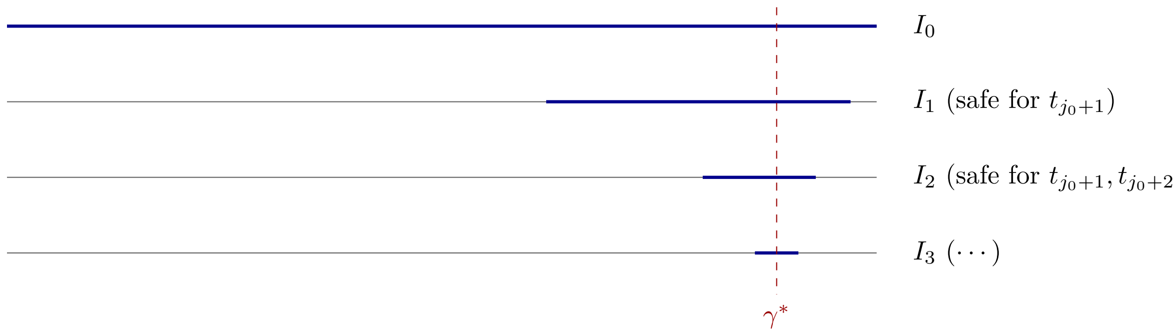


FIGURE 9. The nest, schematically. The widths are wildly not to scale, and cannot be: at $d_2 = 4$ each interval is 768 times narrower than its parent, so a true-scale second round would already be thinner than the ink. That violence is the message — the free-gap lemma spends a factor $192\, d_2$ of width every round, and the Adversary's forced growth refills exactly that factor before the next target arrives. Completeness of the reals guarantees a point γ^* surviving all trims.

which is precisely (4). *This is the entire ledger of the constant:* $192 = 4 \times 48$, the free-gap lemma's input demand $(4d_2^2/t)$ divided by its output guarantee $(d_2/(48t))$. The lacunarity ratio of B is exactly what refills the width the previous round spent. The lemma returns a closed $I_{j+1} \subseteq I_j$, safe also for t_{j_0+j+1}, of width $\geq d_2/(48\, t_{j_0+j+1})$ — the inductive hypothesis for the next round.

The bookkeeping, as a ledger — each row is one round:

Round	target	width needed	width delivered	who pays the gap
1	t_{j_0+1}	$4d_2^2/t_{j_0+1}$	$\geq d_2/(48\, t_{j_0+1})$	$\lvert I_0 \rvert = \frac{1}{4}$: ample
$j{+}1$	t_{j_0+j+1}	$4d_2^2/t_{j_0+j+1}$	$\geq d_2/(48\, t_{j_0+j+1})$	lacunarity: $t_{j_0+j+1} \geq 192 d_2\, t_{j_0+j}$

Each round *spends* a factor $192\, d_2$ of width (input demand over output guarantee) and the Adversary's forced growth *refills* exactly that factor before the next target arrives. Figure 9 draws the nest.

The limit. The rounds produce nonempty closed bounded nested intervals $I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots$; the nested interval theorem yields a real γ^* in all of them. For every $i > j_0$, the slope γ^* lies in I_{i-j_0}, which is safe for t_i; so γ^* is safe for every target simultaneously. By fact (3) of §6.2, the Beatty walk $A_{\gamma^*,c}$ is a legal (d_1, d_2)-sequence with $(kA_{\gamma^*,c}) \cap B = \emptyset$. (This proves Theorem 6.1.) □

6.4. **One win, worked end to end.** Before the summary, let us actually play a round of the game and win it, with numbers throughout. Take

$$k = 3, \qquad (d_1, d_2) = (1, 4), \qquad r = 192\, d_2 = 768,$$

and let the Adversary answer with the powers of 800:

$$B \;=\; \{\, 800,\ 640{,}000,\ 512{,}000{,}000,\ 409{,}600{,}000{,}000,\ \ldots\}, \qquad b_i = 800^i.$$

This is legal — $b_{i+1}/b_i = 800 \geq 768$ — and it is close to the slowest-growing play the ratio permits, hence the densest legal adversary, so the bookkeeping will be tight. Now we answer.

The offset. We need $b_{j_0} \geq 4d_2^2 = 64$: already $b_1 = 800$ qualifies, so $j_0 = 1$ and

$$c \;=\; b_1 + 1 \;=\; 801, \qquad kc \;=\; 2403.$$

The single element $b_1 = 800$ is now below $\min kA \geq k(c + d_2 - 1) = 3 \cdot 804 = 2412$, so it is dodged for free, whatever slope we choose. Every later element becomes a shifted target:

$$t_2 = 800^2 - 2403 = 637{,}597, \qquad t_3 = 800^3 - 2403 = 511{,}997{,}597, \qquad t_4 = 409{,}599{,}997{,}597.$$

(Check the threshold: $t_2 = 637{,}597 \geq 32d_2^2 = 512$, amply. And check the ratios survive the shift: $t_3/t_2 \approx 803.0$ and $t_4/t_3 \approx 800.004$, both ≥ 768, as (4) promised.)

The rounds. Start at $I_0 = [\,3.5,\ 3.75\,]$, of width $\frac{1}{4}$, and turn the crank:

Round	target t	width *needed* $\frac{4d_2^2}{t}$	width *we have*	width *delivered* $\lvert I_j \rvert$
1	637,597	1.00×10^{-4}	2.5×10^{-1}	1.47×10^{-6}
2	511,997,597	1.25×10^{-7}	1.47×10^{-6}	1.83×10^{-9}
3	409,599,997,597	1.56×10^{-10}	1.83×10^{-9}	2.29×10^{-12}

Read the middle two columns down: *what we have always exceeds what the next round needs.* Round 1's numbers, in full: $s_1 = \lceil 637597/3.75 \rceil = 170{,}026$, so

$$R = \frac{637597}{170026} \approx 3.749997, \qquad L = \frac{637600}{170027} \approx 3.749993, \qquad |J| = \frac{127519}{28909010702} \approx 4.41 \times 10^{-6},$$

and I_1 is the closed middle third of J — width 1.47×10^{-6}, comfortably above the guaranteed $d_2/(48t_2) \approx 1.31 \times 10^{-7}$. After three rounds the surviving stretch is

$$I_3 \;\approx\; [\,3.7499955633047, \quad 3.7499955633071\,],$$

and the rounds continue forever, each one pinning down a few more digits. The common point is a specific real number $\gamma^* \approx 3.74999556330\ldots$, and the walk

$$A \;=\; \{\, 801 + \lfloor i\,\gamma^* \rfloor \;:\; i \geq 1 \,\} \;=\; \{\, 804,\ 808,\ 812,\ 815,\ 819,\ 823,\ 827,\ 830,\ \ldots \}$$

— steps of 4 and 3, all inside $[1, 4]$ — has $(3A) \cap B = \emptyset$. That is a win, in full.

Where the constant lives. Compare each round's *delivery* against the next round's *demand*:

$$\frac{d_2/(48\,t_j)}{4d_2^2/t_{j+1}} \;=\; \frac{1}{192\,d_2} \cdot \frac{t_{j+1}}{t_j} \;\approx\; \frac{800}{768} \;=\; 1.042.$$

Four percent to spare, every round, forever — and that four percent is exactly the margin by which the Adversary's ratio 800 exceeds the demanded $192d_2 = 768$. Had the ratio been 760, each round's delivery would have fallen short of the next round's demand, and *this* induction would no longer close. (Note precisely what that does and does not say: it is a statement about the bookkeeping of this proof, not about the game. Whether 760 is in fact a winning announcement at these parameters is a question this document does not answer — the constant 192 is what our estimates can pay for, and §6's closing remark already disclaimed any belief that it is optimal.)

Proof of Theorem 6.1: summary. The existence proof in one breath: aim a Beatty walk's slope dial at the top of the permitted window; each element of B forbids a sparse family of dial positions, but the free-gap lemma always finds a safe sub-stretch losing only a factor $192\,d_2$ of width; lacunarity of B regains exactly that factor before the next element arrives; completeness of the reals hands us a single slope surviving every round.

Two closing remarks, and then a promised debt about irrationality. First, nothing was random and nothing was irrational-by-fiat: the winning slope γ^* is whatever real number the nest closes on, and no argument above ever asked anything else about it. Second, the constant $192\,d_2$ is an artifact of our bookkeeping (the middle third, the choice $\frac{15}{32}$, the crude bound (3)); we make no claim that it is anywhere near optimal.

Why the surviving slope must be irrational. *Daylight is necessary but nowhere near sufficient — the true resource is the continuum.* An arithmetic progression with step $d_2 \geq k+1$ has splendid daylight, yet it loses: its gap word is periodic, §10 will show that every eventually-periodic walk's sumset contains a full congruence-class tail *whatever the parameters*, and §8 builds a single legal B — available at any prescribed ratio, including our own $192\, d_2$ — that meets every such sumset. More: a *rational* slope $\gamma = u/v$ makes the Beatty walk's gap word periodic with period v (Lemma 3.2(2) gives $\lfloor (i+v)\gamma \rfloor = \lfloor i\gamma \rfloor + u$), so when the Adversary plays that certificate B, the slope γ^* our nest produces *cannot be rational.* (Two citations forward, both legitimate: Lemma 10.1 is proved without using $d_2 \leq k$ — see Remark 10.2 — and what we borrow from §8 is the *construction* of the certificate B, which is likewise parameter-free, not the conditional statement of Lemma 8.6(2).) The nested-interval theorem is therefore not a bookkeeping convenience: the completeness of the real line — an entire interval of dial positions, almost all of them irrational — is precisely the resource the existence proof spends, and the threshold $d_2 \geq k+1$ is precisely the condition under which that interval exists.

State of play. We now know: above the threshold, $r = 192\, d_2$ wins — part (1) of Theorem A is proved. What remains: part (2), non-existence below the threshold, in its strong form. Next: convert "defeat every walk" into a property checked one walk at a time.

Remark 6.2 (Only the tail of B matters)*.* Lacunarity entered solely through (4), one ratio per consecutive pair, and only for indices beyond j_0. The argument therefore survives unchanged if B satisfies $b_{i+1} \geq 192\, d_2\, b_i$ merely *eventually*: choose j_0 past the index where the ratio bound begins, which is the only step in §6.2 that needs adjusting.

Part III. **Non-existence: the reduction**

7. Interlude: a mini-course on remainders

Why does the mathematics change character here? Part II lived on the real number line: when $d_2 \geq k+1$ the windows leave daylight, and a continuum of dial positions was the resource we spent. Below the threshold that resource is gone — the windows crush into a solid block, and no real-valued tuning can help. But look *inside* the block: the integers a walk's sums actually reach arrange themselves into rigid repeating patterns — arithmetic progressions, congruence classes. Tracking rigid repetition is a job for the arithmetic of clock faces, and that is the toolkit we now build.

From here to the end, the natural language of the proof is the arithmetic of *remainders* — what mathematicians call modular arithmetic. Part II never needed it; Parts III and IV use it constantly. This interlude builds all of it, with proofs, in a few pages. If you have seen modular arithmetic before, skim the statements — especially the covering lemma (Lemma 7.6), which the rest of the document uses more than a dozen times — and move on.

7.1. **Divisibility, and division with remainder.** For integers $d \neq 0$ and x, say d *divides* x, written $d \mid x$, if $x = d\,c$ for some integer c. So $7 \mid 42$, $7 \nmid 40$, every d divides 0, and 1 divides everything.

Lemma 7.1 (Division with remainder)**.** *Let $m \geq 1$ and let x be any integer. There are unique integers q (the quotient) and r (the remainder) with*

$$x \;=\; q\,m + r, \qquad 0 \leq r < m.$$

Proof. Existence: take $q := \lfloor x/m \rfloor$ and $r := x - qm$; then $q \le x/m < q+1$ (Definition 3.1) gives $0 \le r < m$ after multiplying by m. Uniqueness: if $qm + r = q'm + r'$ with both r, r' in $[0, m)$, then $m(q - q') = r' - r$ is a multiple of m of absolute value $< m$, hence 0. ☐

7.2. **Greatest common divisors, and Bézout's lemma.** The *greatest common divisor* $\gcd(a, b)$ of integers a, b (not both 0) is the largest integer dividing both. Call a and b *coprime* if $\gcd(a, b) = 1$: they share no factor beyond 1. For a finite set G, $\gcd(G)$ is the largest integer dividing every element.

The single deepest fact we need about gcds — and it is not very deep — is that the gcd can be *built* out of a and b:

Lemma 7.2 (Bézout)**.** *Let a, b be integers, not both 0. Then there exist integers x, y with*
$$a\,x + b\,y \;=\; \gcd(a, b).$$

Proof. Consider all integers of the form $ax + by$ (x, y ranging over all integers, positive and negative) and let d be the *smallest positive* one. Two things justify that phrase: at least one such integer is positive (take $x = a$, $y = b$, giving $a^2 + b^2 > 0$), and *every nonempty set of positive integers has a least element* — the well-ordering principle of Appendix A, which is what makes "the smallest" meaningful. We claim $d = \gcd(a, b)$.

First, $d \mid a$: divide with remainder, $a = qd + r$ with $0 \le r < d$; then
$$r \;=\; a - q\,d \;=\; a - q\,(ax + by) \;=\; a\,(1 - qx) + b\,(-qy)$$
is itself of the form $ax' + by'$, nonnegative and smaller than d — by minimality of d, r cannot be positive, so $r = 0$. Likewise $d \mid b$. So d is a common divisor. Finally, any common divisor c of a and b divides $ax + by = d$; in particular $\gcd(a, b) \le d$ and $\gcd(a, b) \mid d$, while $d \le \gcd(a, b)$ since d is a common divisor. Hence $d = \gcd(a, b)$. ☐

Lemma 7.3 (Euclid's cancellation)**.** *If $m \mid u\,\alpha$ and $\gcd(m, \alpha) = 1$, then $m \mid u$.*

Proof. Bézout gives $\alpha x + my = 1$. Multiply by u: $u = u\alpha\,x + um\,y$. Both terms on the right are divisible by m (the first because $m \mid u\alpha$), so $m \mid u$. ☐

7.3. **Congruence.**

Definition 7.4. Fix $m \ge 1$. Integers x and y are *congruent modulo* m, written
$$x \equiv y \pmod{m}, \qquad \text{if} \qquad m \mid x - y.$$

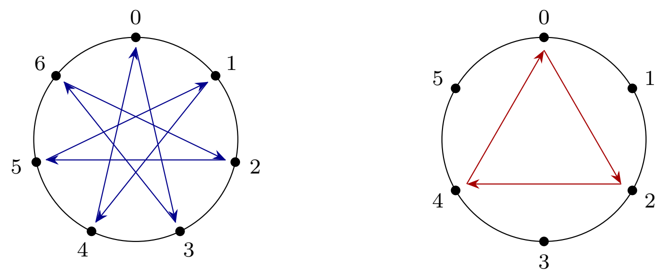
Equivalently: x and y leave the same remainder upon division by m (if $x = qm + r$ and $y = q'm + r'$ with $r, r' \in [0, m)$, then $m \mid x - y$ iff $m \mid r - r'$ iff $r = r'$, since $|r - r'| < m$). Congruence behaves like a looser equality; two operations are always legal:

Lemma 7.5. *If $x \equiv y$ and $u \equiv v \pmod{m}$, then $x + u \equiv y + v$ and $x\,u \equiv y\,v \pmod{m}$.*

Proof. $(x + u) - (y + v) = (x - y) + (u - v)$: a sum of multiples of m. And $xu - yv = (x - y)u + y(u - v)$: likewise. ☐

Division, by contrast, is *not* generally legal: $6 \equiv 16 \pmod{10}$, i.e. $2 \cdot 3 \equiv 2 \cdot 8$, yet $3 \not\equiv 8 \pmod{10}$. The correct substitute for division is coming in Lemma 7.7.

Fix m. The *congruence class* (or *residue class*) of r modulo m is the set of all integers congruent to r:
$$\{\,\dots,\; r - 2m,\; r - m,\; r,\; r + m,\; r + 2m,\; \dots\}$$
— an arithmetic progression with gap m, extending both ways. There are exactly m classes (one per remainder $0, 1, \dots, m - 1$), and every integer belongs to exactly one (Lemma 7.1).



mod 7, steps of 3:
every peg hit once

mod 6, steps of 2:
trapped in a sub-orbit

FIGURE 10. The covering lemma on two clock faces. Left: $\gcd(3,7) = 1$, and the multiples $0, 3, 6, 9, \ldots$ visit all seven pegs before returning — the star polygon is the lemma. Right: $\gcd(2,6) = 2$, and the multiples of 2 never leave the even pegs. Coprimality is exactly the difference.

Two reflexes to internalize: *a class is a progression*, so it contains arbitrarily large numbers; and *knowing x's class* means knowing x up to adding multiples of m.

7.4. **The covering lemma, and inverses.** Here is the workhorse. It will be cited constantly in Part IV, and silently used whenever we say "the multiples of α sweep out all residues".

Lemma 7.6 (Covering lemma)**.** *Let $m \geq 1$ and let α be coprime to m. Then the m numbers*

$$0 \cdot \alpha, \quad 1 \cdot \alpha, \quad 2 \cdot \alpha, \quad \ldots, \quad (m-1) \cdot \alpha$$

lie in m different *congruence classes modulo m — hence in* all *m classes, each exactly once.*

Proof. Suppose $i\alpha \equiv j\alpha \pmod{m}$ with $0 \leq i, j \leq m-1$. Then $m \mid (i-j)\alpha$, so $m \mid i-j$ by Euclid's cancellation (Lemma 7.3); since $|i-j| < m$, this forces $i = j$. So the m listed numbers occupy m distinct classes; as there are only m classes in total, every class is hit exactly once. □

Lemma 7.7 (Modular inverse)**.** *If* $\gcd(\alpha, m) = 1$*, there is an integer $\bar{\alpha} \in [0,\ m-1]$ with $\alpha\,\bar{\alpha} \equiv 1 \pmod{m}$; it is unique in that range, and we call it the* inverse *of α* modulo *m. Multiplying by $\bar{\alpha}$ undoes multiplying by α — the legal substitute for division.*

Proof. By the covering lemma, some $i \cdot \alpha$ with $i \in [0, m-1]$ lies in the class of 1: take $\bar{\alpha} := i$. Uniqueness: if $\alpha i \equiv \alpha i' \equiv 1 \pmod{m}$, then $i \equiv i \cdot (\alpha i') = (i\alpha) \cdot i' \equiv i' \pmod{m}$ (using Lemma 7.5 twice), and two congruent numbers in $[0, m-1]$ are equal. □

Example 7.8 (Solving a congruence)**.** Solve $5\,s \equiv 2 \pmod{7}$. The inverse of 5 modulo 7 is 3 (as $5 \cdot 3 = 15 \equiv 1$), so multiply both sides by 3: $s \equiv 3 \cdot 2 = 6 \pmod{7}$. The solutions are exactly the class $\{\ldots, 6, 13, 20, 27, \ldots\}$ — one class, an arithmetic progression. This "pin the variable to one class" move is used at a crucial point in §12 and §14, and throughout Part IV.

8. THE CERTIFICATE TRICK: ONE B AGAINST EVERY A

Part III begins the non-existence half. **Standing assumptions for Parts III and IV:** k, d_1, d_2 are integers with

$$k \geq 3, \qquad 1 \leq d_1 < d_2, \qquad d_2 \leq k.$$

The goal is part (2) of Theorem A in its full strength: for *any* prescribed sequence $R(1), R(2), \ldots$ of ratio lower bounds, a single B with $b_{i+1} \geq R(i)\, b_i$ whose intersection with kA is nonempty for *every* (d_1, d_2)-sequence A.

This section performs the first great simplification: it converts "defeat every A" — a statement about all walks at once — into a property to be checked *one walk at a time*.

Remark 8.1 (Where $k \geq 3$ is spent, and what happens at $k = 2$). Part II proudly worked from $k = 2$; Parts III–IV will not. The hypothesis $k \geq 3$ is consumed in two specific places: the width dichotomy's even case needs $d_2 \leq k - 1$ or a boundary argument requiring $k \geq 4$ (§§12–14), and the slot lemma needs a tail alphabet of three letters, which $d_2 = 2$ forbids outright. At $k = 2$, $d_2 = 2$ this paper's machinery genuinely does not decide the question, the problem as posed excludes it, and we make no claim there.

8.1. **Tail-covering.**

Definition 8.2 (Tail-covering)**.** A (d_1, d_2)-sequence A is *tail-covering* (for our fixed k) if there are integers $m \geq 1$, ρ, and X_0 such that

$$kA \supseteq \{\, x \geq X_0 \,:\, x \equiv \rho \pmod{m} \,\}:$$

the sumset contains *every* sufficiently large number in some fixed congruence class.

We met the phenomenon in Example 2.3: for an arithmetic progression with gap d, the sumset kA is a full tail of a class modulo d. The definition allows any modulus m, and this generosity is necessary — the modulus that works can exceed d_2, and kA need not contain all large integers.

Example 8.3. Let the gaps of A alternate $3, 5, 3, 5, \ldots$ and take $k = 5$ (so $d_2 = 5 \leq k$). The prefix sums (partial sums of the gaps) are $0, 3, 8, 11, 16, 19, 24, \ldots$: the numbers $8m$ and $8m + 3$. A 5-fold sum of members of A is $5a_1$ plus a sum of five prefix sums, and modulo 8 such a sum is $3j$ where $j \in \{0, 1, \ldots, 5\}$ counts how many of the five had the form $8m+3$. Since $3j \bmod 8$ takes the values $\{0, 3, 6, 1, 4, 7\}$, the sumset $5A$ lives in six of the eight classes modulo 8 and misses the classes $5a_1 + 2$ and $5a_1 + 5$ entirely. It *is* tail-covering — it eventually contains all of $5a_1 + 8\mathbb{Z}$, for instance — but with modulus $8 > d_2$, and it is far from containing all large integers.

Example 8.4 (The gap word $3, 5, 3, 5, \ldots$, in detail)**.** Take the walk of Example 8.3. Its prefix sums are exactly the numbers $8m$ and $8m + 3$: a prefix is m complete $(3, 5)$-blocks, each summing to 8, possibly preceded by a single 3. Choosing all five summand indices from the $8m$ family gives $5a_1 + 8(m_1 + \cdots + m_5)$, so $5A$ contains every integer $5a_1 + 8N$, $N \geq 0$ — a full class tail modulo 8. It also misses two classes outright: a 5-fold sum is $5a_1 + 8m + 3j$ with $j \leq 5$, and $3j \bmod 8$ runs over $\{0, 3, 6, 1, 4, 7\}$, never 2 or 5. The modulus here exceeds d_2 and the sumset is far from cofinite; both are why Definition 8.2 permits an arbitrary modulus.

8.2. **One sequence to meet them all.**

Suppose — this is what the rest of Part III will prove — that *every* admissible walk is tail-covering. Each walk hands us a triple (m, ρ, X_0): its sumset owns the class ρ modulo m from X_0 onward. The Adversary does not know which walk you will play, hence does not know which class; but the candidate pairs (m, ρ) can be arranged in a single infinite list, and the Adversary has infinitely many moves. So the Adversary hedges: build B to visit *every* class of *every* modulus, infinitely often, while growing as fast as anyone demands. Whatever walk you play, its class eventually contains an element of B beyond its X_0 — and that element is a collision.

To organize the hedging we need a list of all pairs.

Lemma 8.5 (The recurring list)**.** *There is a sequence of pairs $(m_1, \rho_1), (m_2, \rho_2), \dots$ in which every pair (m, ρ) with $m \geq 1$ and $0 \leq \rho < m$ appears infinitely often.*

Proof. List in blocks. Block N (for $N = 1, 2, 3, \dots$) lists all pairs with $m \leq N$, say in order of m then ρ: block 2 is $(1, 0), (2, 0), (2, 1)$; block 3 is $(1, 0), (2, 0), (2, 1), (3, 0), (3, 1), (3, 2)$; and so on. Concatenate the blocks. Each block is finite, so the concatenation is a legitimate infinite list, and a given pair (m, ρ) appears in every block from the m-th onward — infinitely often. $\square$

Lemma 8.6 (The certificate)**.** *Fix the recurring list of Lemma 8.5 and let $R : \mathbb{N} \to \mathbb{N}$ be an arbitrary prescribed ratio sequence — growing as fast as you like, with no regularity assumed. Define B by $b_1 := 1$ and, for $i \geq 1$,*

$$b_{i+1} \;:=\; \textit{the least integer} \;>\; \max\big(R(i)\, b_i,\; b_i,\; i\big) \;\; \textit{that is} \;\equiv \rho_{i+1} \pmod{m_{i+1}}.$$

Then:

(1) *B is a strictly increasing sequence of positive integers with $b_{i+1} \geq R(i)\, b_i$ for every i;*
(2) *if every (d_1, d_2)-sequence is tail-covering, then $(kA) \cap B \neq \emptyset$ for every (d_1, d_2)-sequence A.*

Proof. (1) Each b_{i+1} exists: the numbers $\equiv \rho_{i+1} \pmod{m_{i+1}}$ form an arithmetic progression, which contains numbers beyond any bound. By construction $b_{i+1} > b_i$ (strictly increasing, hence positive throughout) and $b_{i+1} > R(i)\, b_i$. Note also $b_{i+1} > i$, so $b_i \geq i$ for every $i \geq 2$: the sequence grows at least linearly, a small fact we need in a moment.

(2) Let A be any (d_1, d_2)-sequence, and let (m, ρ, X_0) witness its tail-covering. We must find an element of B that is $\equiv \rho \pmod{m}$ *and* $\geq X_0$ — such an element lies in kA by Definition 8.2. The pair (m, ρ) appears infinitely often in the recurring list, say at positions $i^{(1)} < i^{(2)} < \cdots$; at each such position $i \geq 2$ the construction made $b_i \equiv \rho \pmod{m}$. Since $b_i \geq i \to \infty$, some such position has $b_i \geq X_0$ as well. That b_i belongs to both B and kA. $\square$

Certificate mechanism. Sparseness cannot save the walker, because congruence classes are everywhere: however fast B is forced to grow, it can always take its next step *inside* any prescribed class. The certificate B cycles through all classes of all moduli forever, so it eventually lands, arbitrarily late, in whichever class a given walk's sumset owns. One sequence, built blind, defeats every tail-covering walk.

Notice what the certificate does *not* depend on: k, d_1 and d_2 never entered the construction — only the ratio sequence R and the recurring list (Figure 11). So the very same B works simultaneously for all parameter triples below the threshold. This is stronger than the theorem demands; it comes for free.

The non-existence half is now reduced to a statement about individual walks:

> **Goal of Part III.** *Every (d_1, d_2)-sequence with $d_2 \leq k$ (and $k \geq 3$) is tail-covering.*

The rest of Part III proves this, walk by walk, according to how the walk wobbles; one case will lean on the subset-sum lemma of Part IV.

State of play. We now know: one B, built blind, defeats every tail-covering walk — gap (2) of the ledger is closed. What remains: proving that below the threshold *every* admissible walk is tail-covering (gap (3)). Next: coordinates adapted to a single walk, and the case split that organizes the rest of the part.

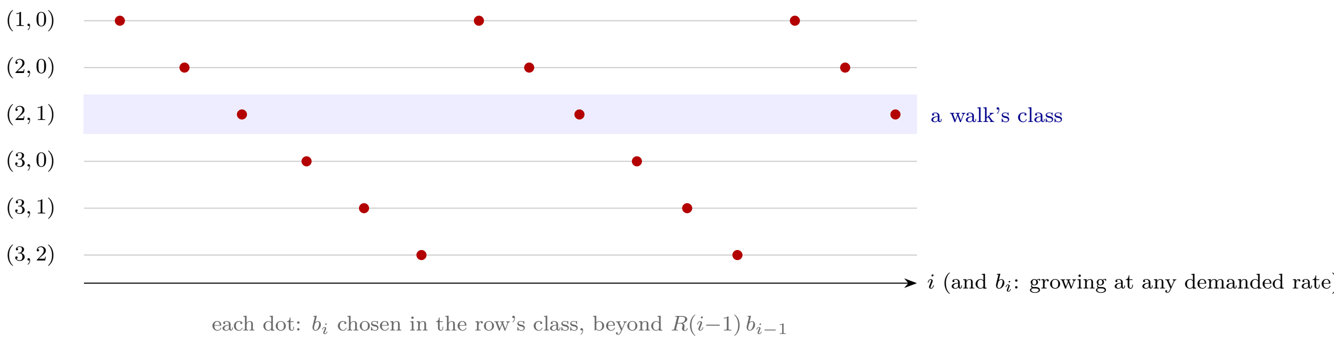


FIGURE 11. The certificate's weave. Rows are the pairs (m, ρ) of the recurring list; the Adversary's terms $b_1, b_2, \ldots$ (dots, marching rightward ever faster) visit every row again and again, forever. Whatever walk you play, its sumset owns some row from some point X_0 on (shaded) — and that row still receives visits beyond any X_0. Each such visit is a collision.

9. Gap words, tail alphabets, and the plan

To analyze a single walk we need coordinates adapted to its steps. This section sets them up, dispatches the two easiest kinds of walk, and lays out the battle plan for the rest.

9.1. **Prefix sums, and the sumset in step coordinates.** Let A be a (d_1, d_2)-sequence with gaps $g_n = a_{n+1} - a_n \in [d_1, d_2]$. Define the *prefix sums*

$$p_0 := 0, \qquad p_n := g_1 + g_2 + \cdots + g_n \quad (n \geq 1),$$

so that $a_{n+1} = a_1 + p_n$ for every $n \geq 0$: each term of the walk is the start plus the steps taken so far.

Lemma 9.1 (The sumset in step coordinates)**.**

$$kA \;=\; k\, a_1 \;+\; \{\, p_{i_1} + p_{i_2} + \cdots + p_{i_k} \;:\; i_1, \ldots, i_k \geq 0 \,\}.$$

Proof. A member of kA is $a_{j_1} + \cdots + a_{j_k}$ with each $j_t \geq 1$; substituting $a_{j_t} = a_1 + p_{j_t - 1}$ and setting $i_t := j_t - 1 \geq 0$ gives one direction, and the substitution reverses. ☐

From now on we study the set $kA - ka_1$ of *sums of k prefix sums*. Note that tail-covering is insensitive to the shift: kA contains a class tail if and only if $kA - ka_1$ does (with the residue shifted by ka_1).

9.2. **The gap word and the tail alphabet.** The *gap word* of A is the infinite sequence $g_1, g_2, g_3, \ldots$ read as a word — a string of "letters" drawn from the finite alphabet $\{d_1, \ldots, d_2\}$ (Figure 12 shows the coordinate systems this part will move between). Since the alphabet is finite, some letters occur infinitely often in the word and some (perhaps) only finitely often. Define the *tail alphabet*

$$G_\infty \;:=\; \{\, \delta \;:\; g_n = \delta \text{ for infinitely many } n \,\},$$

which is nonempty, and fix an index T beyond which *only* letters of G_∞ occur (possible: each of the finitely many non-recurring letters has a last occurrence; take T past all of them).

It pays to name the object the rest of this part actually studies.

Definition 9.2 (G-walk)**.** Let G be a finite set of positive integers. A *G-walk* is a strictly increasing sequence of positive integers all of whose consecutive gaps lie in G.

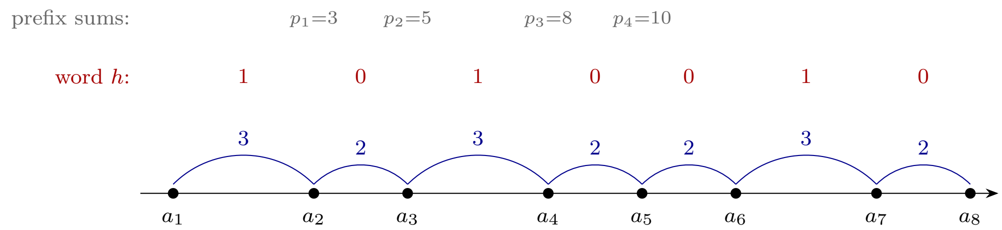


FIGURE 12. Four coordinate systems for one walk (gaps $3, 2, 3, 2, 2, 3, 2, \dots$; letters $\delta = 2$, $\delta' = 3$, so $e = 1$): the terms a_n on the line, the gaps g_n as arches, the binary word h (large step = 1), and the prefix sums $p_n = a_{n+1} - a_1$. §§9–14 translate between these constantly; this figure is the dictionary.

Every (d_1, d_2)-sequence is a G-walk with $G = \{d_1, d_1 + 1, \dots, d_2\}$, and, after passing to the tail, a G_∞-walk.

Naming this is not bookkeeping for its own sake. The point is that d_1 *now retires*: from here to the end of Part III, no argument refers to it again. What those sections use about a walk is its alphabet — how many letters, their gcd, and the largest of them — and nothing else. That matters immediately, because the rescaling of Lemma 9.5 divides every gap by $g > 1$ and so destroys the lower bound d_1. The rescaled object is emphatically *not* a (d_1, d_2)-sequence, and a theorem stated for (d_1, d_2)-sequences could not legitimately be applied to it. It is a G-walk whose letters are still $\leq d_2$, and that is every hypothesis the machinery ever consumes. We therefore state the results of §§12–14 for walks with a bound on their gaps, never for (d_1, d_2)-sequences.

Lemma 9.3 (Tail reduction)**.** *Let $A' := \{a_{T+1} < a_{T+2} < \cdots\}$ be the tail of A beyond T — itself a G_∞-walk, and in particular a walk with all gaps $\leq d_2$. If A' is tail-covering, so is A.*

Proof. Every sum of k members of A' is a sum of k members of A, so $kA \supseteq kA'$; a class tail inside kA' is a class tail inside kA. □

We will invoke this silently: "pass to the tail" means replace A by A', so that every gap lies in G_∞ and every letter of G_∞ still occurs infinitely often.

The analysis now splits on the *size* of the tail alphabet, and the first case is immediate:

Lemma 9.4 (One letter)**.** *If $G_\infty = \{\delta\}$, then A is tail-covering.*

Proof. Pass to the tail: all gaps equal δ, so the tail is an arithmetic progression with gap δ, and Example 2.3 (which is now official: the computation there is exact) shows its k-fold sumset is a full class tail modulo δ. □

One more normalization. If all the recurring letters share a common factor, we can divide it out:

Lemma 9.5 (Rescaling)**.** *Suppose (after passing to the tail) every gap of A lies in G_∞, and let $g := \gcd(G_\infty) > 1$. Define the* rescaled walk *A' by*

$$a'_n := \frac{a_n - a_1}{g} + 1 \qquad (n \geq 1),$$

a (G_∞/g)-walk, where $G_\infty/g := \{\delta/g : \delta \in G_\infty\}$, with $\gcd(G_\infty/g) = 1$ and maximum letter $\leq d_2/g \leq d_2$. If A' is tail-covering modulo m', then A is tail-covering modulo $g\,m'$.

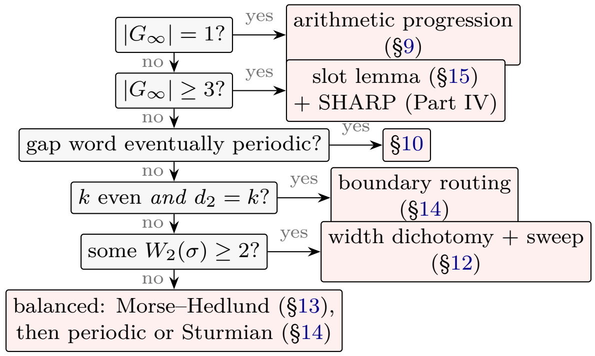


FIGURE 13. Routing a walk, after passing to the tail and rescaling to $\gcd(G_\infty) = 1$. Every admissible walk enters at the top and leaves through exactly one branch, each of which ends in *tail-covering*. The two-letter branches (the lower four) are where the work is; the assembly table of §25 is this picture in words.

Proof. Every gap of A is a multiple of g, so each $a_n - a_1$ is one too and the a'_n are integers, strictly increasing from $a'_1 = 1$; their gaps are g_n/g by construction. Unwinding the definition, $a_n = (a_1 - g) + g\, a'_n$, so a sum of k members of A is $k(a_1 - g) + g \cdot$(a sum of k members of A'):

$$kA \;=\; k(a_1 - g) \;+\; g \cdot kA'.$$

Now suppose $kA' \supseteq \{x \geq X'_0 : x \equiv \rho' \pmod{m'}\}$. Consider any $y \equiv k(a_1 - g) + g\rho' \pmod{g\,m'}$ with y large, and put $w := y - k(a_1 - g)$. Three small deductions. First, the congruence says $w = g\rho' + g\,m'\,c$ for some integer c — both terms visibly multiples of g — so $g \mid w$. Second, $x := w/g = \rho' + m'c \equiv \rho' \pmod{m'}$. Third, x is large along with y. So $x \in kA'$, and $y = k(a_1 - g) + g\,x \in kA$. Hence kA contains a full class tail modulo $g\,m'$. □

So from now on we may and do assume $\gcd(G_\infty) = 1$ (rescaling first if necessary, and remembering that the final modulus picks up a factor g — harmless, since tail-covering permits any modulus).

9.3. **The battle plan.** Here is the case tree for the Goal; each node names the section that closes it.

Tail alphabet (after rescaling)	Closed by
$\lvert G_\infty\rvert = 1$	Lemma 9.4 (arithmetic progression)
$\lvert G_\infty\rvert = 2$, gap word eventually periodic	§10
$\lvert G_\infty\rvert = 2$, k even and $d_2 = k$ (the boundary)	§14 (its own routing)
$\lvert G_\infty\rvert = 2$, otherwise:	
— irregular enough (wide band)	the sweep (§12)
— maximally regular (balanced)	Morse–Hedlund (§13),
	then the Sturmian endgame (§14)
$\lvert G_\infty\rvert \geq 3$	§15 (slot lemma) + Part IV

This is the *canonical* tree, drawn as a flowchart in Figure 13: the assembly in §25 will retrace exactly these branches, in exactly this order, with the closing lemma numbers filled in. (The third row may look unmotivated now — it is the one configuration the width machinery of §12 cannot reach, and it gets a dedicated argument.)

The two-letter case is the deep one: a two-letter walk has exactly one degree of freedom per step, and the walks that use that freedom *most evenly* — neither settling into a period nor drifting irregularly — are precisely the Sturmian walks, where the hardest analysis lives. Three or more letters, counterintuitively, is *easier*: more letters means more kinds of step to combine, and a bounded combinatorial gadget (the slot lemma) plus one finite theorem (Part IV) covers all such walks at once.

10. Eventually periodic gap words

A gap word is *eventually periodic* if beyond some point it repeats with a fixed period: there are $N_0 \geq 0$ and $\ell \geq 1$ with

$$g_{n+\ell} = g_n \qquad \text{for all } n > N_0.$$

Lemma 10.1 (Eventually periodic)**.** *If the gap word of A is eventually periodic, then A is tail-covering.*

Remark 10.2 (This one escapes the standing assumption). Part III runs under the standing hypothesis $d_2 \leq k$, but the proof below never uses it — no relation between d_2 and k is invoked anywhere in it. Lemma 10.1 is therefore the one result of this part that holds *above* the threshold as well, which is exactly what the closing remark of §6 needed when it argued that a rational slope loses whatever the parameters. Worth flagging, because everywhere else in Parts III and IV the standing assumption is genuinely in force.

Proof. Pass to the tail beyond N_0 (Lemma 9.3); relabel so that the gap word satisfies $g_{n+\ell} = g_n$ for *all* $n \geq 1$. Let

$$P \ := \ g_1 + g_2 + \cdots + g_\ell$$

be the *period sum* — the distance the walk advances in each full period. Then the prefix sums repeat with drift P: for $0 \leq j < \ell$ and $m \geq 0$,

$$p_{m\ell+j} \ = \ mP + p_j,$$

by grouping the first $m\ell$ gaps into m full periods (each summing to P) plus the leftover prefix.

Now build sums of k prefix sums using only the multiples of the period: choosing $i_t = m_t\ell$ (i.e. $j = 0$) for all t gives

$$p_{m_1\ell} + \cdots + p_{m_k\ell} \ = \ (m_1 + \cdots + m_k)\, P,$$

and $m_1 + \cdots + m_k$ ranges over every integer ≥ 0: each m_t is an arbitrary nonnegative integer, so the sum realizes 0 (all $m_t = 0$) and every larger value N (take $m_1 = N$ and the rest 0). By Lemma 9.1, $kA \supseteq ka_1 + P\,\{0, 1, 2, \dots\}$: a full tail of the class of ka_1 modulo P. □

Note how little of the period structure we used — only the sub-walk at the period marks, which is an arithmetic progression with gap P. The other residues $j \neq 0$ would give further classes (as in Example 8.3, which is the case $\ell = 2$, $P = 8$ of this proof), but one class is all tail-covering asks.

Example 10.3 (A periodic walk can cover every class)**.** For the repeating gap block $(2, 3, 2)$ — period $\ell = 3$, period-sum $P = 7$ — the prefix sums modulo 7 are $\{0, 2, 5\}$, and sums of three of them realize every residue modulo 7. So at $k = 3$ this walk's sumset eventually contains all large integers, not merely one class. Lemma 10.1 claims only one class because one is all that is needed.

11. Interlude: a mini-course on words, and a little analysis

The two-letter case will occupy us for four sections, and it runs on a small amount of theory — about *words* (infinite strings of symbols) and about *limits* — that deserves its own unhurried introduction. Nothing here mentions the problem; it is pure toolbox.

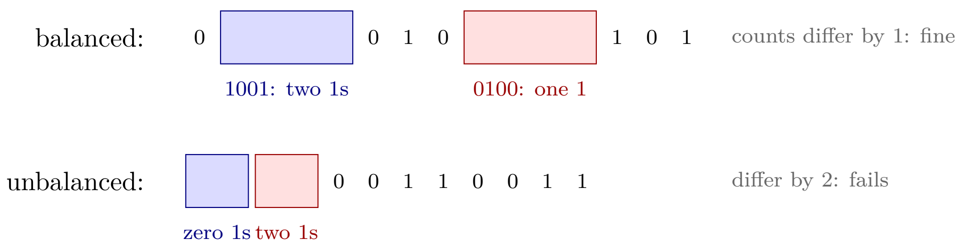


FIGURE 14. Balance, by sliding windows. Above, a balanced word: any two windows of the same length hold 1-counts differing by at most one, wherever you place them. Below, $0011\,0011\cdots$ fails at once — the windows 00 and 11 differ by two. Balance is the strongest possible "spread the 1s evenly" condition short of periodicity, and §13 shows it forces the word to track a straight line.

11.1. **Words.** An *alphabet* is a finite set of symbols; a *word* is a finite or infinite string of them. We care about *binary* words, alphabet $\{0,1\}$, written $h = h_1h_2h_3\cdots$ with $h_n \in \{0,1\}$. A *window* of h is a run of consecutive letters $h_{i+1}h_{i+2}\cdots h_j$ (which we address by its endpoints $i<j$; its *length* is $j-i$).

The *counting function* of h is

$$q_0 := 0, \qquad q_n := h_1 + h_2 + \cdots + h_n \quad \text{(the number of 1s among the first } n \text{ letters).}$$

The count of 1s in the window from i to j is then $q_j - q_i$. Two properties tie h and q together: q is nondecreasing with steps $q_n - q_{n-1} = h_n \in \{0,1\}$, and conversely any such staircase function comes from exactly one word.

Definition 11.1 (Balanced)**.** A binary word h is *balanced* if any two windows of equal length contain numbers of 1s differing by at most one:

$$j - i = j' - i' \quad \Longrightarrow \quad \big|(q_j - q_i) - (q_{j'} - q_{i'})\big| \le 1.$$

Balance is a strong uniformity requirement: wherever you cut out two equal stretches of the word, they must contain almost exactly the same number of 1s (Figure 14).

Example 11.2. (a) The constant words $000\cdots$ and $111\cdots$ are balanced, as is the alternating word $010101\cdots$ (a window of length $2m$ always has exactly m ones; length $2m+1$ has m or $m+1$). (b) The periodic word $001001001\cdots$ is balanced (check windows of each length modulo 3). (c) The periodic word $0011\,0011\cdots$ is *not* balanced: the windows 00 and 11 have 1-counts 0 and 2. (d) The *Fibonacci word* $0100101001001010010\cdots$ — built by iterating the substitution $0 \mapsto 01$, $1 \mapsto 0$ — is balanced and *not* eventually periodic. It will reappear in §14 as the paradigm of the hardest case. (Proving these two claims about it is not needed for our proof; the classification below works from balancedness alone, whatever word exhibits it.)

11.2. **Suprema, and limits.** Two notions from analysis, in the exact strength we need.

Supremum. A set S of real numbers that is nonempty and bounded above has a *least upper bound* $\sup S$: an upper bound of S that is $\le$ every other upper bound. (Its existence is, again, the completeness of the real line — the same axiom behind the nested interval theorem of §6; indeed each implies the other.) Likewise a nonempty set bounded below has a greatest lower bound $\inf S$. Two reflexes to internalize: every element of S is $\le \sup S$; and for any $\varepsilon > 0$, some element of S exceeds $\sup S - \varepsilon$ (else $\sup S - \varepsilon$ would be a smaller upper bound).

One warning that will matter: *a supremum need not belong to its set.* The set $\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots\}$ — the numbers $1 - \frac{1}{n}$ — has supremum 1, yet no element equals 1. When §13 defines a certain slope as a supremum, this distinction is load-bearing: read "sup" as *least upper bound*, never as *maximum*.

To see the exact supremum that section will use, take the alternating word $0101\cdots$, whose counting function we tabulated above, and list the quantities $(q_n - 1)/n$:

$$\tfrac{-1}{1},\quad \tfrac{0}{2},\quad \tfrac{0}{3},\quad \tfrac{1}{4},\quad \tfrac{1}{5},\quad \tfrac{2}{6},\quad \tfrac{2}{7},\quad \tfrac{3}{8},\ \dots\ =\ -1,\ 0,\ 0,\ 0.25,\ 0.2,\ 0.333,\ 0.286,\ 0.375,\ \dots$$

These creep upward toward $\frac{1}{2}$ without ever reaching it (the even terms are $\frac{n/2-1}{n} = \frac{1}{2} - \frac{1}{n}$). So the supremum is $\frac{1}{2}$ — the density of 1s in the word, exactly as it should be — and it is attained by no term. That is why §13 must build the slope as a supremum and only afterwards prove it is the limit.

Limits. A sequence $x_1, x_2, \dots$ of reals *converges* to x (written $x_n \to x$) if for every $\varepsilon > 0$, all but finitely many n have $|x_n - x| < \varepsilon$. We need only three facts, each immediate from the definition: (i) $1/n \to 0$; (ii) if $x_n \to x$ and $x_n \le C$ for all n, then $x \le C$ (a limit cannot escape a closed bound); (iii) if $x_n \to x$ and $x_n \ge c$ for all n, then $x \ge c$.

11.3. The discrete intermediate value theorem.

Lemma 11.3 (Discrete IVT)**.** *Let $v_0, v_1, \dots, v_N$ be integers with $|v_{i+1} - v_i| \le 1$ for every i. Then every integer between v_0 and v_N (inclusive) equals some v_i.*

Proof. Say $v_0 \le v_N$ (else reverse the list) and let u be an integer with $v_0 \le u \le v_N$. Let j be the *first* index with $v_j \ge u$ (one exists: $j = N$ qualifies). If $j = 0$, then $v_0 \ge u \ge v_0$, so $v_0 = u$. Otherwise $v_{j-1} < u$, so $v_{j-1} \le u - 1$; and $v_j \le v_{j-1} + 1 \le u$, which with $v_j \ge u$ gives $v_j = u$. □

A sequence of integers that never jumps by more than 1 cannot step over an integer. We will use this constantly: it is the discrete shadow of the "no-skipping" arguments that ran through Part II.

12. Two letters: the reachable band, and the sweep

We now take up the main case. **Standing setup for §§12–14:** after tail-passing and rescaling (§9), A is a walk with all gaps $\le d_2$ — a G_∞-walk in the sense of Definition 9.2, *not* in general a (d_1, d_2)-sequence — with tail alphabet $G_\infty = \{\delta < \delta'\}$ and

$$\gcd(\delta, \delta') = 1, \qquad e := \delta' - \delta \ge 1, \qquad \delta' = \delta + e \le d_2 \le k,$$

and both letters occur infinitely often. (The bound $\delta' \le d_2$ holds because every gap is $\le d_2$; after rescaling it could only improve. Note $\gcd(\delta, e) = \gcd(\delta, \delta') = 1$.) Encode the gap word in binary: write

$$g_n \ =\ \delta + e\, h_n, \qquad h_n \in \{0, 1\},$$

so small steps are 0s and large steps are 1s, and let q_n be the counting function of the binary word h.

12.1. The reachable band.

The prefix sums split along the same lines: $p_n = \sum_{j \le n}(\delta + e h_j) = \delta n + e\, q_n$. So a sum of k prefix sums, with indices $i_1, \dots, i_k \ge 0$ summing to s, is

$$p_{i_1} + \cdots + p_{i_k} \ =\ \delta\, s\ +\ e\left(q_{i_1} + \cdots + q_{i_k}\right),$$

and by Lemma 9.1,

$$kA - k\,a_1 \;=\; \{\,\delta s + e\,w \;:\; s \ge 0,\; w \in W_s\,\}, \qquad W_s := \Big\{\sum_{t=1}^{k} q_{i_t} \;:\; i_t \ge 0,\; \sum_{t=1}^{k} i_t = s\Big\}. \quad (5)$$

Everything now hangs on the sets W_s: for each "index budget" s, which totals w of 1-counts are achievable? Before the general lemma, compute a few by hand. Take the alternating word $h = 0101\cdots$ — so $h_1 = 0$, $h_2 = 1$, $h_3 = 0$, ..., and the counting function runs

$$q_0 = 0,\quad q_1 = 0,\quad q_2 = 1,\quad q_3 = 1,\quad q_4 = 2,\quad q_5 = 2,\quad q_6 = 3,\ \ldots$$

— and let $k = 3$. At budget $s = 4$ the configurations (unordered, since order never matters) and their values $q_{i_1} + q_{i_2} + q_{i_3}$ are

$$(4,0,0) \mapsto 2,\quad (3,1,0) \mapsto 1,\quad (2,2,0) \mapsto 2,\quad (2,1,1) \mapsto 1,\quad (1,1,2) \mapsto 1,$$

so $W_4 = \{1,2\}$. Running the same computation at each budget gives the band:

s	0	1	2	3	4	5	6	7	8
W_s	$\{0\}$	$\{0\}$	$\{0,1\}$	$\{0,1\}$	$\{1,2\}$	$\{1,2\}$	$\{2,3\}$	$\{2,3\}$	$\{3,4\}$
$w^-(s)$	0	0	0	0	1	1	2	2	3
$w^+(s)$	0	0	1	1	2	2	3	3	4

Two features leap out, and they are exactly the two halves of the next lemma: *no row has a hole*, and *each endpoint row climbs by* 0 *or* 1 *at a time.* Neither is an accident of this example.

Lemma 12.1 (The band is an interval)**.** *For every $s \ge 0$, the set W_s is a full interval of integers: $W_s = [\,w^-(s),\ w^+(s)\,] \cap \mathbb{Z}$ for some integers $w^-(s) \le w^+(s)$. Moreover both endpoints climb gently:*

$$w^{\pm}(s+1) - w^{\pm}(s) \;\in\; \{0,1\}.$$

Proof. Call a list $(i_1, \ldots, i_k)$ of nonnegative integers with sum s a *configuration* (order will never matter — what counts is which indices appear, with repetition allowed), and its *value* the total $\sum_t q_{i_t}$. A *move* takes one unit from one slot and gives it to another: it replaces some pair (i_a, i_b) by $(i_a - 1,\ i_b + 1)$ (allowed when $i_a \ge 1$). A move keeps the sum s and changes the value by

$$(q_{i_b+1} - q_{i_b}) - (q_{i_a} - q_{i_a-1}) \;=\; h_{i_b+1} - h_{i_a} \;\in\; \{-1,\ 0,\ 1\}.$$

Any configuration can be walked to any other by such moves. Let $(i_1, \ldots, i_k)$ and $(i'_1, \ldots, i'_k)$ be two configurations, and measure their disagreement slot by slot: $D := \sum_t |i_t - i'_t|$. Suppose $D > 0$. Since both configurations have the same total s, the slots where i_t exceeds i'_t and the slots where it falls short must both exist: an excess somewhere forces a deficit elsewhere. Pick a slot a with $i_a > i'_a$ and a slot b with $i_b < i'_b$. The move $(i_a, i_b) \mapsto (i_a - 1, i_b + 1)$ is legal ($i_a > i'_a \ge 0$, so $i_a \ge 1$), and it shrinks both $|i_a - i'_a|$ and $|i_b - i'_b|$ by one: D drops by 2. After finitely many such moves $D = 0$ and the configurations agree.

In particular there is a chain of moves from a minimum-value configuration to a maximum-value one; the values along the chain form an integer sequence with steps in $\{-1, 0, 1\}$ from $w^-(s)$ to $w^+(s)$, so by the discrete IVT (Lemma 11.3) every integer in $[w^-(s), w^+(s)]$ is a value. That is the interval claim, and Figure 15 is the mechanism.

For the endpoint increments, compare budgets s and $s+1$ in both directions. Two elementary moves are available. *Adding* a unit to some slot of a configuration for s produces a configuration for $s+1$ whose value has grown by $h_{i_a+1} \in \{0,1\}$ — it never falls and never

FIGURE 15. A configuration move. The k index-slots hold a total budget s; shifting one unit from a high slot to a low one keeps the total fixed and changes the value by a single letter's difference, hence by at most one. Since any configuration reaches any other through such moves, the achievable values step by at most one along the way — so they cannot skip an integer, and the reachable set W_s is a solid interval.

gains more than one. *Removing* a unit from a positive slot of a configuration for $s+1$ produces one for s whose value has dropped by $h_{i_a} \in \{0, 1\}$. Apply each move to each kind of extremal configuration:

Upper endpoint. Adding a unit to a maximizer for s exhibits a configuration for $s+1$ of value $\geq w^+(s)$, so $w^+(s+1) \geq w^+(s)$: the top never falls. Removing a unit from a maximizer for $s+1$ exhibits a configuration for s of value $\geq w^+(s+1) - 1$, so $w^+(s) \geq w^+(s+1) - 1$: the top never rises by more than one. Together, $w^+(s+1) - w^+(s) \in \{0, 1\}$.

Lower endpoint. Symmetrically: removing a unit from a minimizer for $s+1$ gives $w^-(s) \leq w^-(s+1)$ (the floor never falls), and adding a unit to a minimizer for s gives $w^-(s+1) \leq w^-(s) + 1$ (it never rises by more than one). Together, $w^-(s+1) - w^-(s) \in \{0, 1\}$. □

So for each s the reachable pairs (s, w) form a vertical integer segment, and as s grows these segments drift upward, each endpoint by 0 or 1 per step: a *band* across the (s, w)-plane. Write

$$\mathrm{width}(s) \;:=\; w^+(s) - w^-(s)$$

for the band's height. One more elementary bound, used twice below: since $q_i \leq i$ always, every configuration's value is at most $\sum_t i_t = s$, so

$$w^+(s) \;\leq\; s; \tag{6}$$

and taking the configuration $(s, 0, \dots, 0)$ shows $w^+(s) \geq q_s$, which tends to infinity because the letter 1 occurs infinitely often.

12.2. **The sweep.** Now the payoff. Recall from (5) that realizing a target integer x means writing $x = \delta s + ew$ with $w \in W_s$. If the band is *wide* — as wide as the alphabet spread requires — then every large x is realized:

Lemma 12.2 (Sweep)**.** *Suppose there is an S_0 with* $\mathrm{width}(s) \geq d_2 - 1$ *for all $s \geq S_0$. Then $kA - ka_1$ contains every sufficiently large integer; in particular A is tail-covering with modulus $m = 1$.*

Proof. Fix a large integer x (thresholds at the end). Which budgets s could realize x? From $x = \delta s + ew$ we need $\delta s \equiv x \pmod{e}$, and since $\gcd(\delta, e) = 1$ this pins s to a single residue class modulo e; call such s *admissible*, and for admissible s set

$$w(s) \;:=\; \frac{x - \delta s}{e} \;\in\; \mathbb{Z},$$

the height at which the band would have to be hit. As s steps up by e (to the next admissible budget), $w(s)$ steps *down* by exactly δ. Meanwhile, by Lemma 12.1, each band endpoint rises

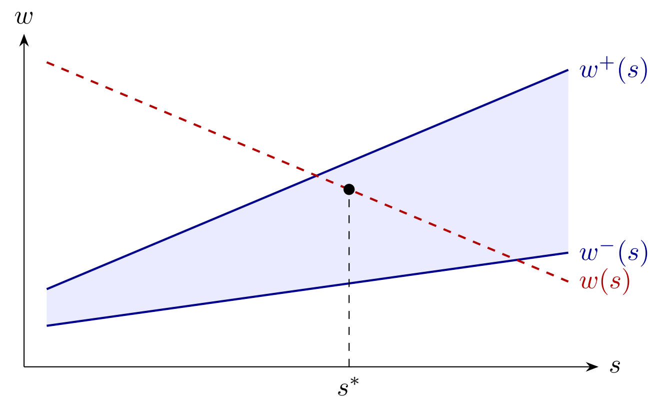


FIGURE 16. The sweep. The reachable band (shaded, endpoints $w^{\pm}$) rises gently as the budget s grows; the required height $w(s)$ for a fixed target x descends along the admissible budgets. If the band is wide, the descending line cannot skip it: some admissible s^* lands inside, and there x is realized.

by between 0 and e over those e unit steps. A descending target against a rising band: the picture from Figure 16 is that they must cross, and that a wide band cannot be jumped over in a single stride. We make both halves precise.

The target starts above the band and ends below it. At the small end: let $s_{\min}$ be the smallest admissible $s \geq S_0$; then $w(s_{\min}) \geq (x - \delta(S_0 + e))/e$, which for large x exceeds $s_{\min}$, hence exceeds $w^+(s_{\min})$ by (6) — explicitly, $x > (\delta + e)(S_0 + e)$ suffices. At the large end: enlarge S_0 once and for all so that $w^+(s) \geq \delta$ for every $s \geq S_0$ (possible since $w^+(s) \geq q_s \to \infty$; this depends only on the word, not on x). For the largest admissible s with $w(s) \geq 0$, the next admissible step would take w negative, so $0 \leq w(s) < \delta \leq w^+(s)$: at that budget the target sits at or below the band's top.

No skipping. Suppose, for contradiction, that no admissible $s \geq S_0$ has $w(s) \in [w^-(s),\ w^+(s)]$. The target descends (by δ per admissible step) from above the band to below its top; since it never lands inside, there must be two *consecutive* admissible budgets s' and $s' + e$ with

$$w(s') \ > \ w^+(s') \qquad \text{and} \qquad w(s'+e) \ < \ w^-(s'+e):$$

the leap over the band happens between them. First, s' is large: from $w(s'+e) \leq w^-(s'+e) \leq w^+(s'+e) \leq s'+e$ (again (6)),

$$\frac{x - \delta(s'+e)}{e} \ \leq \ s' + e \quad \Longrightarrow \quad s' + e \ \geq \ \frac{x}{\delta + e},$$

so $s' \geq S_0$ holds automatically once $x \geq (\delta + e)(S_0 + e)$ — the same threshold as before, so no new constraint. Now stack the inequalities. The target drops by exactly δ across the leap, and the band's floor rises by at most e:

$$w^+(s') \ < \ w(s') \ = \ w(s'+e) + \delta \ \leq \ w^-(s'+e) - 1 + \delta \ \leq \ w^-(s') + e - 1 + \delta.$$

Hence

$$\mathrm{width}(s') \ = \ w^+(s') - w^-(s') \ \leq \ \delta + e - 2 \ = \ \delta' - 2 \ \leq \ d_2 - 2,$$

contradicting the hypothesis $\mathrm{width}(s') \geq d_2 - 1$. So some admissible $s^* \geq S_0$ has $w(s^*) \in [w^-(s^*), w^+(s^*)]$; by Lemma 12.1 the integer $w(s^*)$ is actually attained, and $x = \delta s^* + e\, w(s^*) \in kA - ka_1$. $\square$

Look where the alphabet bound was spent: the contradiction needed width $\geq \delta' - 1$, and the hypothesis supplies $d_2 - 1 \geq \delta' - 1$. A band exactly one narrower could be leapt over — the sweep operates with no room to spare, a first taste of how tight the corner $d_2 = k$ will be.

12.3. **Wide or balanced: the width dichotomy.** The sweep shifts the burden: we must now show the band is wide. Width comes from *irregularity* of the word h — if some stretches are 1-rich and others 1-poor, configurations can exploit both. The cleanest measure uses just two indices. For $\sigma \geq 0$ define the *two-index width*

$$W_2(\sigma) \;:=\; \max\{\, q_i + q_j \;:\; i + j = \sigma,\; i, j \geq 0\,\} \;-\; \min\{\, q_i + q_j \;:\; i + j = \sigma,\; i, j \geq 0\,\}.$$

One bookkeeping lemma first — trivial to prove, used four times in this part, and worth having on the record rather than waving at.

Lemma 12.3 (Splitting)**.** *Split the k slots of a configuration into groups, and prescribe for each group the subtotal of its indices (subtotals summing to s). The groups then choose their indices independently, so among prescribed configurations the largest (smallest) achievable value is the sum of the groups' separate maxima (minima). And since prescribed configurations are particular configurations with total s,*

$$\operatorname{width}(s) \;\geq\; \sum_{\textit{groups}} (\textit{group max} - \textit{group min}).$$

Proof. The value $\sum_t q_{i_t}$ is the sum of the groups' values, and no constraint couples one group's index choices to another's — so extremes add. For the display: $w^+(s)$ is a maximum over *all* configurations, hence at least the prescribed maximum; symmetrically $w^-(s)$ is at most the prescribed minimum. $\square$

Lemma 12.4 (Width dichotomy)**.**

(a) *Suppose $W_2(\sigma_0) \geq 2$ for some σ_0, and suppose k is odd or $d_2 \leq k-1$. Then* $\operatorname{width}(s) \geq d_2 - 1$ *for all sufficiently large s — and the sweep applies.*

(b) *If instead $W_2(\sigma) \leq 1$ for* every *σ, then the word h is balanced.*

Proof. (a) Both parities are one application of the splitting lemma. Note that a two-slot group with prescribed subtotal σ_0 has group max minus group min exactly $W_2(\sigma_0) \geq 2$.

k odd. Write $k = 2u + 1$. For any $s \geq u\,\sigma_0$, split into u two-slot groups prescribed at σ_0 and one single-slot group at $j := s - u\sigma_0 \geq 0$ (a single slot has spread 0). By Lemma 12.3,

$$\operatorname{width}(s) \;\geq\; u\,W_2(\sigma_0) \;\geq\; 2u \;=\; k - 1 \;\geq\; d_2 - 1,$$

using $d_2 \leq k$.

k even, $d_2 \leq k - 1$. Write $k = 2u$. For $s \geq (u-1)\sigma_0$, split into $u - 1$ two-slot groups at σ_0 and one two-slot group at $\tau := s - (u-1)\sigma_0 \geq 0$, whose spread $W_2(\tau)$ is at least 0. By Lemma 12.3,

$$\operatorname{width}(s) \;\geq\; (u-1)\,W_2(\sigma_0) + W_2(\tau) \;\geq\; 2u - 2 \;=\; k - 2 \;\geq\; d_2 - 1.$$

(b) Take two equal-length windows, from i to i' and from j' to j (so $i' - i = j - j'$, i.e. $i + j = i' + j' =: \sigma$). Then

$$\bigl|(q_{i'} - q_i) - (q_j - q_{j'})\bigr| \;=\; \bigl|(q_{i'} + q_{j'}) - (q_i + q_j)\bigr| \;\leq\; W_2(\sigma) \;\leq\; 1,$$

which is precisely balancedness. $\square$

Part (a)'s parity split is not laziness. When k is even and $d_2 = k$, the budget of k indices is exactly wrong: u full pairs use all k indices and give width $\geq k$ — but only when s has the right form — while $u-1$ pairs guarantee only $k-2 = d_2 - 2$, one short of what the sweep needs. This single missing unit of width is a genuine boundary phenomenon, and §14 will close it with a separate argument (Lemma 14.5) rather than a sharper pairing.

Where we stand. If some $W_2(\sigma_0) \geq 2$ and (k odd or $d_2 \leq k-1$): done, by (a) and the sweep. If every $W_2(\sigma) \leq 1$: the word is balanced, and the next section classifies it — eventually periodic (done, by §10) or Sturmian (handled in §14). The one remaining configuration, k even with $d_2 = k$ and the word unbalanced, is also settled in §14. The ledger stays honest.

Example 12.5 (An unbalanced word has two-index width)**.** For $h = 0011\,0011\cdots$ the counting function runs $q = 0, 0, 1, 2, 2, 2, 3, 4, \ldots$, and at $\sigma_0 = 4$ we have $q_0+q_4 = 2$ while $q_2+q_2 = 0$, so $W_2(4) = 2$. At $k = 3$, $s = 12$ the separation is visible in whole configurations: $(4, 4, 4)$ has value 6, $(2, 2, 8)$ has value 4. This is exactly the input Lemma 12.4(a) converts into a band wide enough for the sweep.

Remark 12.6 (Two-index width bounds the full width)*.* Since $q_0 = 0$, any two-index configuration can be padded with $k-2$ zero indices without changing its value; hence $\mathrm{width}(s) \geq W_2(\sigma)$ whenever $s \geq \sigma$ and $k \geq 2$. The inequality is often strict — for $h = 0011\,0011\cdots$ at $k = 3$, $s = 12$, three free indices separate values further than two can.

13. Balanced words: the Morse–Hedlund classification

The width dichotomy left us with the balanced words, and this section proves the structure theorem that tames them: *a balanced binary word is either eventually periodic, or — after discarding at most one initial stretch — is "mechanical": its counting function is $q_n = \lfloor n\alpha + \beta \rfloor$ for a fixed irrational slope α.* This is one direction of a celebrated theorem of Morse and Hedlund [MH40]; a modern textbook account, with the full equivalence between balanced, Sturmian and mechanical words, is [Lo02, Ch. 2].

A reader content to cite that theorem may take Theorem 13.8 on authority and pass directly to §14. We prove it instead for three reasons: we need only the one direction stated above, so the full classification is more than the argument requires; the route below is elementary, importing nothing beyond the completeness of the reals (through sup) and a single pigeonhole; and it is the route the Lean development carries, so proving it here keeps the formalization and the text in step. The section closes with the one approximation tool (Dirichlet, by pigeonhole) that the endgame will need.

Throughout, h is a balanced binary word with counting function q ($q_0 = 0$).

13.1. Step 1: a balanced word is nearly additive.

Lemma 13.1 (Near-additivity)**.** *For all $n, m \geq 0$:* $\; |\, q_{n+m} - q_n - q_m \,| \;\leq\; 1.$

Proof. $q_{n+m} - q_n$ counts the 1s in the window of length m starting after position n; q_m counts them in the window of length m at the very start. Balance bounds the difference by 1. ☐

Iterating pins down long-range behavior:

Lemma 13.2. *For all $n \geq 0$, $t \geq 1$:* $\; t\,q_n - (t-1) \;\leq\; q_{tn} \;\leq\; t\,q_n + (t-1).$

Proof. Induct on t. Trivial at $t = 1$. If it holds at t, then by Lemma 13.1 (with the pair tn, n), $q_{(t+1)n} \leq q_{tn} + q_n + 1 \leq (t+1)q_n + t$, and similarly from below. ☐

13.2. **Step 2: the slope.** The slope we are after *ought* to be the density of 1s in the word — the limit of q_n/n. But we cannot speak of "the limit" before proving one exists, so we build the number as a supremum and only afterwards verify that it deserves the name. For orientation, the values on our running examples: the alternating word $0101\cdots$ will get $\alpha = \frac{1}{2}$; the word $001001\cdots$ gets $\alpha = \frac{1}{3}$; the Fibonacci word gets $\alpha = (3-\sqrt{5})/2 \approx 0.382$.

Lemma 13.3 (The slope exists)**.** *There is a real number $\alpha \in [0,1]$ such that*

$$|D_n| \;\leq\; 1 \quad \textit{for every } n \geq 0, \qquad \textit{where } D_n := q_n - n\alpha.$$

Moreover $q_n/n \to \alpha$.

Proof. The key inequality: for all $n, m \geq 1$,

$$m\,q_n - n\,q_m \;\leq\; n + m - 2. \tag{7}$$

The idea behind it: *count the* 1*s among the first nm letters two ways* — as m blocks of length n, and as n blocks of length m — and let near-additivity certify that both counts are accurate to small errors. Formally, bound q_{nm} twice by Lemma 13.2: viewing nm as m copies of n, $q_{nm} \geq mq_n - (m-1)$; viewing it as n copies of m, $q_{nm} \leq nq_m + (n-1)$. Chaining the two gives (7).

Dividing (7) by nm:

$$\frac{q_n - 1}{n} \;-\; \frac{q_m + 1}{m} \;=\; \frac{mq_n - nq_m - (n+m)}{nm} \;\leq\; \frac{-2}{nm} \;<\; 0.$$

So *every* number of the form $(q_n - 1)/n$ is below *every* number of the form $(q_m + 1)/m$. The set $\{(q_n - 1)/n : n \geq 1\}$ is therefore bounded above; let

$$\alpha \;:=\; \sup_{n\geq 1} \frac{q_n - 1}{n}.$$

By construction $(q_n - 1)/n \leq \alpha$ for every n; and $\alpha \leq (q_m + 1)/m$ for every m, because $(q_m + 1)/m$ is an upper bound of the set. The two together read $q_n - 1 \leq n\alpha \leq q_n + 1$, i.e. $|D_n| \leq 1$ (also at $n = 0$, where $D_0 = 0$). Since $0 \leq q_n \leq n$, we get $(q_n - 1)/n < 1$ for all n, so $\alpha \leq 1$; and $\alpha \geq (q_n - 1)/n \geq -1/n$ for every n, so $\alpha \geq 0$. Finally $|q_n/n - \alpha| = |D_n|/n \leq 1/n \to 0$. □

The number α is the *slope* of h: the limiting density of 1s. The inequality $|D_n| \leq 1$ says the word tracks the line $y = \alpha x$ with error never exceeding 1 — already a strong rigidity statement, extracted from balance alone, and worth seeing (Figure 17).

13.3. **Step 3: the discrepancy barely oscillates.**

Lemma 13.4 (Oscillation bound)**.** *For all $n, m \geq 0$:* $\;|D_n - D_m| \leq 1$.

Proof. Say $m = n + \Delta$ with $\Delta \geq 1$. For $j \geq 0$ let $W(j) := q_{j+\Delta} - q_j$ count the 1s in the length-Δ window starting after j; balance gives $|W(j) - W(n)| \leq 1$ for every j. Sum W over the t consecutive disjoint windows starting at $0, \Delta, 2\Delta, \ldots, (t-1)\Delta$; the sum telescopes to $q_{t\Delta}$, and each summand lies within 1 of $W(n)$:

$$t\,(W(n) - 1) \;\leq\; q_{t\Delta} \;\leq\; t\,(W(n) + 1).$$

Divide by $t\Delta$ and let $t \to \infty$: the middle tends to α (Lemma 13.3), so by the limit facts of §11,

$$\frac{W(n) - 1}{\Delta} \;\leq\; \alpha \;\leq\; \frac{W(n) + 1}{\Delta}, \qquad \text{i.e.} \qquad |W(n) - \Delta\alpha| \;\leq\; 1.$$

And $D_{n+\Delta} - D_n = (q_{n+\Delta} - q_n) - \Delta\alpha = W(n) - \Delta\alpha$. □

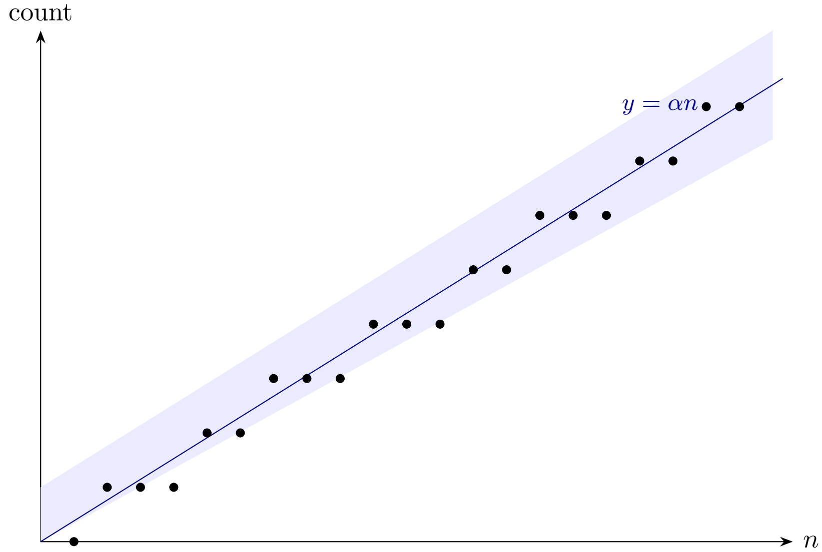


FIGURE 17. The counting function of the Fibonacci word ($\alpha = (3 - \sqrt{5})/2$), trapped forever in the shaded band $\alpha n \pm 1$: Lemma 13.3 drawn. The dots drift within the band — that drift is the discrepancy D_n — but Lemma 13.4 will confine even the drift to a window of height 1.

So all the discrepancies $D_0, D_1, D_2, \ldots$ live in a band of height 1 — they may drift within it, but no two ever differ by more than 1. Everything below flows from this single fact.

13.4. **Step 4: rational slope forces eventual periodicity.**

Proposition 13.5. *If α is rational, h is eventually periodic.*

Proof. Write $\alpha = u/v$ *in lowest terms*: u, v integers, $v \geq 1$, $\gcd(u, v) = 1$ (divide numerator and denominator by their gcd). We avoid the letters p, q here — q_n is already the counting function. Set

$$c_n \;:=\; v\, q_n - u\, n \;=\; v\, D_n,$$

an *integer* with $|c_n| \leq v$ (Lemma 13.3). Let $w_n := q_{n+v} - q_n$ (the 1-count of the length-v window after n), so that

$$c_{n+v} - c_n \;=\; v\,(w_n - u).$$

Balance confines the w_n to two consecutive values $\{W, W+1\}$ (all length-v windows agree to within one). We claim $u \in \{W, W{+}1\}$: if every $w_n \geq u + 1$, then $c_{n+v} - c_n \geq v$ for every n, so following the indices $n, n+v, n+2v, \ldots$ the integer c climbs by at least v per step, exiting $[-v, v]$ — impossible; symmetrically, $w_n \leq u - 1$ for all n is impossible. So either $w_n \in \{u, u+1\}$ for all n, or $w_n \in \{u-1, u\}$ for all n.

In the first case, $c_{n+v} - c_n \in \{0, v\}$: along each of the v index classes modulo v, the sequence c is nondecreasing and bounded, hence eventually constant; from some point on, no step of $+v$ ever occurs, i.e. $w_n = u$ for all large n. The second case is the mirror image (nonincreasing), with the same conclusion. So $q_{n+v} = q_n + u$ for all large n, and then

$$h_{n+v+1} \;=\; q_{n+v+1} - q_{n+v} \;=\; (q_{n+1} + u) - (q_n + u) \;=\; h_{n+1}$$

for all large n: the word repeats with period v from some point on. $\square$

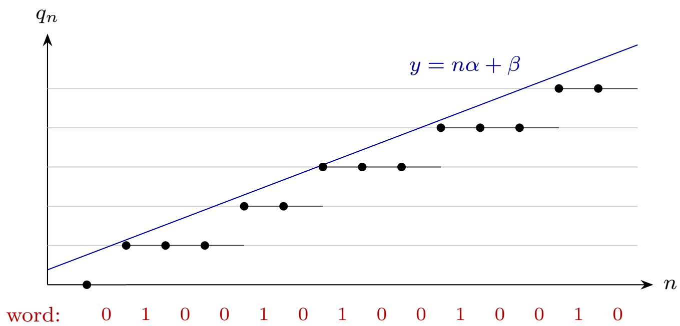


FIGURE 18. The second costume. The staircase $q_n = \lfloor n\alpha+\beta \rfloor$ tracks the line, and the word beneath records where it *steps* (a 1) versus where it *waits* (a 0). Figure 6 showed the same object as a walk on the number line: a Beatty walk's gaps and a mechanical word's letters are two readings of one digitized line.

13.5. **Step 5: irrational slope forces a mechanical tail.**

Definition 13.6 (Mechanical word)**.** A binary word h is *mechanical* with slope α and intercept β if

$$q_n \;=\; \lfloor n\alpha + \beta \rfloor \qquad \text{for every } n \geq 1.$$

Mechanical words with irrational slope are called *Sturmian*.

A mechanical word is the "digitized straight line" of slope α: its n-th letter records whether the line $y = \alpha x + \beta$ crosses an integer height in the n-th unit interval. Beatty walks and mechanical words are two costumes of the same actor — compare Definition 3.3 and Figure 6 with Figure 18. In Part II we *chose* such a line and rode it to victory; here the Adversary's walk turns out to be riding one whether it meant to or not, and that rigidity is what we will exploit against it.

Proposition 13.7. *If α is irrational, then there is an index $n_0 \geq 0$ such that the shifted word $h'_j := h_{n_0+j}$ is mechanical with slope α (and some intercept β').*

Proof. Set $\beta := \sup_n D_n$, finite since $|D_n| \leq 1$. By the oscillation bound, every $D_n \geq \beta - 1$: for any m, $D_m - D_n \leq 1$, and taking the supremum over m gives $\beta - D_n \leq 1$.

When does the mechanical formula hold at n? Unwinding the floor's definition,

$$q_n = \lfloor n\alpha + \beta \rfloor \iff n\alpha + \beta - 1 < q_n \leq n\alpha + \beta \iff \beta - 1 < D_n \leq \beta.$$

The upper half is the definition of β; the lower half holds at least weakly ($D_n \geq \beta - 1$), so the formula can fail only where $D_n = \beta - 1$ exactly — equivalently where $n\alpha + \beta = q_n + 1$ is an integer. If two distinct indices $n_1 \neq n_2$ both had $n_i\alpha + \beta \in \mathbb{Z}$, subtracting would give $(n_1 - n_2)\alpha \in \mathbb{Z}$, impossible for irrational α. So there is at most one exceptional index; call it n_0 (and set $n_0 := 0$ if there is none).

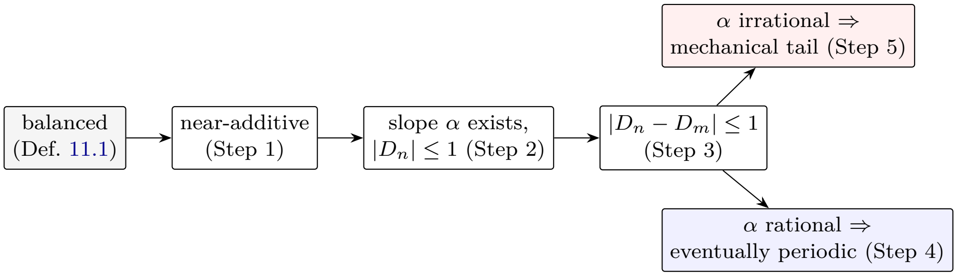
Discard everything through n_0: the word $h'_j := h_{n_0+j}$ has counting function $q'_j = q_{n_0+j} - q_{n_0}$, and for every $j \geq 1$ the index $n_0 + j$ is unexceptional, so

$$q'_j \;=\; \lfloor (n_0 + j)\alpha + \beta \rfloor - q_{n_0} \;=\; \lfloor j\alpha + (n_0\alpha + \beta - q_{n_0}) \rfloor \;=\; \lfloor j\alpha + \beta' \rfloor, \qquad \beta' := \beta - D_{n_0},$$

where the middle equality moves the *integer* q_{n_0} through the floor (Lemma 3.2(2)). $\square$

Assembling Steps 4 and 5:

Theorem 13.8 (Balanced classification)**.** *A balanced binary word is either eventually periodic, or agrees beyond some index with a Sturmian word: a mechanical word of irrational slope $\alpha \in (0,1)$.*



The five steps at a glance. Balance alone forces the word to track a line; the line's slope then decides everything, and the two verdicts are exactly the two branches §12 needs — one already closed by §10, the other handed to §14.

Proof. Steps 2–5; when α is irrational it lies in $(0,1)$, being in $[0,1]$ and not equal to 0 or 1. $\square$

For our purposes the theorem is a perfect fork: the periodic branch was closed in §10, so *the only balanced words still standing are the Sturmian tails*, with an explicit formula $q_n = \lfloor n\alpha + \beta \rfloor$ to attack. The attack needs one lever: the ability to find indices n where the fractional part $\{n\alpha + \beta\}$ lands in any prescribed arc. That is a statement about *rotations of a circle*, and it closes this section.

13.6. **Rotations, and how to hit an arc on schedule.** For a real x write $\{x\} := x - \lfloor x \rfloor \in [0,1)$ for its *fractional part*. Picture the interval $[0,1)$ bent into a circle of circumference 1; the map $x \mapsto \{x+\alpha\}$ is rotation by α. The numbers $\{n\alpha+\beta\}$ are the successive positions of a point rotating by α per tick. Write $\|x\|$ for the distance from x to the nearest integer — the circle distance from $\{x\}$ to 0.

Lemma 13.9 (Dirichlet's approximation theorem)**.** *For every real α and integer $Q \ge 1$ there are integers n_0, m with $1 \le n_0 \le Q$ and*

$$|\, n_0\alpha - m \,| \;<\; \frac{1}{Q}.$$

Proof. Pigeonhole. The $Q+1$ numbers $\{0\cdot\alpha\}, \{1\cdot\alpha\}, \dots, \{Q\alpha\}$ all lie in $[0,1)$, which we cut into Q boxes $[i/Q,\ (i+1)/Q)$. Two of them share a box: $\{j\alpha\}$ and $\{j'\alpha\}$ with $j < j'$, differing by less than $1/Q$. Set $n_0 := j' - j \in [1, Q]$; then $n_0\alpha = (j'\alpha - \lfloor j'\alpha \rfloor) - (j\alpha - \lfloor j\alpha \rfloor) + (\lfloor j'\alpha \rfloor - \lfloor j\alpha \rfloor) = \{j'\alpha\} - \{j\alpha\} + m$ for the integer $m := \lfloor j'\alpha \rfloor - \lfloor j\alpha \rfloor$, and $|n_0\alpha - m| = |\{j'\alpha\} - \{j\alpha\}| < 1/Q$. $\square$

Lemma 13.10 (Uniform hitting)**.** *Let $\alpha \in (0,1)$ be irrational and $\beta \in \mathbb{R}$. For every $\varepsilon > 0$ there is an integer $L = L(\alpha, \varepsilon)$ — depending only on α and ε,* not *on β and not on the arc's position — such that for every arc $J \subseteq (0,1)$ of length $\ge \varepsilon$ and every $N \ge 0$, some index $n \in \{N+1, \dots, N+L\}$ has $\{n\alpha + \beta\} \in J$.*

Proof. An explicit *circle walk*. Choose an integer $Q > 2/\varepsilon$ and apply Dirichlet: there are $1 \le n_0 \le Q$ and m with $\theta := |n_0\alpha - m| < 1/Q < \varepsilon/2$; and $\theta \ne 0$ since α is irrational. Advancing the index by n_0 rotates the point $\{n\alpha + \beta\}$ by exactly θ around the circle — always in the same direction (the sign of $n_0\alpha - m$ is fixed).

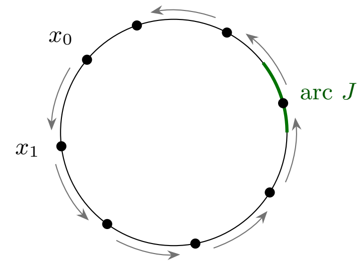


FIGURE 19. The circle walk of Lemma 13.10. Successive points x_j advance by the same short step θ in the same direction; after $\lceil 1/\theta \rceil$ steps the total distance walked is at least a full lap, and a step shorter than the arc cannot cross it without landing in it.

Now fix N, and follow the $\lceil 1/\theta \rceil + 1$ indices

$$n_j \;:=\; N + (j+1)\,n_0 \qquad (j = 0, 1, \ldots, \lceil 1/\theta \rceil),$$

all lying in $\{N{+}1, \ldots, N{+}L\}$ for $L := n_0\big(\lceil 1/\theta \rceil{+}1\big)$. The successive positions $x_j := \{n_j\alpha{+}\beta\}$ each step a fixed distance θ around the circle, in a fixed direction, and over $\lceil 1/\theta \rceil$ steps the total distance walked is at least 1: the walk laps the whole circle. If none of the x_j lay in the arc J, then some single step would have to jump clean over J — but each step has length $\theta < \varepsilon/2 < |J|$, too short to clear it. So some $x_j \in J$ (Figure 19).

Finally, audit the dependence: Q came from ε; n_0, m, θ from α and Q; and $L = n_0(\lceil 1/\theta \rceil + 1) \le Q(\lceil 1/\theta \rceil{+}1)$ from these. Neither β, nor N, nor where J sits was used — only $|J| \ge \varepsilon$. $\square$

The uniformity is the point: in the endgame the arc J will *move* as the target changes, and we need a single window length L serving every position. A subtler equidistribution theorem would say more; this crude walking bound, powered by one pigeonhole, says exactly enough.

Remark 13.11 (Where Step 5 uses irrationality). Twice, and both uses are essential. It gives at most one exceptional index: two indices with $n_i\alpha + \beta \in \mathbb{Z}$ would force $(n_1 - n_2)\alpha \in \mathbb{Z}$. And it places α in the open interval $(0,1)$ rather than at an endpoint. For rational α the exceptional indices recur periodically and no single tail repair is possible — as it must be, since Step 4 has already shown such words are eventually periodic, and a periodic word is not mechanical with irrational slope.

14. The Sturmian endgame

This section finishes the two-letter case: first the Sturmian walks (the hardest single proof in this paper — take it slowly), then the boundary configuration that the width dichotomy could not reach, then the assembled two-letter theorem.

14.1. **The Sturmian lemma.** We keep the two-letter coordinates of §12 — letters $\delta < \delta' = \delta + e$ with $\gcd(\delta, e) = 1$ and $\delta' \le d_2 \le k$, binary word h, counting function q — and now assume, after Theorem 13.8 and a final tail-pass, that the word is Sturmian:

$$q_n \;=\; \lfloor n\alpha + \beta \rfloor \qquad (n \ge 1)$$

for an irrational $\alpha \in (0,1)$ and some real β.

Lemma 14.1 (Sturmian)**.** *In this situation $kA - ka_1$ contains every sufficiently large integer; in particular A is tail-covering with modulus* 1*.*

This is stronger than the sweep gave us — *all* large integers, no residue class needed — and it has to be, because Sturmian words are exactly the words for which the band of §12 is too narrow for the sweep. The proof replaces "wide band" with something better: *exact control.* For a mechanical word the value of a configuration is not merely trapped in a window; it can be steered onto a precise target, because choosing indices means choosing points of an irrational rotation, and the hitting lemma lets us place those points wherever we like.

Proof. Recall (Lemma 9.1 and (5)) that it suffices to realize each large integer x as $x = \delta s + e\,w$ where w is a sum of k counting-function values $q_{i_1} + \cdots + q_{i_k}$ with $i_t \geq 1$ and $i_1 + \cdots + i_k = s$. The proof has four movements.

Step 0: where the values live.[3] Write, for $n \geq 1$,

$$q_n \;=\; n\alpha + \beta - \{n\alpha + \beta\}$$

(floor = number minus fractional part). Summing over k indices with total s:

$$\sum_{t=1}^{k} q_{i_t} \;=\; X_s - \sum_{t=1}^{k} \{i_t\alpha + \beta\}, \qquad X_s \;:=\; s\,\alpha + k\,\beta. \tag{8}$$

Each fractional part lies in $[0, 1)$, so the sum of k of them lies in $[0, k)$: every achievable value sits in the real window $(X_s - k,\ X_s]$ — the two-letter avatar of Part II's window picture. But now we run the logic *forward*: to achieve a prescribed value w at depth $\theta := X_s - w$ below the window's top, it suffices to make the fractional parts sum to *exactly* θ.

Step 1: the tool. Lemma 13.10: for any arc length $\varepsilon > 0$ there is a window length $L(\alpha, \varepsilon)$ such that every L consecutive indices contain one whose rotation point $\{n\alpha + \beta\}$ lies in any prescribed arc of length ε — uniformly in the arc's position.

Step 2: hitting any depth in the middle of the window. Fix a margin $\zeta \in (0, \frac{1}{2})$, to be chosen in Step 3. **Claim:** there is a threshold S_0 (depending on ζ, α, β, k only) such that for every $s \geq S_0$ and every integer w whose depth $\theta = X_s - w$ lies in $[\zeta,\ k - \zeta]$, there are *distinct* indices $i_1 < \cdots < i_k$ (all ≥ 1, summing to s) with

$$\sum_{t=1}^{k} \{i_t\alpha + \beta\} \;=\; \theta \quad \text{exactly,} \qquad \text{hence} \qquad \sum_{t=1}^{k} q_{i_t} = w.$$

The mechanism: place the first $k - 1$ points of the rotation in a narrow arc positioned so their sum falls just below θ, then let the forced last index close the books — with an integrality argument snapping the total onto θ exactly. Because this step introduces several objects in quick succession, here is the cast before the play:

[3] One convention shifts here, and it is deliberate. Equation (5) let the summand indices i_t range over *all* integers ≥ 0; from this proof onward we use only indices $i_t \geq 1$. The reason is that the Sturmian formula $q_n = \lfloor n\alpha + \beta \rfloor$ was established for $n \geq 1$ only (Proposition 13.7 says nothing at $n = 0$), so index 0 is not covered by it. Restricting is harmless: we are *constructing* representations, so using fewer indices than we are entitled to can only make the conclusion stronger.

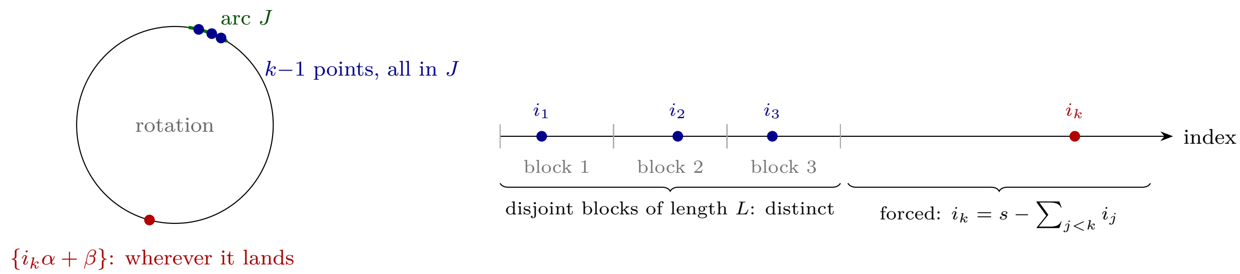


FIGURE 20. How Step 2 chooses its indices. *Right:* the $k-1$ free indices are drawn one from each of $k-1$ disjoint blocks of length L — disjointness makes them distinct — and the last index is then forced by the budget s. Taking $s \ge S_0$ pushes i_k past all of them. *Left:* the hitting lemma is what lets each free index be chosen with its rotation point inside the short arc J, so their fractional parts sum to within κ of the target; the forced point's fractional part is uncontrolled, and integrality (Figure 21) cleans up the remainder.

Object	Job	Depends on
ζ	margin: how deep inside the window we work	fixed in Step 3
$\kappa = \zeta/4$	safety padding inside every estimate	ζ
$\mathcal{I}$	interval of legal per-point targets	θ
T^*	the chosen per-point target	θ
J	the arc where the $k-1$ free points must land	T^*
$\varepsilon_0 = \frac{2\kappa}{k-1}$	J's length — position moves, length never	ζ, k
L	hitting window for arcs of length ε_0	α, ε_0
S_0	budget threshold making the last index behave	L, k

Set $\kappa := \zeta/4$, and consider the open interval of candidate "per-point targets"

$$\mathcal{I} \;:=\; \big(\,\max(0,\ \theta-1)+\kappa,\ \ \min(k-1,\ \theta)-\kappa\,\big).$$

Its length is at least $\zeta/2$ whatever the regime of θ — three cases, by which of the max/min clauses bind:

- $\zeta \le \theta \le 1$: $\quad |\mathcal{I}| = \theta - 2\kappa \ge \zeta - \frac{\zeta}{2} = \frac{\zeta}{2}$;
- $1 < \theta \le k-1$: $\quad |\mathcal{I}| = 1 - 2\kappa = 1 - \frac{\zeta}{2} > \frac{\zeta}{2}$ (as $\zeta < 1$);
- $k-1 < \theta \le k-\zeta$: $\quad |\mathcal{I}| = (k-\theta) - 2\kappa \ge \zeta - \frac{\zeta}{2} = \frac{\zeta}{2}$.

Pick any $T^* \in \mathcal{I}$ and set the arc

$$J \;:=\; \Big(\frac{T^*}{k-1} - \frac{\kappa}{k-1},\ \frac{T^*}{k-1} + \frac{\kappa}{k-1}\Big) \;\subseteq\; (0,1),$$

of length $\varepsilon_0 := 2\kappa/(k-1)$, and let $L := L(\alpha, \varepsilon_0)$ be its hitting constant. The arc's *position* depends on θ (through T^*), but its *length* does not — and the hitting lemma is uniform in position, so this single L serves every target and every s. (This is exactly why we insisted on uniformity in §13.)

Choosing the free indices. For $j = 1, \dots, k-1$, the hitting lemma finds an index i_j in the block $\{(j-1)L+1, \dots, jL\}$ with $\{i_j\alpha + \beta\} \in J$ (Figure 20). Different blocks, so

$i_1 < i_2 < \cdots < i_{k-1}$, with

$$i_j \le jL, \qquad \sum_{j=1}^{k-1} i_j \;\le\; L\,\frac{k(k-1)}{2}, \qquad i_{k-1} \le (k-1)L. \tag{9}$$

The forced last index. What must the threshold S_0 accomplish? The last index $i_k := s-(i_1+\cdots+i_{k-1})$ is forced, and we need it both legal (≥ 1) and distinct from the others, for which $i_k > i_{k-1}$ suffices. By (9) the free indices consume at most $L\,k(k-1)/2$ of the budget, and the largest of them is at most $(k-1)L$ — so it suffices to start s strictly above the sum of those two bounds. Accordingly, set

$$S_0 \;:=\; 1 + L\,\frac{k(k-1)}{2} + (k-1)L \;=\; 1 + L\,(k-1)\Big(\frac{k}{2}+1\Big)$$

(an integer, as $(k-1)(k+2)/2$ has one even factor). For $s \ge S_0$, by (9),

$$i_k \;\ge\; S_0 - L\frac{k(k-1)}{2} \;=\; 1 + L(k-1)\Big(\frac{k}{2}+1-\frac{k}{2}\Big) \;=\; 1+(k-1)L \;>\; i_{k-1},$$

so all k indices are distinct and ≥ 1, with total exactly s.

The snap. Let $F := \sum_{j\le k-1}\{i_j\alpha+\beta\}$; since each term lies in J,

$$F \;\in\; \big(T^*-\kappa,\; T^*+\kappa\big) \;\subseteq\; \big(\max(0,\theta-1),\; \min(k-1,\theta)\big),$$

the inclusion because $T^* \in \mathcal{I}$ keeps distance $> \kappa$ from both endpoints.

[Instance, with Example 14.2's Fibonacci data ($\alpha \approx 0.382$, $\zeta = \alpha/4 \approx 0.095$, $\kappa \approx 0.024$): for the target of that example, $\theta \approx 0.647$, so $\mathcal{I} \approx (0.024,\ 0.623)$; choosing $T^* = 0.3$ puts the arc at $J \approx (0.138,\ 0.162)$, and the two free rotation points must land in that short arc.] Add the last point: $\Theta := F + \{i_k\alpha+\beta\}$, with the fractional part in $[0,1)$, satisfies

$$\Theta \;>\; \max(0,\theta-1) \;\ge\; \theta-1 \qquad\text{and}\qquad \Theta \;<\; \min(k-1,\theta)+1 \;\le\; \theta+1.$$

On the other hand, by (8) applied to our indices,

$$\Theta \;=\; X_s - \sum_{t\le k} q_{i_t},$$

and both $X_s - \theta = w$ and $\sum_t q_{i_t}$ are integers, so

$$\Theta - \theta \;=\; \big(X_s - \textstyle\sum_t q_{i_t}\big) - \big(X_s - w\big) \;=\; w - \textstyle\sum_t q_{i_t} \;\in\; \mathbb{Z}.$$

Now finish in three short steps. The difference $\Theta-\theta$ is an integer. From $\theta-1<\Theta<\theta+1$ we get $-1<\Theta-\theta<1$. The only integer strictly between -1 and 1 is 0. Therefore $\Theta-\theta=0$, i.e. $\Theta=\theta$ — and consequently $\sum_t q_{i_t} = X_s-\theta = w$, exactly as required. The Claim is proved.

Key step: the integrality snap. *The snap* (Figure 21) is the step worth slowing down for, so do not let it pass in four lines: soft estimates confined Θ to an open interval of length 2, and hard integrality confined $\Theta-\theta$ to the integers; the only point obeying both masters is θ exactly. Analysis narrows the suspects to a lineup; arithmetic makes the identification.

(One aside before the ladder: why did Step 2 bother producing *distinct* indices, when Definition 1.5 happily permits repeats? Because distinctness came free — the $k-1$ free indices live in disjoint blocks — and the argument for pinning down i_k was easiest to state as $i_k > i_{k-1}$. Nothing was lost and nothing extra was needed; distinctness is simply the stronger conclusion the construction happened to deliver.)

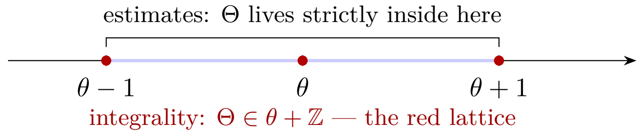


FIGURE 21. The snap. The open interval $(\theta-1,\ \theta+1)$ contains exactly one point of the lattice $\theta+\mathbb{Z}$: its center. So $\Theta=\theta$, with no error at all — the estimates never had to be sharp, only sharper than the lattice spacing.

Step 3: the ladder. Now fix the margin

$$\zeta \ := \ \tfrac{1}{4}\min(\alpha,\ 1-\alpha) \ \in \ (0, \tfrac{1}{8}],$$

and let S_0 be the resulting threshold. Let x be a large integer. As in the sweep, the budgets s able to represent $x=\delta s+ew$ with integer w form one residue class modulo e (the *admissible* s), and for admissible s we set $w(s):=(x-\delta s)/e$ and measure the depth

$$\theta(s) \ := \ X_s - w(s) \ = \ s\alpha + k\beta - \frac{x-\delta s}{e}.$$

Per admissible step $s \mapsto s+e$, the value X_s grows by $e\alpha$ and $w(s)$ falls by δ, so the depth climbs by exactly

$$\psi \ := \ e\,\alpha + \delta \ > \ 0$$

— an arithmetic ladder. Step 2 accepts any depth in the strip $[\zeta,\ k-\zeta]$, of height $k-2\zeta$. The ladder cannot step over the strip, because its rung spacing is smaller:

$$\psi \ = \ \delta + e\alpha \ = \ \delta' - e(1-\alpha) \ \le \ k-(1-\alpha) \ < \ k-2\zeta,$$

using $\delta' \le k$, $e \ge 1$, and $2\zeta \le \frac{1}{2}(1-\alpha) < 1-\alpha$. Recognize this inequality: the rung $\psi=\delta+e\alpha$ is precisely the walk's *average step*, and the strip height is essentially k — "rungs shorter than the strip" is the triptych of Figure 4 all over again, stride against window width, now *proved* for a wobbling walk rather than pictured for a steady one. Since $\theta(s)$ is very negative for small admissible s (there $w(s)\approx x/e$ dominates) and climbs unboundedly by steps $\psi < k-2\zeta$, some admissible s^* lands in the strip: $\theta(s^*) \in [\zeta,\ k-\zeta]$ (Figure 22).

The thresholds, explicitly. Multiplying the definition of θ by e and rearranging: $s\psi = x + e\,\theta(s) - ek\beta$. So any s^* with $\theta(s^*)\in[0,k]$ satisfies

$$\frac{x-ek|\beta|}{\psi} \ \le \ s^* \ \le \ \frac{x+ek(1+|\beta|)}{\psi}.$$

Two requirements: (a) $s^* \ge S_0$, so Step 2 applies — since $\psi \le \delta+e=\delta' \le k$, the left bound gives this once $x \ge kS_0 + ek|\beta|$. (b) $w(s^*)\ge 0$, i.e. $\delta s^* \le x$ — by the right bound this holds once $\delta\,(x+ek(1+|\beta|)) \le (\delta+e\alpha)\,x$, i.e. once $x \ge \delta k(1+|\beta|)/\alpha$ (here $\alpha>0$ earns its keep). So for

$$x \ \ge \ X_0 \ := \ 1+\max\Big(kS_0+ek|\beta|,\ \ \frac{\delta k(1+|\beta|)}{\alpha}\Big),$$

Step 2 realizes $w(s^*)\in W_{s^*}$ with legal indices, and

$$x \ = \ \delta\, s^* + e\, w(s^*) \ \in \ kA - ka_1. \qquad \square$$

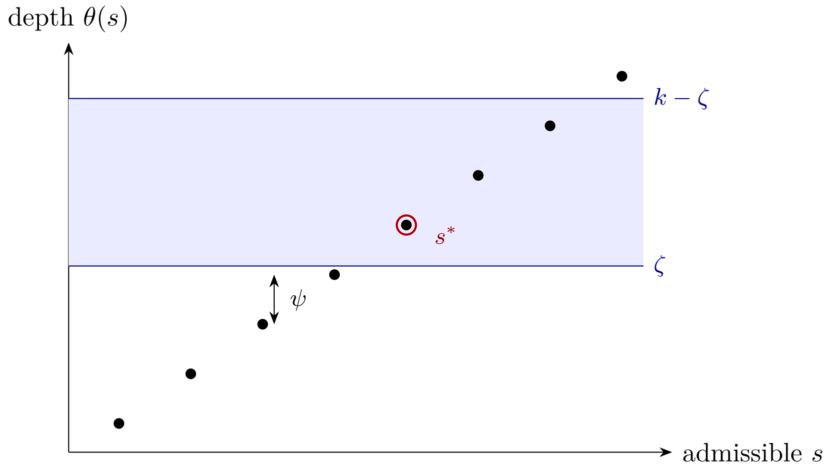


FIGURE 22. The ladder against the strip, drawn with the Fibonacci numbers of Example 14.2: rung spacing $\psi = 1+\alpha \approx 1.38$, strip height $k-2\zeta \approx 2.81$. A ladder whose rungs are closer together than the strip is tall cannot pass from below it to above it without a rung landing inside.

Sturmian mechanism. For a Sturmian walk the window picture stops being an estimate and becomes an equation: the depth of a value below the window's top *equals* the sum of k rotation points on the circle. The hitting lemma lets us place $k-1$ of those points at will; integrality snaps the total onto the exact target; and the ladder's rung $\psi = \delta + e\alpha$ is strictly shorter than the acceptance strip, so no target escapes.

Example 14.2 (The Fibonacci walk, traced)**.** Take $k = 3$, letters $\{1, 2\}$ (so $\delta = e = 1$, every s admissible), and the Fibonacci gap word: $q_n = \lfloor n\alpha + \beta \rfloor$ with $\alpha = (3-\sqrt{5})/2 \approx 0.382$ and $\beta = \alpha$. Then $\zeta = \alpha/4 \approx 0.095$ and the ladder climbs by $\psi = 1 + \alpha \approx 1.382$ per step — comfortably below the strip height $3 - 2\zeta \approx 2.81$. To represent $x = 100$: $\theta(s) = (s+3)\alpha - 100 + s$ climbs by ψ and first enters $[\zeta,\ 3-\zeta]$ at $s^* = 72$, where $\theta(72) = 75\alpha - 28 \approx 0.647$. So $w = w(72) = 100 - 72 = 28$, and what is wanted is three distinct indices with $i_1 + i_2 + i_3 = 72$ and $q_{i_1} + q_{i_2} + q_{i_3} = 28$ — three rotation points summing to depth 0.647. One such triple is $i = (10, 26, 36)$, whose counting values are $4 + 10 + 14 = 28$ (check them against $q_n = \lfloor (n+1)\alpha \rfloor$), so indeed $100 = 1 \cdot 72 + 1 \cdot 28 \in 3A - 3a_1$.

An honest caveat about this example. Step 2 guarantees such a triple only for budgets $s \geq S_0$, and S_0 is built from the hitting constant L of Lemma 13.10 — which, for these parameters, is enormous: the arc has length $\varepsilon_0 \approx 0.024$, Dirichlet's pigeonhole with $Q = 85$ yields a step $\theta \approx 0.0081$ and hence $L \approx 6800$, giving $S_0 \approx 34{,}000$. So $s^* = 72$ is far *below* the threshold at which the theorem's construction promises anything, and the triple above was found by direct search rather than produced by Step 2. That is exactly what one should expect: the proof's constants are chosen for uniformity across all large x, not for sharpness at small ones. The example is here to show you the *ladder* — $\theta(s)$ climbing by ψ until it lands in the strip — with numbers small enough to check by hand, not to demonstrate Step 2 firing.

14.2. **The boundary: k even, $d_2 = k$.** One configuration escaped the width dichotomy: k even at the exact boundary $d_2 = k$, where the pairing argument came up one unit of

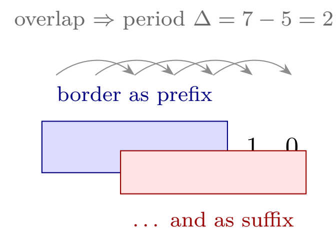


FIGURE 23. Border–period duality (Lemma 14.3) on $u = 0101010$. A prefix that is also a suffix forces the word to repeat with period "length minus border". In Lemma 14.5 this is what converts two palindromic prefixes into a period — and arbitrarily long prefixes of the same period into global periodicity, which is the contradiction sought.

width short. The rescue is a lemma that squeezes width out of *unbalancedness itself*, with a beautiful detour through palindromes.

First, a two-line classic:

The next fact is classical in combinatorics on words (see [Lo02, Ch. 1]); we give its one-line proof to keep the section self-contained.

Lemma 14.3 (Border–period duality)**.** *Let u be a finite word of length τ with a* border *of length ℓ: its prefix of length ℓ equals its suffix of length ℓ. Then u has* period $\tau-\ell$*: $u_i = u_{i+\tau-\ell}$ for all $1 \le i \le \ell$.*

Proof. The border condition says $u_i = u_{(\tau-\ell)+i}$ for $1 \le i \le \ell$ — which *is* the period condition, re-read. □

Example 14.4. $u = 0101010$, of length 7, has the border 01010 of length 5 (first five letters = last five letters). The lemma awards period $7 - 5 = 2$: indeed $u_i = u_{i+2}$ throughout. Long border, short period — the general rule of thumb.

Lemma 14.5 (Boundary width)**.** *Let $k \ge 4$ be even. Suppose the binary word h uses both letters infinitely often, is not eventually periodic, and has the property that* every *tail of h is unbalanced. Then* $\operatorname{width}(s) \ge k-1$ *for all sufficiently large s.*

Proof. First, unbalancedness of every tail yields two-index width ≥ 2 infinitely often. Suppose instead $W_2(\sigma) \le 1$ for all $\sigma \ge \sigma^*$. Take any two equal-length windows of the tail of h beyond position σ^*, say from i to i' and from j' to j; then $\sigma := i + j = i' + j' \ge \sigma^*$, and exactly as in Lemma 12.4(b), $W_2(\sigma) \le 1$ bounds the difference of their 1-counts by 1. So that tail would be balanced — contradiction. Hence $W_2(\sigma) \ge 2$ for infinitely many σ; fix two such, $\sigma_0 < \sigma_1$, and set $\Delta := \sigma_1 - \sigma_0 \ge 1$.

Second, suppose the width bound fails at some large s: $\operatorname{width}(s) \le k-2$. Play the splitting lemma twice, with a twist. Split one: $k/2 - 1$ two-slot groups at σ_0 plus one two-slot group at $\tau := s - (k/2-1)\sigma_0$; Lemma 12.3 forces $\operatorname{width}(s) \ge (k-2) + W_2(\tau)$, so $W_2(\tau) = 0$. Split two ($k \ge 4$ makes room): $k/2 - 2$ groups at σ_0, one at σ_1, and one at $\tau - \Delta$; it forces $W_2(\tau - \Delta) = 0$ likewise.

Third, zero two-index width means palindrome. $W_2(\tau) = 0$ says $q_i + q_{\tau-i}$ is constant over $0 \le i \le \tau$; differencing consecutive i gives $h_{i+1} = h_{\tau-i}$ for $0 \le i \le \tau - 1$ — the prefix $h[1..\tau]$ reads the same forwards and backwards. So both $h[1..\tau]$ and $h[1..\tau-\Delta]$ are palindromes. A palindromic prefix of a palindrome is also its suffix (reverse the whole word: it is unchanged, and the prefix's mirror image — itself — appears at the end). So $h[1..\tau-\Delta]$ is a border of $h[1..\tau]$, and by Lemma 14.3, $h[1..\tau]$ has period Δ. The chain in one line:

$$W_2 = 0 \longrightarrow \text{palindrome} \longrightarrow \text{border} \longrightarrow \text{period } \Delta \longrightarrow \text{(next paragraph) global periodicity.}$$

Finally, infinitely many failures would force global periodicity. Suppose $\mathrm{width}(s) \le k-2$ for infinitely many s. The offset Δ is fixed, while $\tau(s) = s-(k/2-1)\sigma_0 \to \infty$ along the failing s: so we obtain prefixes $h[1..\tau]$ of period Δ with τ arbitrarily large. Now fix any position n and choose a failing s with $\tau(s) \ge n+\Delta$; the periodic prefix contains both positions n and $n+\Delta$, so its period relation gives $h_n = h_{n+\Delta}$. As n was arbitrary, the entire word has period Δ — in particular it is eventually periodic, contradicting the hypothesis. Hence $\mathrm{width}(s) \ge k-1$ for all large s. ☐

Proposition 14.6 (Boundary routing)**.** *Let k be even, $d_2 = k$, and let A have two-letter tail alphabet (the standing setup of §12). Then A is tail-covering.*

Proof. Route by the first case that applies; each closes the matter. If the gap word is eventually periodic: Lemma 10.1. Otherwise, if *some* tail of the gap word is balanced: pass to that tail (Lemma 9.3) and classify it (Theorem 13.8) — eventually periodic (Lemma 10.1 again) or Sturmian on a further tail (Lemma 14.1); note neither route used any width argument. Otherwise *every* tail is unbalanced and the word is not eventually periodic (and $k \ge 4$, being even and ≥ 3): Lemma 14.5 gives $\mathrm{width}(s) \ge k-1 = d_2-1$ for large s, and the sweep (Lemma 12.2) finishes. ☐

14.3. **The two-letter theorem.**

Theorem 14.7 (Two letters)**.** *Let $k \ge 3$ and $d_2 \le k$. Every walk with all gaps $\le d_2$ whose tail alphabet has at most two letters is tail-covering. (In particular every (d_1, d_2)-sequence, for every d_1; and, crucially, every walk obtained from one by the rescaling of Lemma 9.5.)*

Proof. One letter: Lemma 9.4. Two letters: rescale to gcd $= 1$ (Lemma 9.5). If k is even and $d_2 = k$: Proposition 14.6. Otherwise k is odd or $d_2 \le k-1$, and we consult the two-index widths: if $W_2(\sigma_0) \ge 2$ for some σ_0, then Lemma 12.4(a) plus the sweep (Lemma 12.2) conclude; if $W_2(\sigma) \le 1$ for all σ, the word is balanced (Lemma 12.4(b)), hence eventually periodic (Lemma 10.1) or Sturmian on a tail (Theorem 13.8, then Lemma 14.1). Every branch ends in tail-covering. ☐

State of play. We now know: every walk with a tail alphabet of at most two letters is tail-covering. What remains: alphabets of three or more letters — and there the band machinery fails outright, so a new weapon is needed. Next: the slot lemma, and the exact finite statement it costs.

Remark 14.8 (Why the sweep cannot be reused here). Step 0 confines W_s to a real window of length k, so $\mathrm{width}(s) \le k-1$ always. At the critical corner $\delta' = d_2 = k$ the sweep needs width $\ge d_2 - 1 = k-1$, which would require both extreme integers of that window to be attained — and that needs the k fractional parts $\{i_t\alpha + \beta\}$ to pile up near 0 and near 1 simultaneously, which balancedness forbids for all large s. The band is genuinely too narrow, and this is why §14 replaces a width argument with exact control.

15. Three or more letters: the slot lemma

One case of the battle plan remains: tail alphabets with three or more letters. The band-and-sweep machinery does not survive the trip — with three letters, a single move between configurations changes the value by a *difference of letters*, which can exceed 1, so the reachable set W_s can have holes and Lemma 12.1 fails. Instead of repairing the band we change weapons entirely: a small combinatorial gadget — the *slot lemma* — that spends the k summands like a budget, and whose exact price is the finite theorem of Part IV.

15.1. **Subset sums of multisets.**

Definition 15.1. A *multiset* S drawn from a set G is a collection of elements of G with repetitions allowed ($|S|$ counts repetitions). Its *subset sums* are

$$\mathrm{sums}(S) \;:=\; \Big\{ \sum_{x\in T} x \;:\; T \text{ a sub-multiset of } S \Big\}$$

(including the empty sum 0). For a finite set G of positive integers with $M := \max G$, define

$$m(G) \;:=\; \min \{\, |S| \;:\; S \text{ a multiset drawn from } G \text{ whose subset sums contain } M \text{ consecutive integers} \,\},$$

with the convention $m(G) := \infty$ if no such multiset exists.

That convention is not idle. If $\gcd(G) = d > 1$ then *every* subset sum is a multiple of d, so no two consecutive integers — let alone M of them — can both be sums, and the minimum really is over an empty set. So $m(G) = \infty$ whenever $\gcd(G) > 1$. The converse — finiteness when $\gcd(G) = 1$ and $|G| \geq 3$ — is exactly what Theorem 16.1 will establish, and is why the coprimality hypothesis appears in every statement below rather than being an afterthought.

Example 15.2. $G = \{3, 4, 5\}$, $S = \{3, 3, 4, 5\}$: the subset sums are $\{0, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 15\}$, which contain the run $3, 4, \ldots, 12$ — ten consecutive integers, amply covering $M = 5$. So $m(\{3, 4, 5\}) \leq 4$. (Part IV will show 4 is optimal here.)

Part IV is devoted to proving:

> **The bounded subset-sum covering lemma (SHARP).** *Every finite set G of at least three positive integers with* $\gcd(G) = 1$ *and* $\max(G) = M$ *has* $m(G) \leq M - 1$.

For now we take it on credit and show exactly how the non-existence proof spends it.

15.2. **The slot lemma.**

Lemma 15.3 (Slot lemma)**.** *Let A have tail alphabet G_∞ with $|G_\infty| \geq 3$ and* $\gcd(G_\infty) = 1$ *(after rescaling), and put* $M := \max G_\infty \leq d_2$*. If $k - 1 \geq m(G_\infty)$, then A is tail-covering.*

Proof. Pass to the tail (Lemma 9.3): all gaps lie in G_∞, all letters recur infinitely often, and we work with prefix sums via Lemma 9.1. Let S be a multiset realizing $m(G_\infty) =: m \leq k-1$: $|S| = m$, drawn from G_∞, with

$$\mathrm{sums}(S) \;\supseteq\; \{c,\ c+1,\ \ldots,\ c+M-1\}$$

for some c. Enumerate S as $\delta_1, \ldots, \delta_m$ (values in G_∞, possibly repeating).

The fine slots. Because each letter of G_∞ occurs infinitely often in the gap word, we can choose m *distinct* positions $n_1 < n_2 < \cdots < n_m$ (all beyond the tail start) with

$$g_{n_t} = \delta_t \qquad (t = 1, \ldots, m).$$

Summand t of our k-fold sum will be confined to two adjacent choices: index $n_t - 1$ ("off"), contributing the prefix sum p_{n_t-1}, or index n_t ("on"), contributing $p_{n_t} = p_{n_t-1} + \delta_t$. Toggling slot t from off to on adds exactly δ_t. Over all 2^m on/off patterns, the m slots jointly contribute

$$\Big(\sum_{t=1}^{m} p_{n_t-1}\Big) + \mathrm{sums}(S) \;\supseteq\; \mathrm{base}_0 + [\,c,\ c+M-1\,], \qquad \mathrm{base}_0 := \sum_t p_{n_t-1}\,:$$

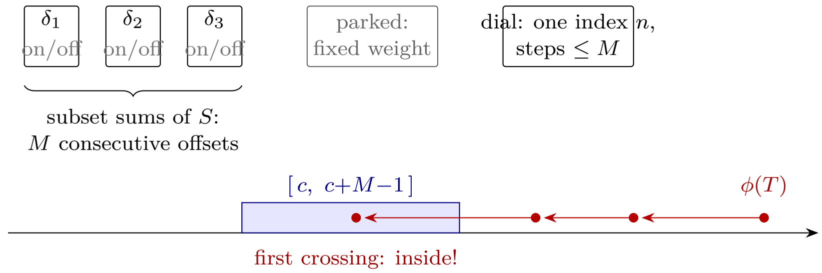


FIGURE 24. The slot machine. The m toggles buy a block of M consecutive achievable offsets; the parked summands are dead weight folded into the base; and the coarse dial walks the remaining requirement $\phi(n)$ *down* in jumps of at most M. A descent by jumps $\le M$ cannot leap over a block of width M: the first value $\le c+M-1$ is automatically $\ge c$.

a full block of M consecutive achievable offsets — this is where the multiset's covering run is cashed in.

The parked summands. We have used m of our k summands. Park the next $k-1-m$ at any fixed tail index (repetition of indices is legal in the sumset), folding their fixed contribution into the base: base := base$_0$ + (parked total). Now $k-1$ summands realize every value of base $+\,[c,\ c+M-1]$.

The coarse dial. The final summand is an index n of our choosing, contributing p_n. In the tail, consecutive prefix sums differ by a gap $g_{n+1} \in G_\infty \subseteq [1, M]$, so $n \mapsto p_n$ is strictly increasing with steps of size at most M, tending to infinity: a coarse dial that ascends forever but can jump by as much as M at a time.

The first-crossing argument. Fix a large target x and let

$$\phi(n) \;:=\; x - \mathrm{base} - p_n,$$

which decreases strictly in n, in steps of at most M, from a starting value $\phi(T) > c+M-1$ (true once x is large, the base and T being fixed) down toward $-\infty$. Let n^* be the *first* index past T with $\phi(n^*) \le c+M-1$. By firstness, $\phi(n^*-1) > c+M-1$; and one step drops ϕ by at most M, so

$$\phi(n^*) \;>\; (c+M-1)-M \;=\; c-1, \qquad \text{i.e.} \qquad \phi(n^*) \;\ge\; c.$$

So $\phi(n^*)$ lands *inside* the achievable block $[c,\ c+M-1]$: some on/off pattern realizes the offset $\phi(n^*)$, and then

$$x \;=\; \mathrm{base} + \phi(n^*) + p_{n^*}$$

exhibits x as a sum of k prefix sums — m slots, $k-1-m$ parked, one dial. (The dial index may coincide with a slot or parked index; harmless, repetitions are allowed.) Hence $kA - ka_1$ contains every large integer, and A is tail-covering (with the factor g restored by Lemma 9.5 if rescaling was applied). □

Watch the accounting at the binding corner (Figure 24). The dial can step by M, so the achievable block must be at least M long — a run of $M-1$ consecutive offsets would let the dial leap clean over it. The block costs $m(G_\infty)$ summands, the dial costs one, and only k are available:

$$m(G_\infty) \;\le\; k-1.$$

When the Adversary's walk uses the largest legal gap infinitely often and the parameters sit at the corner $k = d_2 = M$, this requirement becomes exactly

$$\underbrace{m(G_\infty)}_{\text{fine slots}} + \underbrace{1}_{\text{coarse dial}} \le k = M \qquad \Longleftrightarrow \qquad m(G_\infty) \le M - 1$$

— SHARP, with nothing to spare. The whole weight of the corner case rests on that finite statement, which is why Part IV exists. Away from the corner there is slack: $m(G_\infty) \le M - 1 \le d_2 - 1 \le k - 1$ holds a fortiori.

For perspective — not needed below — a much cruder bound is easy to prove and shows the mechanism in miniature:

Lemma 15.4 (Base grid)**.** *Let G have $|G| \ge 3$, $\gcd(G) = 1$, $a := \min G$ and $M := \max G$. Then $m(G) \le a + M - 1 \le 2M - 3$.*

Proof. First collect residues, greedily. Let $H \subseteq \mathbb{Z}/a$ be the set of residues achievable as subset sums of the multiset built so far, starting from $H = \{0\}$ and the empty multiset. While $H \ne \mathbb{Z}/a$, some $g \in G$ has $H + g \not\subseteq H$: otherwise H would be closed under adding every element of G, hence under the subgroup they generate modulo a, which is all of $\mathbb{Z}/a$ since $\gcd(a, \gcd(G)) = \gcd(a, 1) = 1$ — and as $0 \in H$ that would force $H = \mathbb{Z}/a$. Adjoin one copy of such a g; the achievable set becomes $H \cup (H + g)$, strictly larger. Each step gains at least one residue, so after at most $a - 1$ steps every residue is achievable. Call the resulting multiset R, so $|R| \le a - 1$, and for each residue ρ let $v_\rho \ge 0$ be the smallest subset sum of R in class ρ. Put $S := \max_\rho v_\rho$; since every subset sum of R is at most $(a-1)M$, we have $S \le (a-1)M$.

Now adjoin $x := \lceil (S+M-1)/a \rceil$ copies of a. For any $n \in [S,\ S+M-1]$, take $\rho := n \bmod a$; then $v_\rho \le S \le n$ and $a \mid n - v_\rho$, and $(n - v_\rho)/a \le (S + M - 1)/a \le x$, so $n = v_\rho + a \cdot \frac{n - v_\rho}{a}$ is a subset sum. That is M consecutive integers. Finally $x \le \lceil ((a-1)M + M - 1)/a \rceil = \lceil (aM-1)/a \rceil = M$, so the total size is $|R| + x \le a + M - 1$; and $|G| \ge 3$ squeezes $a \le M - 2$. ☐

The gap between $2M - 3$ and SHARP's $M - 1$ is the gap between an evening's work and Part IV. It is worth being precise about what that gap is worth: the *existence* of an $O(M)$-sized bounded multiset is elementary, as the proof above shows, and we claim no novelty for it. What Part IV buys is the factor of roughly two that brings the budget down to exactly $M - 1$ — and only $M - 1$ is small enough for the slot lemma to afford at the corner $k = d_2 = M$, where $2M - 3$ would be useless.

15.3. **The non-existence half, assembled (modulo SHARP).**

Proposition 15.5 (Goal of Part III)**.** *Assume SHARP. Let $k \ge 3$ and $d_2 \le k$. Then every walk with all gaps $\le d_2$ is tail-covering — in particular every (d_1, d_2)-sequence.*

Proof. Pass to the tail; rescale so $\gcd(G_\infty) = 1$ (Lemmas 9.3, 9.5). If $|G_\infty| \le 2$: Theorem 14.7. If $|G_\infty| \ge 3$: SHARP applies to G_∞ (three or more elements, gcd 1) and gives $m(G_\infty) \le \max(G_\infty) - 1 \le d_2 - 1 \le k - 1$, so the slot lemma (Lemma 15.3) concludes. ☐

Combined with the certificate (Lemma 8.6), this proves the strong non-existence half of Theorem A — *modulo SHARP*. The debt is now fully explicit, and Part IV pays it.

State of play. We now know: both halves of the dichotomy, *provided* $m(G) \le M - 1$ for every finite G with $|G| \ge 3$ and $\gcd(G) = 1$. What remains: exactly that one finite statement — no sequences, no walks, no lacunarity. Next: Part IV, which is about nothing but multisets of integers.

Remark 15.6 (Two details of the slot construction). Parking the surplus summands at index 0 would be legitimate — indices ≥ 0 are permitted by Lemma 9.1 — and changes only the constant base. The requirement that the block have length M rather than $M-1$, by contrast, is not cosmetic: the dial advances in steps of size up to M, and a block of $M-1$ consecutive offsets can be stepped over. Concretely, a dial moving in multiples of 3 passes $6 \mapsto 3$, skipping the two-element block $\{4,5\}$ entirely.

Part IV. **The subset-sum lemma**

16. The subset-sum lemma: statement, sharpness, and the map

Part IV is a self-contained piece of finite mathematics. It never mentions walks, sumsets of sequences, or lacunarity; it proves exactly the statement that Part III left as a debt. Recall $m(G)$ from Definition 15.1: the smallest size of a multiset drawn from G whose subset sums contain $\max(G)$ consecutive integers.

Theorem 16.1 (Bounded subset-sum covering — "SHARP")**.** *Every finite set G of positive integers with $|G| \geq 3$, $\gcd(G)=1$ and $\max(G)=M$ satisfies*

$$m(G) \;\leq\; M-1.$$

Before proving it (§§17–24), let us be precise about what the bound claims, and check that neither hypothesis nor constant can be weakened.

16.1. **First moves, and sharpness.** *If* $1 \in G$, take $M-1$ copies of 1: the subset sums are $0,1,\ldots,M-1$ — exactly M consecutive integers at budget $M-1$. So assume from now on $a := \min G \geq 2$.

The constant cannot be improved to $M-2$. Take $G=\{3,4,5\}$, so $M=5$. Example 15.2 showed $m(G) \leq 4$ via $\{3,3,4,5\}$; we claim no multiset of size ≤ 3 works, so $m(\{3,4,5\}) = 4 = M-1$ exactly. This is a finite check — there are 3 multisets of size one, 6 of size two, and 10 of size three — and the size-three table settles it (smaller multisets have fewer sums and do no better):

Multiset	Subset sums	Longest run
$\{3,3,3\}$	$0,3,6,9$	1
$\{3,3,4\}$	$0,3,4,6,7,10$	2
$\{3,3,5\}$	$0,3,5,6,8,11$	2
$\{3,4,4\}$	$0,3,4,7,8,11$	2
$\{3,4,5\}$	$0,3,4,5,7,8,9,12$	3
$\{3,5,5\}$	$0,3,5,8,10,13$	1
$\{4,4,4\}$	$0,4,8,12$	1
$\{4,4,5\}$	$0,4,5,8,9,13$	2
$\{4,5,5\}$	$0,4,5,9,10,14$	2
$\{5,5,5\}$	$0,5,10,15$	1

The best run of consecutive integers achievable with three elements is 3, short of the required $M=5$. (And smaller multisets genuinely do no better: every multiset of size ≤ 3 sits inside one of size exactly 3, and adding elements only adds subset sums, so runs never shrink — the table's maximum bounds them all.)

The hypothesis $|G| \geq 3$ *cannot be dropped.* Take $G=\{2,3\}$: $\gcd=1$, $M=3$, and the target would be $m(G) \leq 2$. The five nonempty multisets of size ≤ 2 have subset sums

$$\{0,2\}, \quad \{0,3\}, \quad \{0,2,4\}, \quad \{0,2,3,5\}, \quad \{0,3,6\},$$

and none contains 3 consecutive integers. So two-element sets genuinely need more room; three elements is where the uniform budget $M-1$ becomes available.

Finally, a caution about what SHARP does *not* say: it is not a formula for $m(G)$, which is usually far smaller than $M-1$ (for $G=\{1,2,M\}$ it is tiny — Example 16.2). It is a *uniform budget*, tuned to be exactly what the slot lemma's corner $k=d_2=M$ can afford — recall §15: one dial summand plus at most $M-1$ slot summands is all that $k=M$ pays for.

Example 16.2 (A case where $m(G)$ is far below the bound)**.** For $G=\{1,2,M\}$ one has $m(G)\le\lceil M/2\rceil$: one copy of 1 together with $j:=\lceil(M-2)/2\rceil$ copies of 2 has subset sums filling $[0,\ 2j+1]$, a run of $2j+2\ge M$ at size $1+j=\lceil M/2\rceil$. (Equality holds — any covering multiset needs at least one copy of 1, since otherwise every subset sum is even, and then the odd elements available force $i+2j\ge M-1$, whence $i+j\ge M/2$ — but we need only the upper bound.) Theorem 16.1 is a uniform budget, not a formula: $m(G)$ is typically far smaller than $M-1$, and we make no attempt here to determine it in general.

16.2. **The shape of the proof.** The whole of Part IV runs on a single covering mechanism, previewed here and built in §17. To cover M consecutive integers with copies of elements of G:

(1) *Cover the residues.* Using a bounded number of copies of the two largest generators, hit every residue class modulo a (the smallest generator) at least once — a *box* of representatives.
(2) *Slide and stack.* Add $0,1,2,\dots,x$ copies of a: each representative slides up an arithmetic progression of step a, and the progressions interleave into a solid run once every class is represented and the run is long enough.

The cost accounting — how many copies of a suffice, and whether the total stays under $M-1$ — is the *budget*, and the entire difficulty of Part IV is that near the corner the budget has no slack at all.

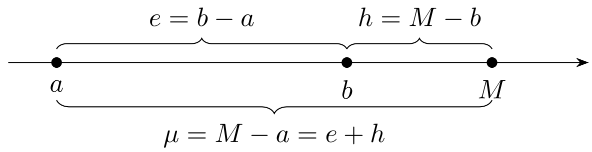
The proof organizes as follows.

- §17: the covering machinery — a two-generator interval lemma, the frame lemma (the slide-and-stack step made exact), the staircase (a refined multi-level version), and the λ-lift (a way to certify infinitely many triples with one finite check).
- §18: reduction of the general statement to *hard-core triples* $\{a,b,M\}$ with $\gcd(a,b)=1$ and b close to M (precisely $1\le M-b\le a-2$).
- §§19–23: the hard core is carved by simple arithmetic invariants into six branches — D, P, L, E, T, B — four closed by explicit formulas valid across their whole range, two (T and B) closed by formulas *plus* a finite list of exceptional triples, each carrying a small computer-verified certificate. §26 explains exactly what was computed and how it is checked.
- §24: assembly, exhaustiveness of the six branches, and the sharpness proposition.

The six-way split is by two divisibility questions and two size questions, routed in order (evaluate top to bottom, take the first that applies):

Branch	Condition	Mechanism	Closure
D (*divisible max*)	$a\mid M$	b spans the residues alone	formula
P (*paired movers*)	$a\mid b+M$	(b,M) pairs, residue-neutral	formula
L (*linear spacing*)	$e=h$	staircase, extended form	formula
E (η-*box*)	$a\ge 12$ and $\mu\ge 12$	the η-box	formula
T (*thin* μ)	$\mu\le 11$	staircase lines; 172 exceptions	formula + table
B (*bounded base*)	$a\le 11$, $\mu\ge 12$	178 bases, lifted	table + lift

The three symbols in the conditions are just the three gaps in the triple, fixed here once so the table can be read now rather than after §17:



So "μ small" means the generators are crowded near M, and "a small" means the smallest generator is tiny. And the 12s and 11s are not deep: 12 is simply the cheapest constant at which Case E's budget estimate closes — see the last line of its proof — and T, B split what E's sizes exclude.) One more reading aid for the whole part: every case section below answers the same three questions — *how are the residues covered? how is the solid run produced? why is the budget* $\leq M-1$*?* If you ever feel lost in a case, locate which of the three questions the current paragraph is answering.

17. Machinery: grids, frames, staircases, and the lift

Everything in Part IV covers runs of consecutive integers by the same two-phase mechanism sketched in §16.2. This section builds it once, with exact constants, so the case sections can spend their attention on budgets.

17.1. **The two-generator grid.** How much of the number line do two coprime generators cover, using boundedly many copies of each? The classic "postage stamp" fact — with stamps of two coprime denominations α, β and unlimited copies, every amount from $(\alpha-1)(\beta-1)$ on is payable — has the following bounded-copies version.

Lemma 17.1 (Two-generator interval)**.** *Let $\alpha < \beta$ be coprime positive integers, and let $x \geq \beta - 1$ and $y \geq \alpha - 1$ be integers. Then, with $C := (\alpha-1)(\beta-1)$,*

$$\{\, i\alpha + j\beta \ :\ 0 \leq i \leq x,\ 0 \leq j \leq y \,\} \ \supseteq\ [\, C,\ \ \alpha x + \beta y - C\,].$$

Note the symmetry of the conclusion: the grid covers everything except a margin of C at each end of its full span $[0,\ \alpha x + \beta y]$.

Proof. Case $\alpha = 1$. Here $C = 0$. For $n \in [0,\ x + \beta y]$ put $j := \min(y,\ \lfloor n/\beta \rfloor)$. If $j = \lfloor n/\beta \rfloor$ then $n - j\beta$ is the remainder, in $[0, \beta-1] \subseteq [0, x]$; if instead $j = y$ then $n - y\beta \leq x$ directly. Either way $i := n - j\beta \in [0, x]$.

Case $\alpha \geq 2$, base $y = \alpha - 1$. Let $n \in [\, C,\ \alpha x + \beta(\alpha-1) - C\,]$. Since $\gcd(\alpha, \beta) = 1$, the multiples $0, \beta, 2\beta, \ldots, (\alpha-1)\beta$ hit all residues modulo α — the covering lemma (Lemma 7.6), in its first of many appearances in this part; choose the unique $j \in [0, \alpha-1]$ with $j\beta \equiv n \pmod{\alpha}$. Then $n - j\beta$ is divisible by α and

$$n - j\beta \ \geq\ C - (\alpha-1)\beta \ =\ -(\alpha-1);$$

a multiple of α that is $\geq -(\alpha-1)$ is ≥ 0. And with $i := (n - j\beta)/\alpha$:

$$n - j\beta \ \leq\ \big(\alpha x + \beta(\alpha-1) - C\big) - j\beta \ \leq\ \alpha x + (\alpha-1)\beta - C \ =\ \alpha x + (\alpha - 1),$$

so $i \leq x + \frac{\alpha-1}{\alpha} < x + 1$, i.e. $i \leq x$.

Induction $y \to y+1$ (for $y \geq \alpha - 1$). (First pass: skippable.) Only the new stretch $n \in (\,\alpha x + \beta y - C,\ \alpha x + \beta(y+1) - C\,]$ needs attention. Subtract one β: certainly $n - \beta \leq \alpha x + \beta y - C$; and $n - \beta > \alpha x + \beta y - C - \beta \geq C$, because

$$\alpha x + \beta y \ \geq\ \alpha(\beta-1) + \beta(\alpha-1) \ =\ 2\alpha\beta - \alpha - \beta \ \geq\ 2C + \beta,$$

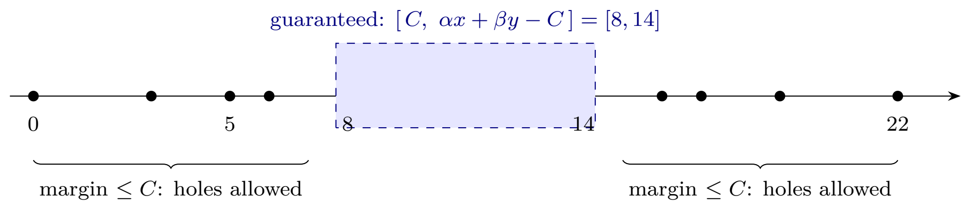


FIGURE 25. Lemma 17.1 for $(\alpha,\beta)=(3,5)$, $x=4$, $y=2$ (so $C=8$): the dots are the grid values $\{3i+5j\}$, solid throughout the guaranteed middle $[8,14]$, gappy only in the two margins of size C — exactly the shape the lemma promises. (Here the coverage happens to extend beyond the guarantee; the margins are worst cases.)

where the last inequality, written out, is

$$2\alpha\beta-\alpha-\beta\ \ge\ 2(\alpha-1)(\beta-1)+\beta\ =\ 2\alpha\beta-2\alpha-\beta+2\qquad\Longleftrightarrow\qquad\alpha\ \ge\ 2.$$

So $n-\beta$ lies in the already-covered interval: $n-\beta=i\alpha+j\beta$ with $i\le x$, $j\le y$, and $n=i\alpha+(j+1)\beta$ with $j+1\le y+1$. $\square$

17.2. **The frame lemma, and budgets.** The second phase — slide and stack — in its general form:

Lemma 17.2 (Frame lemma). *Let ν,g_1,g_2 be positive integers and $L\ge 1$ a target run length. Suppose a* box *of size (Y,Z) covers all residues modulo ν: for every residue ρ there are $j_\rho\le Y$, $k_\rho\le Z$ with $j_\rho g_1+k_\rho g_2\equiv\rho\pmod{\nu}$. Let*

$$S\ :=\ \max_{\rho}\,(j_\rho g_1+k_\rho g_2),\qquad x\ :=\ \Big\lceil\frac{L-1+S}{\nu}\Big\rceil.$$

Then the multiset of Y copies of g_1, Z copies of g_2 and x copies of ν — size $Y+Z+x$ — has subset sums covering the entire integer interval $[\,S,\ \nu x\,]$, which contains $[\,S,\ S+L-1\,]$: a run of length L.

Proof. Let $n\in[S,\nu x]$ and $\rho:=n\bmod\nu$. The representative $r:=j_\rho g_1+k_\rho g_2$ satisfies $r\le S\le n$ and $r\equiv n\pmod{\nu}$, so $t:=(n-r)/\nu$ is a nonnegative integer, and $t\le n/\nu\le x$. Take j_ρ copies of g_1, k_ρ of g_2, and t of ν. Finally $\nu x\ge L-1+S$ by the choice of x, so the covered interval contains $[S,S+L-1]$. $\square$

In every use below, $\nu=a=\min G$, $(g_1,g_2)=(b,M)$, and $L=M$ (Figure 26 shows a complete instance). The *budget* of a box (Y,Z) for the triple (a,b,M) is the resulting multiset size,

$$\mathcal{B}\ =\ \Big\lceil\frac{M-1+S}{a}\Big\rceil+Y+Z,$$

and a box *certifies* the triple when it covers all residues, $Y+Z\le a-1$, and $\mathcal{B}\le M-1$. (The middle condition looks unmotivated, and it is — for now. It does no work at all until §23, where it is exactly what makes the λ-lift free: a box with $Y+Z\le a-1$ costs at most a extra budget per lift, matching what the target gains. Carry it as bookkeeping until then.) The case analysis is, at bottom, a hunt for certifying boxes; the exceptional tables of cases T and B are nothing but lists of such boxes, checkable by the two-line computation in this proof.

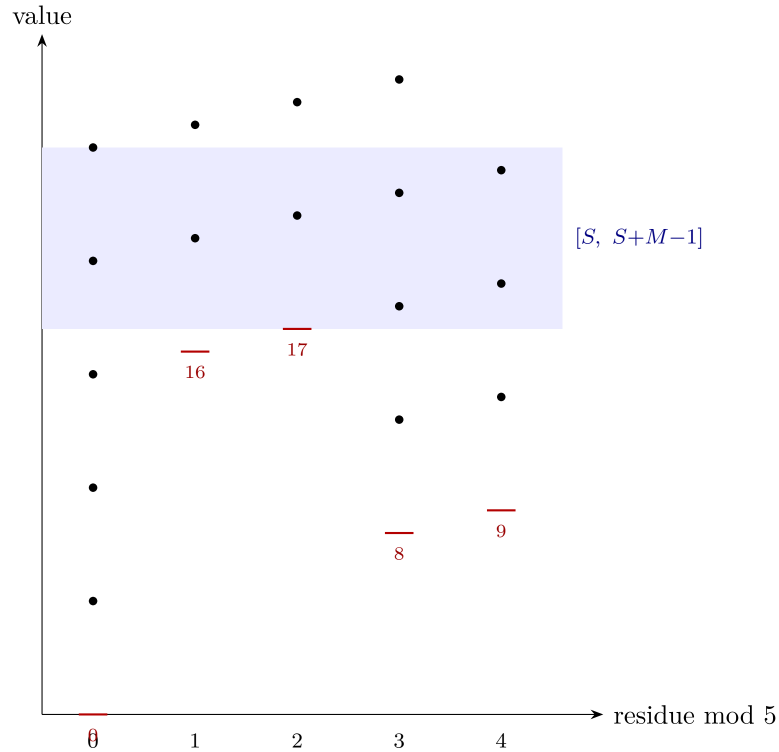
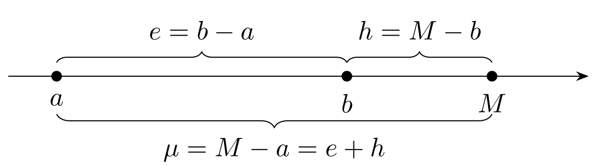


FIGURE 26. Slide and stack, on the triple $(5, 8, 9)$ with box representatives $0, 16, 17, 8, 9$ (values $8j + 9k$, one per residue class modulo 5, marked red). Stacking copies of $a = 5$ on each representative sends five arithmetic progressions climbing in step; from the height $S = 17$ of the *tallest* representative onward, every column is populated, and the shaded band $[S,\ S+M-1] = [17, 25]$ is covered solidly. This one picture is the mechanism of every construction in Part IV.

Mechanism of Part IV. All of Part IV is one mechanism run six ways: *cover the residues modulo a* with a few copies of the big generators, then *slide and stack* with copies of a until the progressions knit into a solid run (Figure 26). The cost is the height of the tallest residue representative, divided by a, plus the box size — and the entire difficulty of the six branches is that near the corner this cost must squeeze under $M - 1$ with zero slack.

17.3. **Standing notation for triples.** *Notation reset.* Part IV reuses a few letters from Part III with new meanings, and from here on the old meanings never reappear: e and h now measure a *triple* (not a two-letter difference, not a binary word), and c will count box copies (not an offset). The picture to keep:



From §18 onward we work with triples $G = \{a, b, M\}$, $a < b < M$, $\gcd(a, b) = 1$. Fix, once and for all,

$$e := b - a, \qquad h := M - b, \qquad \mu := M - a = e + h, \qquad g := \gcd(e, \mu),$$

$$e' := e/g, \qquad \mu' := \mu/g, \qquad C' := (e' - 1)(\mu' - 1).$$

(Note $\gcd(e, \mu) = \gcd(e, h)$, and e', μ' are coprime with $\mu' > e'$, so $\mu' \geq 2$ whenever $h \geq 1$.) One fact used repeatedly:

$$\gcd(a, g) = 1 :$$

a common prime of a and g would divide e and a, hence $b = a + e$, contradicting $\gcd(a, b) = 1$.

17.4. **Levels and staircases.** Consider a multiset of x copies of a, y of b, z of M, with the standing hypotheses

$$y \geq \mu' - 1, \qquad z \geq e' - 1;$$

put $c := y + z$ and $V' := y\, e' + z\, \mu' - C'$. Group its subset sums by the *number of elements used*: choosing i copies of a, j of b, k of M with $i + j + k = t$ gives

$$i\, a + j\, b + k\, M \;=\; t\, a + j\, e + k\, \mu,$$

so *level* t ($0 \leq t \leq x + y + z$) consists of $ta + \mathcal{O}_t$ with offset set

$$\mathcal{O}_t \;=\; \{\, je + k\mu \;:\; j \leq y,\ k \leq z,\ \max(0,\ t - x) \leq j + k \leq t \,\}.$$

Each level is a shifted copy of (part of) the same two-generator grid — a *frame* — and consecutive levels are offset by exactly a. Frames merging into solid runs is the entire game.

Before the general lemma, one instance in full. Take $(a, b, M) = (7, 9, 10)$, so $e = 2$, $h = 1$, $\mu = 3$, $g = 1$, $(e', \mu') = (2, 3)$ and $C' = (2 - 1)(3 - 1) = 2$. Choose $y = 2$ copies of b and $z = 2$ copies of M, so $c = y + z = 4$ and

$$V' \;=\; y\, e' + z\, \mu' - C' \;=\; 4 + 6 - 2 \;=\; 8,$$

and take $x = c + 1 = 5$ copies of a: the multiset $\{7^5,\ 9^2,\ 10^2\}$, of size $9 = M - 1$ exactly. Now watch the levels. The offsets available at any level $t \in [c, x] = [4, 5]$ are all of $\{2j + 3k : j \leq 2,\ k \leq 2\} = \{0, 2, 3, 4, 5, 6, 7, 8, 10\}$, which contains the guaranteed block $[C', V'] = [2, 8]$. So:

$$\text{level } 4: \quad 4 \cdot 7 + [2, 8] = [30, 36], \qquad \text{level } 5: \quad 5 \cdot 7 + [2, 8] = [37, 43].$$

The two frames *abut exactly* — 36 and 37 are consecutive — and their union $[30, 43]$ is a run of $14 \geq M = 10$ consecutive integers. Why did they abut rather than overlap or leave a hole? Because

$$V' - C' \;=\; 6 \;=\; a - 1$$

holds here with equality, which is precisely the merge condition of part (b) below. One unit less of frame and 37 would have been stranded. Keep this picture in mind: *the lemma's clauses are the several ways frames can be made to chain.*

Lemma 17.3 (Staircase)**.** *In the setting above:*

- (a) *For every t with $c \leq t \leq x$: $\mathcal{O}_t \supseteq g \cdot [\,C',\ V'\,]$ (meaning $\{gu : C' \leq u \leq V'\}$).*
- (b) (Two-frame merge; $g = 1$.) *If $x = c + 1$ and $V' - C' \geq a - 1$, the subset sums cover the interval $[\,ca + C',\ (c + 1)a + V'\,]$, of length $a + (V' - C') + 1$.*
- (c) (Short merge; $g = 1$.) *If $x = c$, then $\mathcal{O}_{c+1} \supseteq [\max(C', 1),\ V']$, and if additionally $V' \geq a + \max(C', 1) - 1$ the subset sums cover $[\,ca + C',\ (c + 1)a + V'\,]$.*

(d) (Phase union; any $g \geq 1$.) *If $x = c+g$, the subset sums cover* $[\,(c+g-1)a + gC',\ ca + gV'\,]$ *with* no *further hypothesis (the* base form*). If moreover $x \geq c+g$ and the* frame-merge condition $V' - C' \geq a-1$ *holds, they cover* $[\,(c+g-1)a+gC',\ (x-g+1)a+gV'\,]$ *(the* extended form*).*

Proof. (a) For $t \in [c, x]$, *every* box pair (j,k) $(j \leq y,\, k \leq z)$ is admitted at level t: $j+k \leq c \leq t$ and $i = t-j-k$ lies in $[0,x]$. So $\mathcal{O}_t \supseteq \{je + k\mu\} = g\,\{je' + k\mu'\}$, and Lemma 17.1 applied to the coprime pair (e', μ') — its hypotheses are exactly $y \geq \mu' - 1$, $z \geq e' - 1$ — covers $[C', V']$.

(b) Levels c and $c+1$ both lie in $[c,x]$, so each covers $ta + [C', V']$; two intervals offset by a merge into one exactly when $ca + V' \geq (c+1)a + C' - 1$, i.e. $V' - C' \geq a - 1$.

(c) At level $c+1 = x+1$, the admission constraint is $j + k \geq t - x = 1$: only the pair $(0,0)$ — offset 0 — is lost. Every offset in $[\max(C', 1), V']$ is nonzero, so all its grid representations survive. The merge then needs $ca + V' \geq (c+1)a + \max(C', 1) - 1$.

(d) *Base form.* All levels $t \in [c, c+g]$ satisfy (a). Level t's values $ta + g[C', V']$ lie in the residue class of ta modulo g, and since $\gcd(a,g) = 1$ the map $t \mapsto ta \bmod g$ is a bijection on any g consecutive levels — the covering lemma (Lemma 7.6), with modulus g and multiplier a: each residue class modulo g is served by exactly one $t_0 \in [c,\ c+g-1]$. Given n in the stated interval, let t_0 be its server and $u := (n - t_0 a)/g$; then $u \geq C'$ (since $n \geq (c+g-1)a + gC' \geq t_0 a + gC'$) and $u \leq V'$ (since $n \leq ca + gV' \leq t_0 a + gV'$), so level t_0 realizes n.

Extended form. Within the class served by t_0, the frames at levels $t_0, t_0+g, t_0+2g, \ldots$ (all $\leq x$) are arithmetic progressions of step g offset by ga from one another; consecutive frames leave no class-gap exactly when $(t_0+g)a + gC' \leq t_0 a + gV' + g$, i.e. when $V' - C' \geq a-1$ — the frame-merge condition. Under it, the class is covered solidly from $t_0 a + gC'$ up to $(t_0 + Jg)a + gV'$, where $J := \lfloor (x - t_0)/g \rfloor$. Over $t_0 \in [c, c+g-1]$, all left endpoints are $\leq (c+g-1)a + gC'$, and maximality of J $(t_0 + Jg > x - g)$ makes all right endpoints $\geq (x-g+1)a + gV'$; so every integer of the stated interval is covered within its own class.

Nothing in this argument needs $g \geq 2$. At $g = 1$ the modular bookkeeping simply degenerates: there is one residue class modulo 1, served by the single level $t_0 = c$, the frames at $c, c+1, c+2, \ldots$ chain under the same merge condition $V' - C' \geq a - 1$, and both displayed intervals read correctly. Clause (b) is then the two-frame instance $(x = c+1)$ of the extended form. We state (d) for all $g \geq 1$ because Case L (§19) invokes it on an infinite family with $g = 1$: the triples $(a, a+1, a+2)$ with a even, of which $(4,5,6)$ is the smallest. ☐

Remark 17.4 (The merge condition is real). The frame-merge hypothesis in the extended form cannot be dropped. For $(a,b,M) = (5,7,9)$: $e = 2$, $\mu = 4$, $g = 2$, $(e', \mu') = (1,2)$, $C' = 0$. Take $y = 1$, $z = 0$, so $c = 1$, $V' = 1$, and $x = 3 = c+g$. The merge condition fails $(V' - C' = 1 < 4 = a - 1)$ — and so does the would-be conclusion: the multiset $\{5,5,5,7\}$ has subset sums $\{0,5,7,10,12,15,17,22\}$, missing 11 inside the interval $[10,12]$ that the un-repaired statement would promise. (An early draft of this staircase lemma omitted this hypothesis; the formalization caught it, and the counterexample is now permanently pinned in the Lean development as a guard. A cautionary tale told honestly: machine-checking is not decoration.) Figure 27 shows the merge working and failing, side by side.

17.5. **The λ-lift.** The final tool lets one finite check certify an infinite family. Triples (a,b,M) and $(a, b+a, M+a)$ have the same a, the same $h = M - b$, and the same residues $b, M \bmod a$; climbing this ladder is the *λ-lift*.

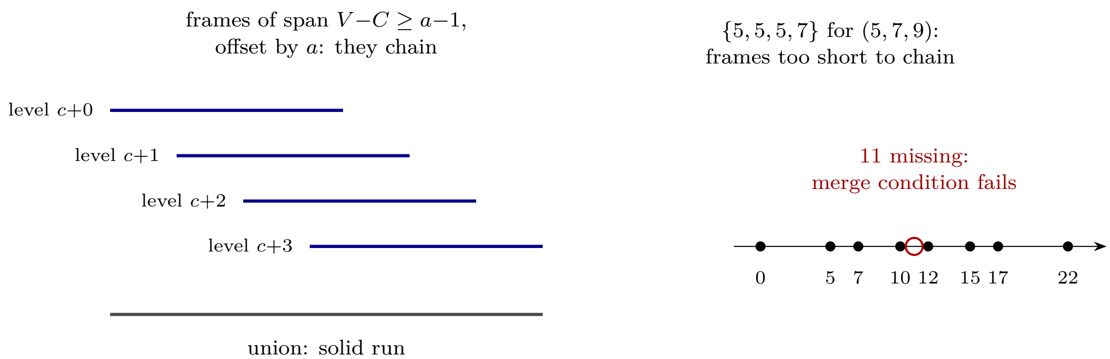


FIGURE 27. Left: the staircase when the frame-merge condition $V'-C' \geq a-1$ holds — consecutive levels' frames overlap and their union is one solid run. Right: the counterexample of Remark 17.4 — subset sums of $\{5,5,5,7\}$, whose frames are too short, leaving the hole at 11.

Lemma 17.5 (λ-lift)**.** *Suppose the box (Y,Z) certifies the triple (a,b,M): it covers all residues modulo a, has $Y+Z \leq a-1$, and budget $\mathcal{B} \leq M-1$. Then the* same *box certifies $(a,\ b+a,\ M+a)$.*

Proof. Modulo a, the generators are unchanged, so the same pairs (j_ρ, k_ρ) still cover all residues. Each representative's height grows by at most $(j_\rho + k_\rho)\,a \leq (Y+Z)\,a$, so the new maximum height satisfies $S' \leq S + (Y+Z)a$, and the new budget is

$$\left\lceil \frac{(M+a)-1+S'}{a} \right\rceil + Y + Z \ \leq\ \left\lceil \frac{M-1+S}{a} \right\rceil + 1 + (Y+Z) + Y + Z \ =\ \mathcal{B} + (Y+Z) + 1.$$

The budget grew by at most $Y+Z+1 \leq a$; the target $M-1$ grew by exactly a. A certificate whose margin was ≥ 0 keeps a margin ≥ 0 — forever, up the whole ladder. $\square$

Remark 17.6 (The merge condition, tested)*.* The subset sums of $\{5,5,5,7\}$ are $\{0,5,7,10,12,15,17,22\}$, and no number of copies of 5 together with a single 7 can produce five consecutive integers: every such sum lies in $\{0,2\}$ mod 5. What is missing is a box covering *all* residues, which is precisely what the frame lemma supplies and what the un-hypothesized staircase statement lacked.

Remark 17.7 (Necessity of the box-size condition)*.* The hypothesis $Y+Z \leq a-1$ caps the budget growth at $Y+Z+1 \leq a$ per lift, matching the target's growth of exactly a. With $Y+Z = a$ the margin leaks one unit per step. Take $a=5$ and the box $(Y,Z) = (3,2)$ on the class base $(5,16,17)$: the representative heights are bounded by $3 \cdot 16 + 2 \cdot 17 = 82$, giving budget $\lceil (16+82)/5 \rceil + 5 = 25$ against a target of 16; one lift to $(5,21,22)$ gives $3 \cdot 21 + 2 \cdot 22 = 107$ and budget $\lceil (21+107)/5 \rceil + 5 = 31$ against 21. The budget rose by $6 = Y+Z+1$, the target by $5 = a$, and the deficit compounds forever.

18. Reduction to the hard core

This section shrinks the problem from all alphabets to a two-parameter family of triples — a funnel with five stages, each stage one lemma:

$$\underset{\text{any } G}{\boxed{|G| \geq 3,\ \gcd = 1}} \xrightarrow{18.2} \boxed{\text{irreducible}} \xrightarrow{18.8} \boxed{\text{triples}} \xrightarrow{18.6} \boxed{\gcd(a,b) = 1} \xrightarrow{18.3,\ 18.4} \underset{\text{hard core}}{\boxed{h \leq a-2}}$$

The tools are one growth lemma of Graham and four reduction lemmas, each an explicit construction.

18.1. Graham's growth lemma.

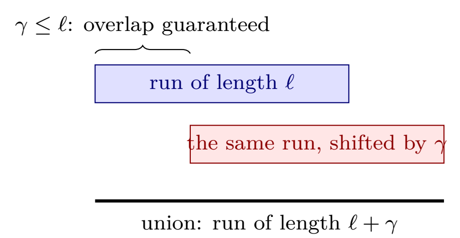
Lemma 18.1 (Graham growth)**.** *Suppose* $\mathrm{sums}(S)$ *contains a run of* ℓ *consecutive integers, and* γ *is a positive integer with* $\gamma \le \ell$*. Then* $\mathrm{sums}(S \cup \{\gamma\})$ *(one new copy of* γ*) contains a run of* $\ell + \gamma$ *consecutive integers.*

Proof. Let $[u,\ u+\ell-1] \subseteq \mathrm{sums}(S)$. The enlarged multiset's sums include both the old run and the old run shifted by γ:

$$[u,\ u+\ell-1] \ \cup\ [u+\gamma,\ u+\gamma+\ell-1].$$

Since $\gamma \le \ell$, the shifted run starts no later than $u+\ell$, so the two runs overlap or abut, forming $[u,\ u+\ell+\gamma-1]$. ☐

A run, once at least as long as a generator, can be extended by that generator forever at a price of one copy per γ gained. This ratchet is what lets us prove SHARP for a *smaller* maximum and then climb.



18.2. **Discarding redundant elements.** Call an alphabet G *irreducible* if $|G| = 3$, or $|G| \ge 4$ and G is *minimal*: no proper subset of size ≥ 3 has gcd equal to 1.

Lemma 18.2 (Graham reduction)**.** *Fix* $M \ge 2$*. If SHARP holds for every irreducible alphabet of maximum* $\le M$*, then it holds for every alphabet of maximum* $\le M$*.*

The grading by M matters. §24 applies this lemma *inside* a strong induction on the maximum, at a stage where SHARP is available only below M and, concurrently, for irreducible alphabets at M. An ungraded statement — "if SHARP holds for all irreducible alphabets" — would have a hypothesis that is not yet in hand at that moment, and invoking it would look circular even though the underlying recursion is not. The proof below is unchanged by the grading: every alphabet it touches has maximum $\le M$.

Proof. Induct on $|G|$, with the maximum bound M fixed throughout. Let $|G| \ge 4$ be non-minimal: some $e^\dagger \in G$ is *redundant*, meaning $G \setminus \{e^\dagger\}$ still has gcd 1 (and size ≥ 3). Two cases.

If $e^\dagger \ne M$: the smaller alphabet has the same maximum M, so its SHARP witness — a multiset of size $\le M-1$ covering M consecutive integers — is literally also a witness for G.

If $e^\dagger = M$: let $M' := \max(G \setminus \{M\})$. By induction, a multiset of size $\le M'-1$ drawn from $G \setminus \{M\}$ covers a run of M' consecutive integers. Now ratchet with $a := \min G$: the run has length $M' > a$, so each added copy of a extends it by a (Lemma 18.1, the hypothesis holding throughout since runs only grow); after $\lceil (M-M')/a \rceil \le M - M'$ copies the run has length $\ge M$. Total size $\le (M'-1) + (M-M') = M-1$. ☐

18.3. **Cheap pairs.** Three lemmas now dispose of every triple whose two smaller elements are arithmetically convenient. Throughout, G has $\min G = a \geq 2$ and $\max G = M$.

Lemma 18.3 (Small coprime pair)**.** *If some coprime pair $\alpha, \gamma \in G$ has $\alpha + \gamma \leq M - 1$, then $m(G) \leq M - 1$.*

Proof. Take $x := M - \alpha$ copies of α and $\alpha - 1$ copies of γ — budget exactly $(M-\alpha)+(\alpha-1) = M - 1$. The representatives $0, \gamma, 2\gamma, \ldots, (\alpha-1)\gamma$ cover all residues modulo α (coprimality), with top height $S = (\alpha-1)\gamma$, so by the frame mechanism (Lemma 17.2 with $\nu = \alpha$) the sums cover $[\,S,\ \alpha x\,]$. Its length:

$$\alpha x - S + 1 \ = \ \alpha(M-\alpha) - (\alpha-1)\gamma + 1 \ \geq \ \alpha(M-\alpha) - (\alpha-1)(M-\alpha-1) + 1 \ = \ M,$$

using $\gamma \leq M - \alpha - 1$; the expansion, line by line, is item (A1) of Appendix C — it lands exactly on M, with nothing to spare. ☐

Lemma 18.4 (Pair at the boundary)**.** *If some coprime pair $\alpha < \beta$ in G has $\alpha + \beta \in \{M,\ M+1\}$, then $m(G) \leq M - 1$.*

Proof. For a parameter $t \geq 0$, take $x := \beta - 1$ copies of α and $y := \alpha - 1 + t$ copies of β. Lemma 17.1 applies and covers $[\,C,\ \alpha x + \beta y - C\,]$ with $C = (\alpha-1)(\beta-1)$; the length works out (item (A2) of Appendix C) to

$$\alpha x + \beta y - 2C + 1 \ = \ \alpha + \beta - 1 + t\beta,$$

at budget $x + y = \alpha + \beta - 2 + t$. If $\alpha + \beta = M + 1$: take $t = 0$; length M, budget $M - 1$. If $\alpha + \beta = M$: $t = 0$ gives length $M - 1$ — *one short* — so take $t = 1$: length $M - 1 + \beta \geq M$, budget $M - 1$ exactly. The budget absorbs precisely one extra copy; nothing is wasted, nothing is spare. ☐

Example 18.5. $(\alpha, \beta, M) = (2, 3, 5)$, the $\alpha + \beta = M$ case. At $t = 0$ the multiset $\{2, 2, 3\}$ has sums $\{0, 2, 3, 4, 5, 7\}$: longest run $[2, 5]$, length $4 < 5$, insufficient exactly as the bookkeeping predicts. At $t = 1$, $\{2, 2, 3, 3\}$ has sums $\{0, 2, 3, 4, 5, 6, 7, 8, 10\} \supseteq [2, 8]$: seven consecutive, at budget $4 = M - 1$.

Lemma 18.6 (Non-coprime pair, $|G| = 3$)**.** *Let $G = \{a, b, M\}$ with $d := \gcd(a, b) \geq 2$. (Then $\gcd(d, M) = 1$ automatically, since $\gcd(G) = 1$.) Then $m(G) \leq M - 1$.*

Proof. Write $a = d\alpha$, $b = d\beta$, $\gcd(\alpha, \beta) = 1$, $\alpha < \beta$. The multiset:

$$x := \beta - 1 \text{ copies of } a, \qquad y := M - d - \beta + 1 \text{ copies of } b, \qquad d - 1 \text{ copies of } M,$$

of size $(\beta - 1) + (M - d - \beta + 1) + (d - 1) = M - 1$ exactly.

The scaled grid. Sums using only a's and b's are d times sums from the (α, β)-grid. The hypotheses of Lemma 17.1: $x = \beta - 1$ exactly, and $y \geq \alpha - 1$ because $M > b = d\beta$ forces $M \geq d\beta + 1$, so $y \geq d\beta + 1 - d - \beta + 1 = (d-1)(\beta-1) + 1 \geq \beta > \alpha - 1$. With $C := (\alpha-1)(\beta-1)$, the covered set is the arithmetic progression $d \cdot [\,C,\ \alpha x + \beta y - C\,]$, of step d and

$$L \ := \ \alpha x + \beta y - 2C + 1 \ \text{ terms.}$$

The grid is long enough: $L \geq M + 1$. (First pass: skippable.) L grows with M (through y), so the worst case is the smallest legal $M = d\beta + 1$, where the claim rearranges (item (A3) of Appendix C, with $u := \beta(d-1)$) to

$$(\beta - 1)\,(u - \alpha + 2) \ \geq \ u + 1,$$

which at $d = 2$ (i.e. $u = \beta$) reduces to $\beta(\beta - \alpha) + \alpha - 3 \geq 0$ — true, since $\beta(\beta - \alpha) \geq \beta \geq 2$ and $\alpha \geq 1$ — and only strengthens as d (hence u) grows, item (A3) again.

Bridging the d residues. The $d-1$ copies of M contribute $0, M, 2M, \dots, (d-1)M$, which — since $\gcd(d, M) = 1$ — run through *all* d residues modulo d. Adding them to the step-d progression fills every gap: for $n \in [\, dC + (d-1)M,\ dC + (L-1)d\,]$, pick $\kappa \in [0, d-1]$ with $\kappa M \equiv n - dC \pmod d$; then $n - \kappa M$ is a step-d progression member (it is $\equiv dC$, and lies in the progression's range). The solid interval has length $(L-1)d - (d-1)M + 1 \geq M$, which is equivalent to $L \geq M + 1$ — just proved; the equivalence, including the small integrality step, is item (A4) of Appendix C. ☐

18.4. Minimal alphabets of four or more elements.

Lemma 18.7 (Structure of minimal alphabets)**.** *Let $G = \{g_1, \dots, g_m\}$ be minimal with $m \geq 4$, and set $d_i := \gcd(G \setminus \{g_i\})$. Then: each $d_i \geq 2$; the d_i are pairwise coprime; $d_j \mid g_i$ whenever $j \neq i$; consequently every $g_i \geq 30$ when $m = 4$ and ≥ 210 when $m \geq 5$; and every pair of elements of G has* $\gcd \geq 2$.

Proof. $d_i \geq 2$ is the definition of minimality (the subset $G \setminus \{g_i\}$, of size ≥ 3, must not have gcd 1). For $i \neq j$, any common divisor of d_i and d_j divides every element of $(G \setminus \{g_i\}) \cup (G \setminus \{g_j\}) = G$, hence divides $\gcd(G) = 1$. That $d_j \mid g_i$ for $j \neq i$ is immediate ($g_i \in G \setminus \{g_j\}$). So g_i is divisible by the product of the $m-1$ pairwise coprime numbers $\{d_j\}_{j \neq i}$, each ≥ 2; the least possible such product is $2 \cdot 3 \cdot 5 = 30$ ($m = 4$) or $2 \cdot 3 \cdot 5 \cdot 7 = 210$ ($m \geq 5$). Finally, a coprime pair inside G would sit inside a proper size-3 subset of gcd 1 (add any third element), contradicting minimality. ☐

Lemma 18.8 (Minimal alphabets)**.** *If G is minimal with $|G| \geq 4$, and SHARP holds for all alphabets of maximum $< M$, then $m(G) \leq M - 1$.*

Proof. (First pass: the two-sentence summary is — rescale the alphabet without its smallest element, apply SHARP inductively at the smaller maximum, then ratchet, scale and bridge as in Lemma 18.6; the rest is the count.)

Descend. Remove $a := \min G$. By minimality $d := \gcd(G \setminus \{a\}) \geq 2$; note $\gcd(a, d) = 1$ (a common factor would divide all of G). The rescaled alphabet $H := (G \setminus \{a\})/d$ has $\gcd(H) = 1$, $|H| \geq 3$, and maximum $M/d < M$ — so SHARP applies to it by hypothesis: a run of M/d consecutive integers within H's subset sums, at cost $\leq M/d - 1$.

Extend. Let $a' := \min H$ and ratchet the run (Lemma 18.1, inside H) up to length

$$\ell \;:=\; \left\lceil \frac{M - 1 + (d-1)a}{d} \right\rceil + 1,$$

at an extra cost of at most $\lceil (\ell - M/d)/a' \rceil$ copies of a'.

Scale and bridge. Multiplying the run's multiset by d (i.e. using the original elements of $G \setminus \{a\}$) turns the run into a step-d progression of ℓ terms; the d residues are bridged by $0, a, 2a, \dots, (d-1)a$ — all residues modulo d, as $\gcd(a, d) = 1$ — yielding a solid interval of length $(\ell - 1)d - (d-1)a + 1 \geq M$ by the choice of ℓ. Total budget:

$$T \;\leq\; \left(\frac{M}{d} - 1\right) + \left\lceil \frac{\ell - M/d}{a'} \right\rceil + (d-1).$$

Count. From $a < \min(G \setminus \{a\}) = d\,a'$, i.e. $a \leq da' - 1$:

$$\ell - \frac{M}{d} \;\leq\; \frac{(d-1)a}{d} + 2 \;<\; (d-1)a' + 2,$$

so the middle term is $\leq d + 1$ and $T \leq M/d + 2d - 1$. The requirement $T \leq M - 1$ rearranges to $M(1 - \frac{1}{d}) \geq 2d$, i.e. $M \geq 2d^2/(d-1)$. For $d \geq 3$ this is at most $3d$, and $M \geq 3d$ because

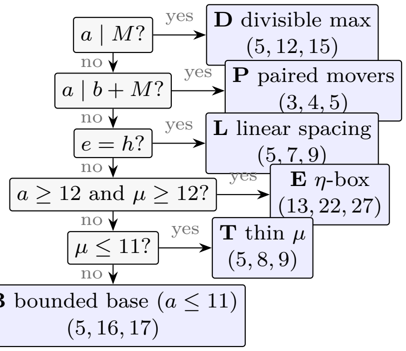


FIGURE 28. Routing a hard-core triple. Conditions are read top to bottom and the first match wins, so by the time you are reading Case E you may assume $a \nmid M$, $a \nmid b+M$ and $e \neq h$ — which is exactly why $\eta \notin \{0, 1, a-1\}$ there. Each branch carries a representative triple; these are the running examples of §§19–23.

$G \setminus \{a\}$ contains three distinct multiples of d. For $d = 2$ it demands $M \geq 8$, — and every element of G is ≥ 30 (Lemma 18.7). □

18.5. **The hard core.** Assemble the reductions. By Lemma 18.2 we may take G irreducible; Lemma 18.8 absorbs the minimal $|G| \geq 4$ case (its appeal to smaller maxima will be organized as a strong induction in §24); so $G = \{a, b, M\}$ is a triple, with $a \geq 2$ after the $1 \in G$ observation of §16. If $\gcd(a, b) \geq 2$: Lemma 18.6. So $\gcd(a, b) = 1$; and now Lemmas 18.3 and 18.4 applied to the pair (a, b) dispose of every triple with $a + b \leq M + 1$. What survives is the *hard core*:

$$G = \{a,\ b,\ M\}, \qquad \gcd(a, b) = 1, \qquad h := M - b \in [\,1,\ a-2\,],$$

(the condition $a + b \geq M + 2$ *is* $h \leq a - 2$), which forces $a \geq 3$. In words: the two large generators crowd within $a - 2$ of the top, and the small generator is at least 3. The standing notation of §17.3 is in force; add to it the mod-a data

$$\bar{e} := e \bmod a \ \in\ [1,\ a-1], \qquad \bar{\mu} := M \bmod a, \qquad \eta := \bar{\mu} \cdot \bar{e}^{\,-1} \bmod a \ \ (\text{when } \bar{\mu} \neq 0),$$

where $\bar{e}^{\,-1}$ is the inverse modulo a (existing since $\gcd(e, a) = \gcd(b - a, a) = \gcd(b, a) = 1$; in particular $\bar{e} \neq 0$). Three observations, each a one-liner worth doing:

$$\eta = 0 \iff a \mid M; \qquad \eta = a - 1 \iff a \mid b + M; \qquad \eta = 1 \text{ is impossible.}$$

(For the last: $\eta = 1$ would mean $\bar{\mu} = \bar{e}$, i.e. $a \mid M - e$; and the complete identity is

$$M - e \ = \ M - (b - a) \ = \ (M - b) + a \ = \ h + a,$$

so $a \mid M - e$ forces $a \mid h$ — impossible, since $1 \leq h \leq a - 2$.) The first two conditions are disjoint on the hard core (both would give $a \mid b$).

Every hard-core triple is now routed through the ordered six-branch tree of §16.2 — first matching condition wins, as Figure 28 shows — and §§19–23 close the branches one by one.

Why the hard core is hard. Why is the hard core hard? Because b and M crowd within $a-2$ of each other at the top: the two big generators are nearly interchangeable, their small difference h and offset e do all the arithmetic work, and whether the residues modulo a can be covered *cheaply enough* comes down to fine divisibility relations between a, e, and h. The six branches are six answers to one question: *who pays for the residue cover?* — b alone (D), residue-neutral pairs (P), an arithmetic progression's regularity (L), the invariant η (E), or a finite certificate where every formula runs out (T, B).

Example 18.9 (The routing, on six triples)**.** Each of the following is hard-core, and together they exercise every branch in order: $(5, 12, 15)$ routes to D ($a \mid M$); $(3, 4, 5)$ to P ($a \mid b+M$); $(5, 7, 9)$ to L ($e = h = 2$); $(13, 22, 27)$ to E (both sizes at least 12); $(5, 8, 9)$ to T ($\mu = 4 \le 11$); and $(5, 16, 17)$ to B ($a = 5 \le 11$, $\mu = 12$). By contrast $(12, 25, 36)$ is not hard-core at all: $a + b = 37 = M + 1$, so Lemma 18.4 disposes of it before the case tree is reached.

19. Cases D and L: two clean formulas

The first two branches are warm-ups with real content: each is closed by one explicit multiset whose budget check is a two-line inequality.

19.1. **Case D:** $a \mid M$**.** *Who lives here:* triples whose smallest generator divides the largest — e.g. $(5, 12, 15)$. *The trick:* when $a \mid M$, the copies of M are just extra copies of a in disguise (each worth $q = M/a$ of them), so b can be left to do the residue work alone. The three questions: *residues* — $a-1$ copies of b sweep them all; *solid run* — copies of a and M pad at "half price", since M is a multiple of a; *budget* — the padding is so cheap that $2a + q - 3 \le M - 1$ with room to spare unless $q = 2$.

Lemma 19.1 (Case D)**.** *Let (a, b, M) be a hard-core triple with $a \mid M$; write $M = qa$ ($q \ge 2$). Put $x_{\text{eff}} := \lceil (M - 1 + (a-1)b)/a \rceil$ and $z := \lceil (x_{\text{eff}} - (q-1))/q \rceil$. Then the multiset of*

$$a - 1 \text{ copies of } b, \qquad q - 1 \text{ copies of } a, \qquad z \text{ copies of } M$$

has size $\le M - 1$ and its subset sums contain M consecutive integers.

Proof. Coverage. The copies of b give the representatives $0, b, 2b, \ldots, (a-1)b$ — all residues modulo a, coprimality again — with top height $S = (a-1)b$. For the sliding phase we need enough "effective copies of a": the pads $i\,a + \kappa\,M = a\,(i + q\kappa)$ with $i \le q-1$, $\kappa \le z$ realize au for every $u \in [\,0,\ (q-1) + qz\,]$ — the blocks $[q\kappa,\ q\kappa + q - 1]$ abut as κ steps — and $(q-1) + qz \ge x_{\text{eff}}$ by the choice of z. Now run the frame argument (Lemma 17.2's proof verbatim, with the effective copies in place of literal ones): every $n \in [S,\ S + M - 1]$ is (residue representative) $+\ au$ for some $u \le \lceil (M - 1 + S)/a \rceil = x_{\text{eff}}$.

Budget. (First pass: skippable.) Since $b \le M - 1$,

$$\frac{M - 1 + (a-1)b}{a} \;\le\; \frac{M - 1 + (a-1)(M-1)}{a} \;=\; M - 1,$$

so $x_{\text{eff}} \le M - 1 = qa - 1$, whence

$$z \;\le\; \left\lceil \frac{qa - 1 - (q-1)}{q} \right\rceil \;=\; \left\lceil \frac{q(a-1)}{q} \right\rceil \;=\; a - 1.$$

The total is then

$$(a-1) + (q-1) + z \;\le\; 2a + q - 3 \;\le\; qa - 1 \;=\; M - 1,$$

the last step being exactly $(a-1)(q-2) \geq 0$: true for all $q \geq 2$, $a \geq 3$, with equality only at $q = 2$ — the tightest members of the branch. $\square$

Example 19.2. $(a, b, M) = (5, 12, 15)$: $q = 3$, $S = 4 \cdot 12 = 48$, $x_{\text{eff}} = \lceil 62/5 \rceil = 13$, $z = \lceil 11/3 \rceil = 4$. Multiset: four 12s, two 5s, four 15s — size $10 \leq 14 = M - 1$. The four 12s hit residues $0, 2, 4, 1, 3$ modulo 5; the pads realize $5u$ for $u \in [0, 14] \ni 13$; and the sums cover $[48,\ 62] \supseteq$ fifteen consecutive integers.

19.2. **Case L: equal spacings,** $e = h$**.** *Who lives here:* triples in arithmetic progression — e.g. $(5, 7, 9)$. *The trick:* regular spacing makes the offsets $je + k\mu$ into consecutive multiples of g, so the staircase's frames are as long as they can be. The three questions: *residues* — handled inside the staircase, whose levels are offset by a; *solid run* — the frame-merge condition holds, with equality at the extremes; *budget* — exactly $a + 2g - 1 = M - 1$, spent to the last unit.

When $e = h$ the triple is an arithmetic progression $\{a,\ a+g,\ a+2g\}$ with $g = e = h$ (so $g \leq a - 2$ and $\gcd(a, g) = 1$), and the staircase machinery of §17 closes it directly.

Lemma 19.3 (Case L)**.** *For* $G = \{a,\ a+g,\ a+2g\}$ *with* $\gcd(a, g) = 1$ *and* $1 \leq g \leq a - 2$, *the multiset*

$$(x, y, z) \;=\; \left(\left\lfloor \tfrac{a-2}{2} \right\rfloor + 2g, \quad 1, \quad \left\lceil \tfrac{a-2}{2} \right\rceil \right)$$

(copies of a*,* b*,* M *respectively) has budget exactly* $a + 2g - 1 = M - 1$ *and covers* M *consecutive integers.*

Proof. Here $e = g$, $\mu = 2g$, so $(e', \mu') = (1, 2)$, $C' = 0$, and with $y = 1$: $V' = 1 + 2z$, $c = z + 1$. The budget is $x + 1 + z = (a-2) + 2g + 1 = M - 1$ in all parity cases. Three sub-cases, one per applicable staircase clause.

a even. Then $z = \frac{a-2}{2}$ and $x - c = 2g - 1 \geq g$, so $x \geq c + g$; the frame-merge condition $V' - C' = 2z + 1 = a - 1 \geq a - 1$ holds *with equality.* (This branch includes $g = 1$ — the triples $(a, a+1, a+2)$ with a even, smallest $(4, 5, 6)$ — which is why Lemma 17.3(d) is stated for all $g \geq 1$.) The extended form (Lemma 17.3(d)) covers a solid interval of length $(x - c - 2g + 2)a + g(2z + 1) + 1 = a + g(a-1) + 1 \geq a + 2g = M$ (i.e. $g(a-3) \geq -1$: always).

a odd, $g \geq 2$. Then $z = \frac{a-1}{2}$, $x - c = 2g - 2 \geq g$, and $V' - C' = a \geq a - 1$ with room to spare. The extended form covers length $ga + 1 \geq a + 2g = M$ (i.e. $(g-1)a \geq 2g - 1$: true for $g \geq 2$, $a \geq 3$).

a odd, $g = 1$. Then $x = c$ exactly, and the short merge (Lemma 17.3(c)) applies: level c covers $ca + [0, a]$ (as $V' = 2z + 1 = a$), level $c + 1$ covers $(c+1)a + [1, a]$, and the merge condition $V' \geq a + \max(C', 1) - 1 = a$ holds with equality; the run $[\,ca,\ (c+1)a + a\,]$ has length $2a + 1 \geq a + 2 = M$. $\square$

Note the pattern, which will repeat: the constructions are not delicate to *find* — they are the obvious staircases — but their budgets close against $M - 1$ with equality at the extremes. The margin throughout Part IV is zero at the corners; that is what "sharp" means concretely.

Example 19.4 (Case L on $(5, 7, 9)$)**.** Here $g = 2$ and $a = 5$ is odd, so the construction takes $(x, y, z) = (5, 1, 2)$ — the multiset $\{5, 5, 5, 5, 5,\ 7,\ 9, 9\}$, of size $8 = M - 1$. Levels $c = 3$ through $c + g = 5$ each contribute a full frame, and their union is solid. The contrast with Remark 17.4 is instructive: the failing multiset there, $\{5, 5, 5, 7\}$, has too few frames for them to chain, and Case L's extra copies are exactly what repairs it.

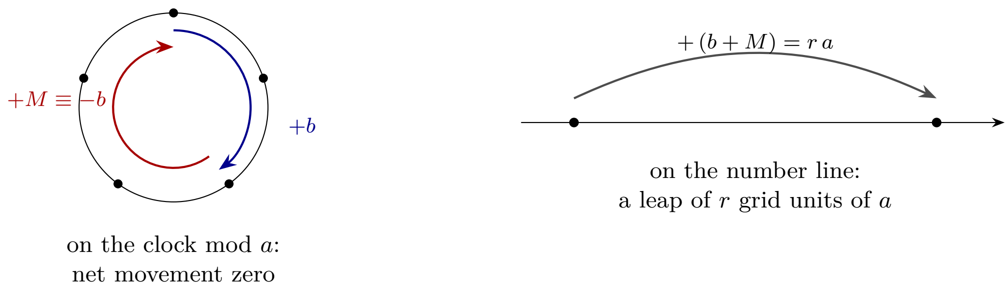


Figure 29. The mover of Case P. One b and one M together go nowhere on the residue clock ($b + M \equiv 0 \bmod a$) while leaping ra up the number line: a pure translation along the walk's own residue class, at a cost of two elements.

20. Case P: pairs as movers

Who lives here: triples whose two big generators sum to a multiple of the small one — e.g. $(3, 4, 5)$, the smallest hard-core triple of all. *The trick:* modulo a the generators b and M cancel, so a (b, M) pair is a pure long-distance *mover*. The three questions: *residues* — unpaired copies of b (or of M) set the class, at most $\lfloor a/2 \rfloor$ of either; *solid run* — movers and copies of a slide within the class; *budget* — exactly $M - 1$, verified through four corner inequalities.

Now the line $a \mid b+M$: here $b+M = ra$ for an integer r, and $r \geq 3$ — because $a < b < M$ gives $b + M > 2a$. This branch has the prettiest mechanism in Part IV. Modulo a, the two big generators are *negatives of each other*: $M \equiv -b$. So one copy of b plus one copy of M is residue-neutral — a *mover*: it costs 2 elements and advances the total by exactly $b+M = ra$, i.e. r grid units of a, without touching the residue. Unpaired copies of b (or of M) set the residue; movers and copies of a then slide the frame (Figure 29).

Lemma 20.1 (Case P)**.** *Let (a, b, M) be a hard-core triple with $b + M = ra$, $r \geq 3$. Then the multiset*

$$(x,\ y,\ z)\ =\ \Big(r-1,\ \ \Big\lceil \tfrac{M-r}{2} \Big\rceil,\ \ \Big\lfloor \tfrac{M-r}{2} \Big\rfloor\Big)$$

(copies of a, b, M), of budget exactly $(r-1)+(M-r) = M-1$, has subset sums containing M consecutive integers.

Proof. Coordinates. A selection of i copies of a, j of b, κ of M is worth

$$i\,a + j\,b + \kappa\,M \ =\ i\,a + w\,b + \kappa\,r\,a, \qquad w := j - \kappa$$

(substituting $M = ra-b$): κ full movers, plus w unpaired copies of b if $w > 0$ (or $|w|$ unpaired copies of M if $w < 0$, by the symmetric substitution). Set

$$n_b := \Big\lfloor \frac{a}{2} \Big\rfloor, \qquad n_M := \Big\lfloor \frac{a-1}{2} \Big\rfloor \qquad (\text{so } n_b + n_M = a - 1)$$

— the most unpaired copies of b, respectively of M, we will ever spend (we avoid the letters p, q, busy elsewhere in this paper) — and declare the run to start at $\mathrm{low} := \max(n_b\,b,\ n_M\,M)$. We must realize every $n \in [\,\mathrm{low},\ \mathrm{low} + M - 1\,]$.

Three preliminary facts.

(P1) $M \geq a + r - 1$. From $b < M$: $ra = b + M < 2M$, so $2M \geq ra + 1$; and $a, r \geq 3$ give $a(r-2) \geq 3(r-2) \geq 2r - 3$, whence $2M \geq ra + 1 = a(r-2) + 2a + 1 \geq 2a + 2r - 2$.

(P2) $M \leq a(r-1) - 1$. From $b > a$: $M = ra - b < a(r-1)$.

(P3) $y \geq n_b$ and $z \geq n_M$. By (P1), $M - r \geq a - 1$, so $z = \lfloor (M-r)/2 \rfloor \geq \lfloor (a-1)/2 \rfloor = n_M$ and $y = \lceil (M-r)/2 \rceil \geq \lceil (a-1)/2 \rceil = n_b$.

The two branches. Fix a target n. Since $\gcd(a,b) = 1$ there is a unique $w \in [0,\ a-1]$ with $w\,b \equiv n \pmod{a}$ (the covering lemma, Lemma 7.6).

Branch 1: $w \leq n_b$ — spend w unpaired bs. Then $wb \leq n_b\, b \leq \mathrm{low} \leq n$, so $A := (n - wb)/a$ is a nonnegative integer. Split $A = i + r\kappa$ with $i := A \bmod r \leq r - 1 = x$ and $\kappa := \lfloor A/r \rfloor$: take i copies of a, κ movers, and the w unpaired bs — that is, i copies of a, $w + \kappa$ copies of b, κ copies of M, worth exactly n (check: $ia + (w+\kappa)b + \kappa M = ia + wb + \kappa\, ra = aA + wb = n$). Availability requires

$$\kappa \leq z \qquad \text{and} \qquad w + \kappa \leq y.$$

For the first, since $aA \leq n \leq \mathrm{low} + M - 1$, it suffices that $\mathrm{low} + M - 1 \leq a(rz + x)$ (then $A \leq rz + x$, so $\kappa \leq z$). For the second, note $a(A + rw) = (n - wb) + w(b + M) = n + wM \leq (\mathrm{low} + M - 1) + n_b M$, so it suffices that

$$\mathrm{low} + M - 1 + n_b\, M \ \leq\ a\,(r\,y + x). \tag{C$_y$}$$

Branch 2: $w > n_b$ — spend $v := a - w \in [1,\ n_M]$ unpaired Ms. Since $M \equiv -b$, $vM \equiv -vb \equiv wb \equiv n \pmod{a}$, and $vM \leq n_M M \leq \mathrm{low} \leq n$: so $A' := (n - vM)/a$ is a nonnegative integer, split as $A' = i' + r\kappa'$, giving i' copies of a, κ' copies of b, $v + \kappa'$ copies of M. Availability: $\kappa' \leq y$, which follows from $\mathrm{low} + M - 1 \leq a(ry + x)$ (weaker than (C$_y$)); and $v + \kappa' \leq z$, for which — from $a(A' + rv) = n + vb \leq (\mathrm{low} + M - 1) + n_M b$ — it suffices that

$$\mathrm{low} + M - 1 + n_M\, b \ \leq\ a\,(r\,z + x). \tag{C$_z$}$$

(C$_z$ also implies the interim condition of Branch 1.)

Four extremes. (First pass: skippable — the mechanism is complete; what follows audits the two conditions at their four corner cases.) Each of (C$_y$), (C$_z$) splits in two according to which term achieves $\mathrm{low} = \max(n_b b,\ n_M M)$; using $n_b + n_M = a - 1$ and $b + M = ra$, the four resulting inequalities are

(E1)	$(a-1)\,b + M - 1 \ \leq\ a\,(rz + x)$	$[\mathrm{C}_z,\ \mathrm{low} = n_b b]$
(E2)	$n_M\, ra + M - 1 \ \leq\ a\,(rz + x)$	$[\mathrm{C}_z,\ \mathrm{low} = n_M M]$
(F1)	$n_b\, ra + M - 1 \ \leq\ a\,(ry + x)$	$[\mathrm{C}_y,\ \mathrm{low} = n_b b]$
(F2)	$a\,M - 1 \ \leq\ a\,(ry + x)$	$[\mathrm{C}_y,\ \mathrm{low} = n_M M]$

(for (E1): $n_b b + n_M b = (a-1)b$; for (F1): $n_b b + n_b M = n_b\, ra$; for (F2): $n_M M + n_b M + M = aM$). Verifying these four suffices for every w and every n, because w entered only through $w \leq n_b$ and $v \leq n_M$ — the extremes used.

(F2). It suffices that $M \leq ry + r - 1$. From $y \geq (M - r)/2$ and $r \geq 3$: $ry + r - 1 \geq \frac{3}{2}(M - r) + r - 1 = \frac{1}{2}(3M - r - 2) \geq M \iff M \geq r + 2$, and (P1) with $a \geq 3$ gives $M \geq a + r - 1 \geq r + 2$.

(F1). $n_b\, ra \leq a\, r\, y$ by (P3), so (F1) reduces to $M - 1 \leq a\, x = a(r-1)$, which follows from (P2).

(E2). Identical: $n_M\, ra \leq a\, r\, z$ by (P3), and the remainder is (P2) again.

(E1) — the binding one. Substitute $b = ra - M$ and use $z \geq (M - r - 1)/2$; it suffices that

$$2\big[(a-1)(ra - M) + M - 1\big] \ \leq\ a\, r\,(M - r - 1) + 2a(r-1).$$

Expand both sides. On the left,

$$2\big[(a-1)(ra - M) + M - 1\big] \ =\ 2ra^2 - 2ra - 2aM + 4M - 2;$$

on the right,

$$ar(M-r-1)+2a(r-1) \;=\; arM-ar^2-ar+2ar-2a \;=\; arM-ar^2+ar-2a.$$

Moving everything to the right and collecting the M terms, the inequality rearranges to

$$0 \;\le\; M\,(ar+2a-4) \;-\; ar^2 \;-\; 2ra^2 \;+\; 3ar \;-\; 2a \;+\; 2.$$

The coefficient of M is positive, so we may insert (P1)'s bound $2M \ge ra+1$; after multiplying through by 2, the right side is at least

$$(ra+1)(ar+2a-4)-2ar^2-4ra^2+6ar-4a+4 \;=\; a\big(ar^2-2ar-2r^2+3r-2\big)$$

(the intermediate expansion is item (A5) of Appendix C), so (E1) holds whenever

$$a\,r\,(r-2) \;\ge\; 2r^2-3r+2. \qquad (*)$$

For $r \ge 4$: $a \ge 3$ gives $ar(r-2) \ge 3r^2-6r$, and $3r^2-6r \ge 2r^2-3r+2 \iff r^2-3r-2 \ge 0$, true from $r=4$ on. For $r=3$: $(*)$ reads $3a \ge 11$, i.e. $a \ge 4$.

So $(*)$ can fail only at $(a,r)=(3,3)$ — where $b+M=9$ with $3<b<M$ forces $(a,b,M)=(3,4,5)$, and we check that one triple directly: $(x,y,z)=(2,1,1)$, and (E1) reads $2\cdot 4+5-1=12 \le 3(3+2)=15$. True. (Concretely the multiset is $\{3,3,4,5\}$ of Example 15.2, whose sums cover the run $[3,12]$ — our very first example was the tightest triple on the P-line.) □

Case P needs no certificate table and no computer: one closed-form multiset, four inequalities, one directly-checked triple. (The Lean formalization proves this case by a slightly different route — instantiating small r separately — a difference of route, not of content.)

Example 20.2 (Case P on $(3,4,5)$, by hand)**.** Here $r=3$, $(x,y,z)=(2,1,1)$, $n_b=n_M=1$ and low $=\max(1\cdot 4,\ 1\cdot 5)=5$, so the targets are $5,\dots,9$. Two are worth tracing. For $n=8$: $8 \equiv 2 \pmod 3$ gives $w=2>n_b$, so Branch 2 spends $v=1$ unpaired copy of M, and $8=5+3$. For $n=9$ the mover itself appears: $9=4+5$, one copy of b paired with one of M, residue-neutral and worth $ra=9$.

21. Case E: the η-box

Who lives here: everything the divisibility branches missed, in the roomy regime where both a and μ are at least 12 — e.g. $(13,22,27)$. *The trick:* one invariant, η, measures how many steps of b it takes to manufacture one step of M modulo a; a box of that shape covers the residues, and roominess pays for it. The three questions: *residues* — the η-box $(t-1,Z)$, whose offsets sweep an interval of length $\ge a$; *solid run* — the frame lemma, as always; *budget* — one inequality, $M(a-\sigma) \ge a\sigma$, which both 12s exist to guarantee.

The remaining triples have $\eta \notin \{0,\ 1,\ a-1\}$ — neither divisibility holds — and split by size. This section handles the roomy regime: both a and $\mu = M-a$ at least 12. Here a single box works, built directly from the invariant η, and the whole case is one budget computation. The case operates entirely modulo a: recall $b \equiv \bar{e}$ and $M \equiv \bar{\mu} = \eta\,\bar{e} \pmod a$.

Lemma 21.1 (Case E)**.** *Let (a,b,M) be a hard-core triple with $a \nmid M$, $a \nmid b+M$, $a \ge 12$ and $\mu \ge 12$. Put*

$$t := \min(\eta,\ a-\eta) \;\in\; [2,\ a/2], \qquad Z := \left\lceil \frac{a-t}{t} \right\rceil, \qquad \sigma := t+Z.$$

Then the box $(Y,Z)=(t-1,\ Z)$ covers all residues modulo a, and its budget is $\le M-1$.

Proof. First, the asserted range $t \in [2,\ a/2]$: $t \le a/2$ is the definition of a minimum of η and $a-\eta$; and $t \ge 2$ because the earlier branches removed $\eta \in \{0,\ a-1\}$ (cases D and P) while $\eta = 1$ is impossible on the hard core (§18.5) — so $2 \le \eta \le a-2$, and both η and $a-\eta$ are at least 2.

Coverage. Work modulo a. If $t = \eta$: a box pair (j,k) contributes $jb + kM \equiv \bar{e}\,(j+\eta k) = \bar{e}\,(j+tk)$, and over $j \le t-1$, $k \le Z$ the quantity $j+tk$ takes every value in $[\,0,\ t(Z+1)-1\,]$ — the t-blocks abut — which has length $t(Z+1) \ge a$ by the choice of Z. If $t = a-\eta$: then $M \equiv -t\,\bar{e}$, the contribution is $\bar{e}\,(j-tk)$, and $j-tk$ covers the interval $[-tZ,\ t-1]$, again of length $\ge a$. Either way $j+tk$ (or $j-tk$) sweeps a full residue system, and multiplying by the *unit* $\bar{e}$ permutes residues: all of $\mathbb{Z}_a$ is covered.

Budget. (First pass: skippable.) Every representative height is at most $S_0 := (t-1)\,b + Z\,M$, and the box size is $K := t-1+Z = \sigma-1$. It suffices to show

$$M - 1 + S_0 \ \le\ a\,(M-1-K),$$

for then $\lceil (M-1+S)/a \rceil \le M-1-K$ and the budget is $\le M-1$. Using $b \le M-1$: $M-1+S_0 \le t(M-1) + ZM \le \sigma M - t$, while the right side is $a(M-\sigma)$; so it suffices that $\sigma M + a\sigma \le aM + t$, and — discarding $t \ge 0$ — that

$$M\,(a-\sigma) \ \ge\ a\,\sigma.$$

Now bound σ: $Z \le (a-1)/t$ gives $\sigma \le t + (a-1)/t =: f(t)$, and f peaks at an endpoint of $[2,\ a/2]$ — no calculus needed: for $s < t$,

$$f(t) - f(s) \ =\ (t-s)\Big(1 - \frac{a-1}{t\,s}\Big),$$

so f decreases as long as $ts < a-1$ and increases once $ts > a-1$; on any interval its maximum is therefore at one of the two ends. Evaluating, $\sigma \le \max\big(f(2),\ f(a/2)\big) \le \max\big(\frac{a+3}{2},\ \frac{a}{2}+2\big) = \frac{a}{2}+2$. (In particular $\sigma < a$, so the quantity $a-\sigma$ we are about to divide by is positive.) Hence

$$\frac{a\,\sigma}{a-\sigma} \ \le\ \frac{a\,(a+4)}{a-4} \ =\ a + 8 + \frac{32}{a-4} \ \le\ a+12 \qquad (a \ge 12),$$

and $M = a + \mu \ge a+12$ delivers exactly this. Both hypotheses $a \ge 12$ and $\mu \ge 12$ are spent in this last line. ☐

Example 21.2. $(a,b,M) = (13,22,27)$: $e = 9$, $h = 5$, $\mu = 14$, $\bar{e} = 9$, $\bar{\mu} = 1$, and $\eta = 9^{-1} \bmod 13 = 3$ (as $9 \cdot 3 = 27 \equiv 1$). So $t = 3$, $Z = \lceil 10/3 \rceil = 4$: box $(2,4)$ — drawn in Figure 30. The offsets $\bar{e}(j+3k)$ with $j \le 2$, $k \le 4$ have $j+3k$ sweeping $[0,14] \supseteq [0,12]$. Corner height $S_0 = 2 \cdot 22 + 4 \cdot 27 = 152$; budget $\le \lceil (26+152)/13 \rceil + 6 = 14 + 6 = 20 \le 26 = M-1$, with plenty to spare — Case E is the *comfortable* regime, which is exactly why it tolerates a one-size-fits-all box.

Remark 21.3 (Why $t = \min(\eta,\ a-\eta)$). Working from η directly would be legitimate but wasteful. On $(13,24,34)$, which routes to Case E with $\eta = 9$, the box built from η is $(8,1)$: top height $8 \cdot 24 + 34 = 226$ and budget $\lceil (33+226)/13 \rceil + 9 = 29$. The box built from $t = \min(9,4) = 4$ is $(3,3)$: top height $3 \cdot 24 + 3 \cdot 34 = 174$ and budget $\lceil (33+174)/13 \rceil + 6 = 22$. Both clear $M-1 = 33$ here, but only the minimum keeps $\sigma \le a/2+2$, which the threshold argument requires.

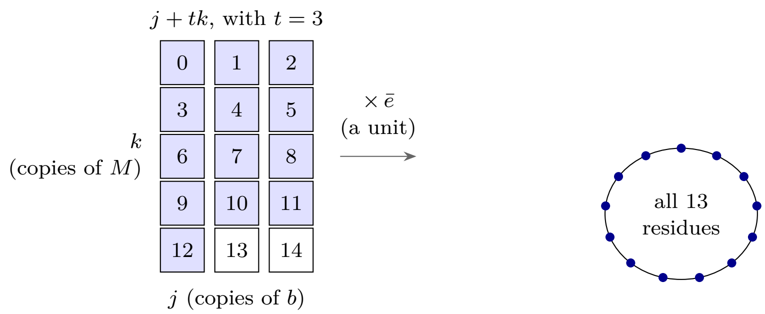


FIGURE 30. The η-box for $(a, b, M) = (13, 22, 27)$, where $\eta = 3$. The box of $Y = t - 1 = 2$ copies of b and $Z = 4$ copies of M realizes the quantities $j + tk$ (left), which sweep the whole run $0, 1, \ldots, 14$ — shaded are the first 13, already a complete set of residues modulo 13. Multiplying by the unit $\bar{e}$ permutes the residue clock without collapsing it, so every class is covered (right). The t-blocks abutting as k steps is the whole mechanism: choosing $t = \min(\eta, a - \eta)$ is what keeps the box small enough to afford.

22. CASE T: THIN μ, AND THE FIRST TABLE

Who lives here: triples whose generators crowd together at the top ($\mu \leq 11$) while a runs free — e.g. $(5, 8, 9)$. *The trick:* finitely many shapes (e, h), each a *line* parameterized by a alone, on which one staircase formula works for every a — and only the budget, growing more slowly than the target, can ever fail. The three questions: *residues* and *solid run* — the staircase, unconditionally, for every a; *budget* — fails for exactly 172 triples, each of which carries a replacement certificate.

Now $\mu = e + h \leq 11$ (with $e \neq h$, neither divisibility, and a unbounded). Since $e \leq 10$ and $h \leq 10$, there are finitely many pairs (e, h); each pair defines a *line* of triples $(a,\ a{+}e,\ a{+}e{+}h)$ parameterized by a alone. On each line we run one explicit staircase; its *coverage* works for every a, and only its *budget* can fail — and only finitely often, which is where the certificates of Table A come in. Note $\mu' \geq 3$ throughout this case: if $\mu' = 2$ then $e' = 1$, giving $e = g$ and $\mu = 2g$, i.e. $h = \mu - e = g = e$ — but $e \neq h$ here.

Construction T.. The multiset will be a staircase, and its z will be a maximum of two branches with two distinct jobs: a *constant branch* maintaining the staircase's standing hypothesis $z \geq e' - 1$, and a *ceiling branch* buying enough run length. With that read ahead of time, take $y := \mu' - 1$ and:

- *Variant A* ($g = 1$; here $e' = e$, $\mu' = \mu$, and write $C = C'$, $V = V'$):

$$z := \max\Big(e - 1,\ \Big\lceil \frac{\max(a, \mu) - 1 + 2C - ye}{\mu} \Big\rceil\Big), \qquad x := c + 1 \quad (c = y + z).$$

- *Variant* $g \geq 2$:

$$z := \max\Big(e' - 1,\ \Big\lceil \frac{a + \lceil (\mu - 1)/g \rceil + 2C' - ye'}{\mu'} \Big\rceil\Big), \qquad x := c + g.$$

Exactly one variant applies to each line, according to its g.

Lemma 22.1 (Case T, coverage)**.** *On every Case-T line, Construction T's subset sums contain M consecutive integers. Its budget is $2c + 1$ (variant A) or $2c + g$ (variant $g \geq 2$).*

Proof. Variant A. The ceiling branch gives $z\mu \geq \max(a,\mu) - 1 + 2C - ye$, so

$$V - C \;=\; ye + z\mu - 2C \;\geq\; \max(a,\mu) - 1 \;\geq\; a - 1:$$

the two-frame merge condition. With $x = c+1$, Lemma 17.3(b) covers $[\,ca+C,\;(c+1)a+V\,]$, of length $a + (V - C) + 1 \geq a + \max(a,\mu) \geq a + \mu = M$. Budget $x + y + z = 2c + 1$.

Variant $g \geq 2$. (First pass: skippable — the same argument as variant A, with the base form of the staircase in place of the two-frame merge.) The ceiling branch gives $V' - C' \geq a + \lceil(\mu - 1)/g\rceil \geq a + (\mu - 1)/g$. With $x = c + g$, the *base form* of Lemma 17.3(d) — no merge hypothesis needed — covers a solid interval of length

$$g\,(V' - C') - (g-1)a + 1 \;\geq\; g\,a + (\mu - 1) - (g-1)a + 1 \;=\; a + \mu \;=\; M.$$

Budget $x + y + z = 2c + g$. □

So coverage is unconditional; everything now rides on the inequality

$$\text{budget} \;\leq\; M - 1 = a + \mu - 1.$$

The budget grows like $2z \approx 2a/\mu'$, i.e. with slope $2/\mu' \leq 2/3$ in a, while the target grows with slope 1 — so far out on every line the margin is positive and widening. Failures can only be *early* on a line.

Example 22.2 (The line $(e,h) = (2,1)$)**.** Here $\mu = 3$, $g = 1$, $(e',\mu') = (2,3)$, $C = 2$: the triples $(a,\; a+2,\; a+3)$. Variant A takes $y = 2$ and $z = \max(1,\;\lceil(a-1)/3\rceil)$, budget $2c+1$ against target $a+2$:

a	z	c	budget $2c{+}1$	target $a{+}2$	margin
7	2	4	9	9	0 (tight)
13	4	6	13	15	2
19	6	8	17	21	4
25	8	10	21	27	6

The margin grows at rate $\approx 1/3$; on this lucky line it never goes negative. Other lines are less lucky: on $(e,h) = (5,1)$, the triples $(4,9,10)$, $(7,12,13)$ and $(8,13,14)$ (among others) have construction budgets *exceeding* $M - 1$, and each needs a replacement certificate.

Proposition 22.3 (T-tail)**.** *Across all Case-T lines, the applicable variant's budget exceeds $M-1$ for precisely* 172 *triples, all with $a \leq 29$; each of these carries an explicit box certificate — a triple (x,Y,Z) with $Y+Z \leq a-1$ and budget $\leq M-1$, verified by the two-line computation of Lemma 17.2 — listed as* Table A *in Appendix E of this paper repository. For $a > 3000$ no failures occur; the range $a \leq 3000$ was checked exhaustively by computer.*

Proof, with the computation's role made explicit. Two separate claims.

The finite range $a \leq 3000$. For each of the finitely many lines and each $a \leq 3000$, the applicable variant's budget is a closed formula; a computer evaluated all of them exactly (integer arithmetic, no floating point) and found the budget to exceed $M - 1$ for exactly the 172 triples recorded in Table A, the largest at $a = 29$. This is a finite computation, redone independently three ways (§26); each failing triple's replacement certificate is itself re-verified mechanically. Nothing here asks you to trust a search heuristic: the claims are "this explicit formula, at these explicit inputs, exceeds that explicit bound" and "this explicit multiset covers that explicit run", both checkable by hand for any single row — 172 rows is merely more than a human wants to do.

The tail $a > 3000$. (First pass: skippable.) Here no computer is needed. On a fixed line, once $a \geq 3000$ the ceiling branch of z dominates its constant branch (the ceiling's argument exceeds $(3000-1-100)/11 > 260$, while the constant branch is at most 9), so the budget equals its ceiling-branch value, which is bounded above by a linear function $\beta(a)$ of slope $2/\mu' \le 2/3$; the target $\tau(a) = a + \mu - 1$ has slope 1. Therefore the margin $\tau - \beta$ is nondecreasing in a (rate $\ge 1/3$), and it suffices that it is positive for every Case-T line at $a = 3000$. That is a finite check, and it can be made uniform rather than line-by-line. Separate β into its a-dependent part and a constant:

$$\beta(a) \;=\; \frac{2a}{\mu'} \;+\; K, \qquad K \;:=\; 2\Big[(\mu'-1) + \frac{\mu + 2C' - (\mu'-1)e'}{\mu'} + 1\Big] + g.$$

A Case-T line is a pair (e,h) with $e \ne h$, $e,h \ge 1$ and $e+h \le 11$, and (as noted above) $\mu' \ge 3$ on every one of them; there are exactly 50 such lines. On each, $g \le \mu \le 11$, $e' \le e \le 10$, $\mu' \le \mu \le 11$ and $C' = (e'-1)(\mu'-1) \le 9 \cdot 10 = 90$, so every term of K is bounded independently of a. Evaluating K over all 50 lines gives $\max K = 435/11 = 39.5\overline{45}$, attained on the single line $(e,h) = (10,1)$ (there $\mu = \mu' = 11$, $e' = 10$, $g = 1$, $C' = 90$); the runners-up are $(9,2)$ at $415/11$ and $(8,3)$ at $395/11$, both below 38. In particular $K \le 40$, with 0.45 to spare on the worst line. Hence for every line and every $a \ge 3000$,

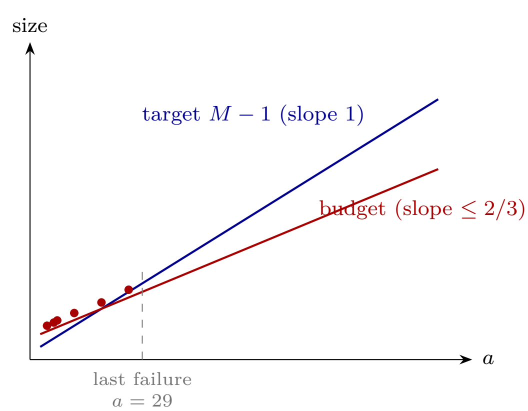
$$\tau(a) - \beta(a) \;=\; a + \mu - 1 - \frac{2a}{\mu'} - K \;\ge\; a - \frac{2a}{3} - K \;=\; \frac{a}{3} - K \;\ge\; 1000 - 40 \;>\; 0,$$

using $\mu' \ge 3$ and $\mu \ge 1$. No cross-line search is needed, and nothing beyond this paragraph is asked of the computer for the tail.

It is worth seeing the tightest line concretely. On $(e,h) = (2,1)$ we have $\mu' = 3$, $e' = 2$, $C' = 2$, $g = 1$, so

$$\beta(3000) = 2\Big[2 + \frac{3000+3+4-4}{3} + 1\Big] + 1 = 2\,[\,3 + 1001\,] + 1 = 2009, \qquad \tau(3000) = 3002,$$

margin $993 > 0$. (Note what is being compared: the target against the *linear over-bound* β, not against the exact budget — the exact budget at $a = 3000$ on this line is 2005, margin 997. Comparing against β costs four units of margin and buys the monotonicity that finishes the proof.) Positive there, nondecreasing after: positive forever. $\square$



The shape of Case T (schematic): the construction's budget grows with slope at most 2/3, the target with slope 1. Early on the budget can poke above the target — those finitely many failures (dots) are Table A — but once the target line is ahead it stays ahead forever. The computer's entire role is confined to the left of the dashed line.

Role of the finite computation. Case T has the logical shape of every computer-assisted step in this work. The *infinite* part is the slope comparison above — a budget growing at rate $\leq 2/3$ against a target growing at rate 1 — and is checkable by hand. Computation enters only to exhaust the finite exceptional range $a \leq 3000$ and to verify the 172 certificates it produces. §26 describes that apparatus in full.

Example 22.4 (Construction T at its tightest)**.** The $a = 7$ row of Example 22.2 is the extreme case on its line. There $y = 2$, $z = 2$, $x = 5$, giving the multiset $\{7^5, 9^2, 10^2\}$ of size $9 = M - 1$ exactly. Levels 4 and 5 each cover a frame, and since $V' - C' = 6 = a - 1$ they abut precisely; the union is a run of $14 \geq M = 10$ consecutive integers. The budget meets the target with nothing to spare, and yet does not exceed it — which is why this line contributes no rows to Table A.

Example 22.5 (A Case-T failure, and its certificate)**.** The triple $(5, 8, 9)$ lies on the line $(e, h) = (3, 1)$, where Construction T's budget evaluates to 11 against a target of $M - 1 = 8$: one of the 172 failures. Its replacement certificate is $(x, Y, Z) = (5, 2, 1)$ — the multiset $\{5, 5, 5, 5, 5,\ 8, 8,\ 9\}$, of size $8 = M - 1$. The parts $8j + 9k$ with $j \leq 2$, $k \leq 1$ take the values $\{0, 8, 16, 9, 17\}$, whose residues modulo 5 are $\{0, 3, 1, 4, 2\}$: all five. Lifting by the copies of 5 gives the run $[17, 25]$, which is $9 = M$ consecutive integers.

23. Case B: finite bases, lifted forever

Who lives here: the mirror image of Case T — a tiny smallest generator ($a \leq 11$) with the others far above it ($\mu \geq 12$) — e.g. $(5, 16, 17)$. *The trick:* bounded a means only finitely many *residue patterns*, and moving up within one pattern costs less than it earns, so one certificate per pattern suffices forever. The three questions: *residues* and *solid run* — a box certificate, verified once per pattern; *budget* — climbs by at most a per lift while the target climbs by exactly a, so a margin once won is never lost.

The last branch: $a \leq 11$ with $\mu \geq 12$. Now a is bounded but b (hence e, hence M) is not — each small a carries infinitely many triples. The finiteness that saves us is *modular*: group the triples into *classes*

$$[\,a;\ \bar{e},\ h\,], \qquad \bar{e} \in [1,\ a-1], \quad h \in [1,\ a-2],$$

by the residue of e modulo a and the exact value of h. Within a class, members differ only in $\lambda := \lfloor e/a \rfloor$, and stepping λ by one is exactly the map $(a, b, M) \mapsto (a,\ b + a,\ M + a)$ of the λ-lift (Lemma 17.5). So:

> *certify the smallest Case-B member of each class by an explicit box; the λ-lift then certifies every larger member, forever.*

There are finitely many classes ($a \leq 11$, $\bar{e} < a$, $h \leq a - 2$), and exactly 178 of them have members reaching Case B. That count is machine-checked, but it is also a short hand computation, which we give — it is the one place in Part IV where an enumeration could otherwise only be taken on trust.

Lemma 23.1 (The Case-B classes, counted)**.** *Exactly* 178 *classes* $[\,a; \bar{e}, h\,]$ *reach Case B.*

Proof. A class is a triple $(a, \bar{e}, h)$ with $3 \leq a \leq 11$, $\gcd(\bar{e}, a) = 1$ and $1 \leq h \leq a - 2$; it reaches Case B exactly when $\eta \notin \{0, 1, a-1\}$, the excluded values being claimed by earlier branches. Now $\eta = 1$ never occurs (§18.5), and the other two exclusions are congruences in h: since $\eta = \bar{\mu}\,\bar{e}^{\,-1}$ and $\bar{\mu} \equiv \bar{e} + h$,

$$\eta = 0 \iff h \equiv -\bar{e} \pmod{a}, \qquad \eta = a - 1 \iff h \equiv -2\bar{e} \pmod{a}.$$

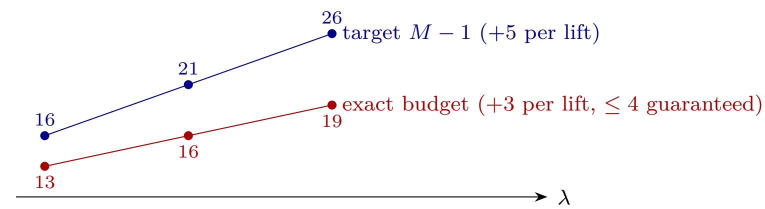
The two are distinct: $-\bar{e} \equiv -2\bar{e}$ would force $a \mid \bar{e}$, contradicting $\gcd(\bar{e}, a) = 1$. So for each admissible $\bar{e}$ at most two of the $a-2$ values of h are discarded — exactly two when both congruences fall in $[1, a-2]$, fewer otherwise. Counting survivors for each a gives

$$0, \quad 1, \quad 6, \quad 5, \quad 20, \quad 17, \quad 32, \quad 25, \quad 72 \qquad (a = 3, 4, \ldots, 11),$$

which sum to 178. At $a = 3$ the only admissible h is 1, and there $\eta \in \{0, 2\} = \{0, a-1\}$ always, so $a = 3$ contributes nothing. ☐

Their certified bases are *Table B* of Appendix E.

Example 23.2 (A class and its lift)**.** The class $[\,5;\ 1,\ 1\,]$: triples with $a = 5$, $e \equiv 1 \pmod 5$, $h = 1$. Its base in Table B is $(a, b, M) = (5, 16, 17)$ with box $(x, Y, Z) = (10, 1, 2)$: the parts $16j + 17k$ ($j \le 1$, $k \le 2$) have residues $\{0, 1, 2, 3, 4\}$ modulo 5 — check: $0, 16 \equiv 1, 17 \equiv 2, 33 \equiv 3, 34 \equiv 4, 50 \equiv 0$ — and the budget is $10 + 1 + 2 = 13 \le 16 = M - 1$. The class members are $(5, 21, 22), (5, 26, 27), (5, 31, 32), \ldots$, and Lemma 17.5 hands the *same* box up the whole ladder: the residues never change, each lift grows the budget by at most $Y + Z + 1 = 4$ while the target $M - 1$ grows by exactly 5. The margin widens forever; one finite check settles an infinite family.



The λ-ladder for the class $[\,5; 1, 1\,]$, with exact budgets computed at each rung: the target climbs by $a = 5$ per lift, the budget by at most $Y + Z + 1 = 4$ (here in fact 3). Lines that never re-cross: one finite check, an infinite family certified.

Where the tables live — and what a row looks like. A document that leans on $172 + 178$ certificates owes the reader the certificates. They are in the same repository as this paper, in two machine-readable files (`paper/data/table-A.csv`, `paper/data/table-B.csv`) and typeset in full in Appendix E; every row is verifiable by the two-line frame-lemma computation, and §26 describes the three independent systems that have verified all of them. To make the format concrete, here are the first rows of each, verbatim:

Table A row $(a, b, M;\ x, Y, Z)$	budget	vs. $M-1$
$(4,\ 9,\ 10;\ 7, 1, 1)$	9	9 (tight)
$(5,\ 8,\ 9;\ 5, 2, 1)$	8	8 (tight; Example 22.5)
$(7,\ 12,\ 13;\ 7, 3, 1)$	11	12
$(8,\ 13,\ 14;\ 7, 3, 2)$	12	13
Table B class $[a; \bar{e}, h] \to$ base $(a, b, M;\ x, Y, Z)$	**budget**	**vs. $M-1$**
$[4; 1, 1] \to (4,\ 17,\ 18;\ 13, 1, 1)$	15	17
$[5; 1, 1] \to (5,\ 16,\ 17;\ 10, 1, 2)$	13	16 (Example 23.2)
$[5; 1, 2] \to (5,\ 16,\ 18;\ 11, 2, 1)$	14	17

Two administrative points, each with a one-line proof idea.

Class membership of cases D and P is λ-invariant. Both conditions ($a \mid M$; $a \mid b + M$) depend only on residues modulo a, which the lift preserves. So a class is "a D-class", "a P-class", or neither, wholesale — no class straddles branches as λ climbs, except for one wrinkle:

The $e = h$ wrinkle. The condition $e = h$ (Case L) is *not* λ-invariant — e grows with λ while h stays fixed — but it can only hold at the class's smallest member ($e \geq a > h$ once $\lambda \geq 1$). Exactly six classes (all on the diagonal $\bar{e} = h$, with $a \in \{9, 10, 11\}$) have a $\lambda = 0$ member that satisfies $\mu \geq 12$ but is diverted to Case L; those classes are simply *re-based* at their first genuine Case-B member, from which the lift proceeds as usual.

Why most bases sit at $\lambda \geq 1$. A class's $\lambda = 0$ member often has $\mu = e + h < 12$ (small e), landing in Case T instead; the class enters Case B only once e has grown past $12 - h$. Example 23.2's base has $\lambda = 2$ for exactly this reason. The enumeration accounts for every class and every member — completeness of the routing is part of what the harnesses check.

Example 23.3 (A Case-B base, and one lift)**.** The class $[\,5; 1, 1\,]$ has base $(5, 16, 17)$ with box $(x, Y, Z) = (10, 1, 2)$. The parts $16j + 17k$ ($j \leq 1$, $k \leq 2$) are $0, 16, 17, 33, 34, 50$, with residues $0, 1, 2, 3, 4, 0$ modulo 5 — a complete system — and the ten copies of 5 lift this to a run of $17 = M$ consecutive integers, at budget $13 \leq 16 = M - 1$. One lift gives $(5, 21, 22)$, where the same box has top height 44 and exact budget $13 + 3 = 16$ against a target of 21: the lemma guarantees only $13 + 4 = 17$, so the margin in practice widens faster than the bound promises.

24. The proof of SHARP, and its sharpness

Proof of Theorem 16.1. The recursion is well-founded on the pair $(M, |G|)$ ordered lexicographically, and it is worth naming that explicitly rather than saying "induction on M", because one of the descents does *not* lower M.

Descents that lower M. Lemma 18.8 passes to the rescaled alphabet H of maximum $M/d < M$; and Lemma 18.2, in its case $e^\dagger = M$, invokes the statement for $G \setminus \{M\}$, of maximum $M' < M$.

The descent that does not. Lemma 18.2, in its case $e^\dagger \neq M$, passes to $G \setminus \{e^\dagger\}$ — the same maximum M, one element fewer. This is why the lemma is stated graded by M and proved by an inner induction on $|G|$: at maximum M the pair drops in its second coordinate.

Every appeal therefore strictly decreases $(M, |G|)$. No hard-core branch invokes the induction hypothesis at all, and no branch re-enters either lemma at a larger maximum, so the recursion terminates. Fix M and assume SHARP for every alphabet with smaller maximum.

If $1 \in G$: done (§16). By Lemma 18.2, reduce to irreducible G; by Lemma 18.8, dispose of minimal $|G| \geq 4$; by Lemma 18.6, assume the triple $\{a, b, M\}$ has $\gcd(a, b) = 1$; by Lemmas 18.3–18.4, assume $a + b \geq M + 2$: the hard core. Route:

if $a \mid M$: D $\to$ else if $a \mid b{+}M$: P $\to$ else if $e = h$: L $\to$ else if $a, \mu \geq 12$: E $\to$ else if $\mu \leq 11$: T $\to$ else B.

Each branch delivers a multiset of size $\leq M{-}1$ covering M consecutive integers: Lemmas 19.1, 20.1, 19.3, 21.1, 22.1 with Proposition 22.3 (Table A), and Table B with Lemma 17.5.

Exhaustiveness and consistency of the routing. D and P are disjoint on the hard core (both would force $a \mid b$). A triple with $e = h$ *and* $a \mid b + M$ would have $b + M = 2a + 3e$, so $a \mid 3e$; with $\gcd(a, e) = 1$ this forces $a \mid 3$, i.e. $a = 3$, $e = h = 1$: the single triple $(3, 4, 5)$, which the ordered routing sends to P (and which L would also handle — no conflict, the router just picks first). Every triple surviving to the last three branches has $\eta \notin \{0, 1, a - 1\}$, and the conditions "$a \geq 12 \wedge \mu \geq 12$", "$\mu \leq 11$", "$a \leq 11 \wedge \mu \geq 12$" partition the remaining (a, μ)-space. Every hard-core triple lands in exactly one branch. $\square$

The debt of Part III is paid: SHARP holds, so Proposition 15.5 holds unconditionally, and with it the non-existence half. It remains only to record sharpness — the checks were exhibited in §16:

Proposition 24.1 (Non-improvability)**.** $m(\{3,4,5\}) = 4 = M-1$*: the constant* $M-1$ *cannot be improved to* $M-2$*. And for* $G = \{2,3\}$ *(*$\gcd 1$*,* $M=3$*), no multiset of size* ≤ 2 *covers* 3 *consecutive integers: the hypothesis* $|G| \geq 3$ *cannot be dropped.* ☐

Remark 24.2 (Overlapping branch conditions are harmless). The routing sends $(3,4,5)$ to Case P, though Case L also applies. The two constructions in fact coincide: Lemma 19.3 at $a=3$, $g=1$ gives $(x,y,z) = (2,1,1)$, the multiset $\{3,3,4,5\}$, which is exactly Case P's witness. Overlaps elsewhere are equally benign — $(13,25,26)$ satisfies both the D-condition and Case E's size conditions — because the tree is evaluated in order and the first matching branch supplies a witness.

Part V. Assembly, and what the computer checked

25. Assembly: proof of the dichotomy

All the pieces exist; this section only has to bolt them together.

Proof of Theorem A. Part (1), existence. This is Theorem 6.1 (proved there for all $k \geq 2$, hence for the problem's $k \geq 3$): if $d_2 \geq k+1$, the announcement $r = 192\, d_2$ wins against every lacunary B of that ratio.

Part (2), strong non-existence. Let $d_2 \leq k$ (with $k \geq 3$) and let $r_1, r_2, \ldots$ be any prescribed sequence of ratio lower bounds. Build the certificate B of Lemma 8.6 for the sequence $R(i) := r_i$; it satisfies $b_{i+1} \geq r_i\, b_i$ for every i. Let A be any (d_1,d_2)-sequence. By Proposition 15.5 — now unconditional, since Part IV proved SHARP (Theorem 16.1) — A is tail-covering. Retracing which case fired, for the record:

The walk's tail alphabet (rescaled)	Case closed by
one letter	Lemma 9.4
two letters, gap word eventually periodic	Lemma 10.1
two letters, k even and $d_2 = k$	Proposition 14.6
two letters, some $W_2 \geq 2$ (k odd or $d_2 \leq k-1$)	Lemmas 12.4(a), 12.2
two letters, balanced	Theorem 13.8, then Lemma 10.1 or 14.1
three or more letters	Lemma 15.3 + SHARP

By Lemma 8.6(2), $(kA) \cap B \neq \emptyset$. As A was arbitrary, the single B defeats every admissible walk; as (r_i) was arbitrary, this is the strong form.

The dichotomy. For any $k \geq 3$ and $1 \leq d_1 < d_2$, exactly one of $d_2 \geq k+1$ and $d_2 \leq k$ holds; part (1) gives a winning strategy in the first case, part (2) denies one in the second (a winning r would in particular defeat the certificate built for the constant sequence $r_i \equiv r$). So the answer to Erdős problem 1112 is: *yes precisely when* $d_2 \geq k+1$. ☐

Every promise of the honest ledger (§2.4) has now been redeemed: gap (1) in Part II, gaps (2) and (3) in Part III, gap (4) in Part IV.

25.1. **What we did not settle.** The dichotomy is settled; the quantitative question behind it is wide open, and so is the question of whether the proof needs to look like this. Four problems, in descending order of how much we would like to know the answer.

1. Is $r_k(d_1,d_2)$ bounded? Theorem 6.1 gives $r_k(d_1,d_2) \leq 192\, d_2$ whenever $d_2 \geq k+1$ — a bound that grows with d_2. Every exact value cited in this paper is 2: Chen [Ch00] shows $r_2(a,b) \leq 2$ off the diagonal $b = 2a$, whence $r_2(a,2a) = 2$ on it by the monotonicity of §1.9, and $r_2(2,3) = 2$ [ErGr80, BHJ97]. We know of no instance where the answer exceeds 2, though we have not surveyed the literature exhaustively for one.

Question. Is there an absolute constant C with $r_k(d_1, d_2) \le C$ for every $k \ge 3$ and every $d_2 \ge k+1$? Is $C = 2$?

Our $192\,d_2$ is not close to optimal and we make no claim that it is. It is what the nested-interval construction of Part II pays for being uniform: the constant absorbs a safety margin at every scale, and the d_2 absorbs the width of the window the walker must be denied. Either could plausibly be pushed hard. The gap is starkest wherever an exact value is known: at $k = 2$ the truth is 2 and our uniform machinery would deliver only $192\,d_2$, so the construction is overpaying by a factor of order d_2 in every case anyone can check. Whether that factor is an artefact of insisting on one construction for all parameters, or whether r_k genuinely grows for $k \ge 3$, is exactly what Question 1 asks.

2. Lower bounds. The gap in Problem 1 is embarrassing from both ends, and the lower end is the neglected one. For $k \ge 3$ we know of no lower bound on $r_k(d_1, d_2)$ beyond the elementary.

Question. Is $r_3(1, 4) = 2$? More generally, is there any (k, d_1, d_2) with $d_2 \ge k+1$ for which $r_k(d_1, d_2) > 2$ can be proved?

A single such example would be more informative than any improvement to 192: it would show the phenomenon is not simply "ratio 2 always suffices above the threshold".

3. Must the finite layer exist? The proof of Theorem 16.1 is uniform on four of its six branches; Cases T and B (§22, §23) are where the uniform constructions run out and 350 printed certificates take over. Nothing we know says that is necessary.

Question. Is there a proof of Theorem 16.1 with no finite case list at all?

We would be glad to see the tables go. They are the part of this work that a reader cannot check by thinking, and §26.3 exists only because they are there. Our own attempts to unify Cases T and B foundered on the same obstruction each time — the η-box argument needs $\mu \ge 12$ and $a \ge 12$ simultaneously, and both small-parameter corners genuinely behave differently — but that is evidence about our ingenuity, not about the problem.

4. Which G are extremal? Sharpness is not confined to small M, and it is worth recording why, since Proposition 24.1 on its own might leave the impression that $m(\{3,4,5\}) = 4$ is a small-parameter accident. It is not: the bound is attained at *every* maximum, by at least two families.

Proposition 25.1 (An extremal family at every maximum)**.** *For every $M \ge 3$, $m(\{1, M-1, M\}) = M - 1$.*

Proof. Upper bound. The $M-1$ copies of 1 have subset sums $\{0, 1, \ldots, M-1\}$.

Lower bound. Let a multiset consist of i copies of 1, j of $M-1$ and k of M. A subset sum taking $p \le i$, $q \le j$ and $r \le k$ of them equals

$$p + q(M-1) + rM \;=\; (q+r)M + (p-q),$$

so every subset sum lies in $sM + [-j,\ i]$ for some integer $s \ge 0$. Suppose $i + j + 1 < M$. Then $[-j, i]$ meets fewer than M residue classes modulo M, so some class ρ is missed entirely — and every one of any M consecutive integers would have to be a subset sum, including the one in class ρ. Hence $i + j + 1 \ge M$, and the multiset has size $i + j + k \ge i + j \ge M - 1$. $\square$

So sharpness is not a small-M accident: it recurs at every maximum, and $M-1$ is the exact worst case rather than a bound that happens to bind once. A second family behaves the same way — $m(\{M-2, M-1, M\}) = M-1$, whose upper bound is Case L at $g = 1$ (Lemma 19.3), where the budget $a + 2g - 1 = M - 1$ is met with equality; we have verified its lower bound exhaustively for $M \le 20$ but have not written out a proof for general M.

Exhaustive computation of $m(G)$ over all G with $|G| = 3$ and $\max G \leq 15$ finds the maximum to be exactly $M-1$ at every single M, and finds many extremals — so "are there large extremal sets" has the dull answer *yes, always*. The live question is which ones.

> *Question.* Characterise the G with $m(G) = M-1$. Are the two families above the basic extremizers, in the sense that every extremal G contains one of the configurations driving them? Is there a stability statement — if $m(G)$ is close to $M-1$, must G be close to an extremal family?

This one is asked out of curiosity rather than need: the slot lemma consumes exactly $m(G) \leq M-1 \leq k-1$, so a sharper bound would not shorten the proof of the dichotomy — and, by the above, no sharper bound exists to be had. It might well shorten the proof of Theorem 16.1.

The remaining section is not part of the proof; it is about how the proof is *checked*.

26. What the computer checked, and how you can too

This proof is computer-assisted, and this paper has flagged every place the assistance occurs: the 172 Table-A certificates and the $a \leq 3000$ scan (Case T), and the 178 Table-B base certificates with the class enumeration (Case B) — those four items are the load-bearing finite layer. (This paper paper's harnesses additionally run corroborating enumerations that this document's proofs never lean on, such as listing all minimal alphabets with $M \leq 130$; corroboration is welcome, but the ledger above is complete without it.) This closing section explains what, exactly, was computed, why you need not take anyone's word for it — and what such checking can and cannot promise. Nothing here is needed to *understand* the proof; it is needed to *trust* it, which is a different and equally honorable activity.

26.1. **Certificates, not searches.** Every computed object in this work is a *certificate*: a small tuple whose correctness is a finite, mindless computation. A Table-A or Table-B row $(a, b, M;\ x, Y, Z)$ asserts two things — the multiset of x copies of a, Y of b, Z of M has size $\leq M-1$, and its subset sums contain M consecutive integers — and *you* can check any row with pencil and paper in minutes (the worked instances in §§22–23 did exactly this). How the rows were *found* is irrelevant to their truth: a certificate severs the proof from the search that produced it. The same holds for the scans: "this formula exceeds that bound at exactly these 172 inputs" is a statement any programmer can re-derive in an afternoon, in any language.

To be precise about packaging: the complete human-readable proof record is *this paper alone*. All 350 rows are printed in Appendix E, generated from the same canonical data that the Lean kernel independently decides, and exported as CSV with recorded digests for readers who would rather compute than read. Nothing load-bearing is held elsewhere.

26.2. **Three checks, and what their independence is worth.** The repository checks the finite layer three separate ways, sharing no code. Before listing them, it is worth being exact about what that buys, because "three independent checks" is a phrase that promises more than any of this delivers. The three implementations are independent *of each other*; they are not independent *of the specification*. All three were written against the same case analysis — the one in §§19–23 — so they are three readings of one text, not three verifications of one theorem. What they can catch is a slip in transcription or arithmetic. What no number of them can catch is an error in the case analysis itself, because that error would be faithfully reproduced by each. Only the proofs address that, and only a reader does.

(1) *Two Python harnesses.* One reconstructs every designated witness across all 83,251 hard-core triples with $M \leq 120$ and verifies each in exact integer arithmetic; the other is a from-scratch re-implementation working only from the lemma statements. Both fail closed — any invalid row, missing class, or deviant count aborts with an error — and both report identical branch tallies (2392 D, 2605 P, 1420 L, 71,421 E, $3254 + 172$ T, 1987 B) — tallies which sum to exactly 83,251, the number of triples: the books visibly close. Agreement between independent implementations is strong evidence against ordinary coding slips — two people are unlikely to mistype the same thing — but it is evidence, not proof, and it says nothing at all about a shared misunderstanding of the mathematical design, since both programs were written from the same case analysis. That is what the proofs are for.
(2) *The table generator.* The typeset tables in the certificate supplement are *generated* from one canonical data file by a script that re-verifies every row (budget and coverage, by exact bitmask arithmetic) on every build, so no printed row can silently diverge from the data that is checked.
(3) *The Lean kernel.* The entire proof — not just the finite layer — is formalized in Lean 4 and machine-checked; the certificate tables are transcribed into the formal development and re-decided by Lean's kernel.

26.3. **The four finite obligations, named.** "The computer checked it" is too coarse to audit. Exactly four finite statements are needed from outside the prose, two for Case T and two for Case B, and they are worth separating because they are not all closed the same way — and one of them is closed by hand.

T1 *Completeness of the T scan.* Every hard-core triple in Case T either satisfies the uniform bound of §22 or appears among the 172 printed lines. *How it is closed:* by hand for the tail — the calculation at §22 bounds $K \leq 40$ over all 50 Case-T lines and so settles every $a \geq 3000$ with margin ≥ 960, uniformly and with no search — and by exhaustive scan below that threshold. This is the one obligation whose completeness genuinely rests on the scan, and a referee spending scepticism should spend it here.

T2 *Validity of the T witnesses.* Each of the 172 printed lines satisfies its budget and covers M consecutive integers. *How it is closed:* exact integer arithmetic on a finite table, by four separate implementations (the two harnesses, the table re-verifier, Lean's kernel) — with the caveat of §26.2 about what their agreement is worth. Here, though, the caveat bites least: validity of a printed row is a self-contained arithmetic claim, not a reading of the case analysis.

B1 *Completeness of the B enumeration.* The 178 printed classes are exactly the residue classes for which Case B must supply a base. *How it is closed: by hand*, in Lemma 23.1 — the two exclusions $h \equiv -\bar{e}$ and $h \equiv -2\bar{e} \pmod{a}$ are always distinct, giving per-a counts $0, 1, 6, 5, 20, 17, 32, 25, 72$ for $a = 3, \ldots, 11$, which sum to 178. No scan is involved in this one; the computation only confirms it.

B2 *Validity of the B witnesses, and the lift.* Each printed base is valid — budget, coverage, and the frame-merge condition $Y + Z \leq a - 1$ that Lemma 17.5 requires and Remark 17.4 shows cannot be dropped — and the λ-lift carries it to every larger member of its class. *How it is closed:* finite arithmetic for the bases (as in **T2**), and a proof — Lemma 17.5 — for the propagation, which is where the "forever" comes from and is not a finite claim at all.

B3 *Minimality of the B bases.* For each of the 178 classes, the printed base is the *first* member of that class routed to Case B. *How it is closed:* finite check. This obligation is easy to miss, and we missed it in an earlier draft: **B1** says which classes need a base and **B2** says each base is valid and lifts *upward*, but neither forbids a base sitting one λ-rung too high — which would leave the genuine Case-B members below it certified by nothing at all. It is a separate finite assertion and is checked as one.

So of the five, one is a theorem with a finite check attached (**B2**), one is a hand proof (**B1**), and three are finite checks (**T2**, **B3**, and the enumeration half of **T1**) — of which only **T1** is an exhaustive search rather than the validation of a printed list.

26.4. **What a Lean formalization is.** Lean is a *proof assistant*: a programming language in which definitions, theorem statements and proofs are all written formally, and a small program — the *kernel* — checks that every proof step follows from the rules of logic and the axioms. The human (or, in this work, largely the AI systems noted below) supplies the proof; the kernel is the referee that cannot be sweet-talked. Three facts summarize the state of this development:

- It is *complete*: all three headline theorems — the dichotomy, the explicit ratio $192\,d_2$, and the strong non-existence form — are proved, over frozen definitions, from start to finish.
- It is *gap-free*: no `sorry` (Lean's "I owe you a proof" placeholder) anywhere, and the axiom audit reports only Lean's three standard foundational axioms.
- Its finite checks run *in the kernel*: no appeal to compiled external code (no `native_decide`), so even the $172 + 178$ certificate rows and the $a \leq 3000$ scan are validated by the same small trusted checker as every logical step.

Anyone can reproduce this: clone the repository, run `lake build`, and watch it compile with no complaints (the repository README gives the exact commands, and links a public log of a complete run).

26.5. **What machine-checking does not promise.** Honesty requires the other half. A formal verification establishes: *these formal statements follow from these axioms.* It does not establish:

- *That the formal statement means what the English problem says.* If the Lean definition of "lacunary" or "gap window" subtly mismatched the problem, the kernel would happily verify the wrong theorem. This correspondence must be audited by humans reading both — which is why §26.6 walks through each modelling choice (for instance: over the natural numbers, careless subtraction truncates at zero, and a naively phrased gap condition would silently admit decreasing walks; the formalization phrases the window additively to dodge exactly this). The definitions are few and short by design, to make that audit as small as possible.
- *Novelty or attribution.* Whether SHARP is genuinely new is a question about the literature, addressed in §1.9 by a documented search — no computer can settle it.
- *That nothing outside the trusted base is wrong.* The kernel, the pinned compiler and library versions — these are trusted. The trust base is small and standard, but it is not zero; no honest account says otherwise.

The layers of trust, drawn as a stack — each layer checks the one above and must itself be trusted or audited:

English theorem (Erdős 1112)	faithfulness: human-audited, §26
↓	
formal statements (frozen Lean definitions)	small by design, so the audit is small
↓	
the Lean development (every proof step)	checked mechanically by the kernel
↓	
kernel + three standard axioms	trusted — small, standard, but not zero
↑	
certificates + two Python harnesses	independent corroboration, outside the stack

One story worth retelling: an early draft of the staircase lemma omitted the frame-merge hypothesis, and the *formalization process* caught it — the counterexample $(5, 7, 9)$ of Remark 17.4 is now permanently enshrined in the development as a guard. The moral runs both ways: human-written mathematics slips, and machine-checking catches real slips — and what the machine certifies is exactly the statement it was given, neither more nor less.

26.6. **The formal statement, and why it is faithful.** A formal proof certifies only the statement it was given, so the statement deserves to be read. The definitions live in a frozen file (`lean/Erdos1112.lean`) that the development imports but never edits, and the three theorems are stated once each, directly over those definitions:

```
def IsLacunaryWith (r : ℕ) (b : ℕ → ℕ) : Prop :=
  0 < b 0 ∧ StrictMono b ∧ ∀ i, r * b i ≤ b (i + 1)

def HasGapsIn (d₁ d₂ : ℕ) (a : ℕ → ℕ) : Prop :=
  0 < a 0 ∧ ∀ i, a i + d₁ ≤ a (i + 1) ∧ a (i + 1) ≤ a i + d₂

def kFoldSumset (k : ℕ) (a : ℕ → ℕ) : Set ℕ :=
  { n | ∃ f : Fin k → ℕ, n = ∑ j, a (f j) }

def RatioWorks (k d₁ d₂ r : ℕ) : Prop :=
  ∀ b : ℕ → ℕ, IsLacunaryWith r b →
    ∃ a : ℕ → ℕ, HasGapsIn d₁ d₂ a ∧
      Disjoint (kFoldSumset k a) (Set.range b)

def Question (k d₁ d₂ : ℕ) : Prop := ∃ r : ℕ, RatioWorks k d₁ d₂ r
```

And here are the theorems themselves, likewise verbatim. They are what the kernel actually certifies, so they — not the definitions alone — are what a faithfulness audit must read:

```
theorem erdos_1112 (k d₁ d₂ : ℕ) (hk : 3 ≤ k) (hd₁ : 1 ≤ d₁) (hd : d₁ < d₂) :
    Question k d₁ d₂ ↔ k + 1 ≤ d₂

theorem erdos_1112_existence_bound (k d₁ d₂ : ℕ) (hk : 3 ≤ k) (hd₁ : 1 ≤ d₁)
    (hd : d₁ < d₂) (h : k + 1 ≤ d₂) :
    RatioWorks k d₁ d₂ (192 * d₂)

theorem erdos_1112_strong_nonexistence (k d₁ d₂ : ℕ) (hk : 3 ≤ k)
    (hd₁ : 1 ≤ d₁) (hd : d₁ < d₂) (h : d₂ ≤ k) (R : ℕ → ℕ) :
    ∃ b : ℕ → ℕ, IsVarLacunaryWith R b ∧
      ∀ a : ℕ → ℕ, HasGapsIn d₁ d₂ a →
        (kFoldSumset k a ∩ Set.range b).Nonempty

theorem question_iff_questionInt (k d₁ d₂ : ℕ) :
    Question k d₁ d₂ ↔ QuestionInt k d₁ d₂
```

```
theorem erdos_1112_int (k d₁ d₂ : ℕ) (hk : 3 ≤ k) (hd₁ : 1 ≤ d₁) (hd : d₁ < d₂) :
    QuestionInt k d₁ d₂ ↔ k + 1 ≤ d₂
```

Four things to check against these, which the definitions alone cannot settle. The hypotheses $3 \le k$, $1 \le d_1$ and $d_1 < d_2$ appear explicitly in every headline theorem, so none is proved on a degenerate parameter range — in particular $1 \le d_1$ is present, which is what forces A strictly increasing. The conclusion of `erdos_1112` is a biconditional with $k + 1 \le d_2$, not an implication. The bound theorem asserts `RatioWorks` at the specific value $192 \cdot d_2$, so the constant is in the statement rather than hidden in a proof. And the strong form quantifies in the order $\forall R\ \exists b\ \forall a$: the sequence b is chosen after the ratio bounds R but before the walk a, which is exactly the strength claimed in Theorem A(2) — the weaker $\forall R\ \forall a\ \exists b$ would let b depend on the walk.

Six modelling choices carry the weight, and each is a place where a formalization could quietly prove the wrong theorem.

(1) **Sets as increasing enumerations.** A and B are infinite sets of positive integers, encoded by the functions that list them in order. `StrictMono` is what makes such a function enumerate a *set* rather than a multiset; it is imposed on B directly and *derived* for A from the gap condition when $d_1 \ge 1$.
(2) **The gap window, phrased additively.** Over the natural numbers, subtraction is truncated at zero — so a subtractive phrasing would be a trap. An isolated upper bound "$a_{i+1} - a_i \le d_2$" would read $0 \le d_2$ on a *decreasing* step, which is true, and would silently admit walks that go backwards. Writing the window as $a_i + d_1 \le a_{i+1} \wedge a_{i+1} \le a_i + d_2$ avoids this entirely. (The lower bound is not the trap: an underflowed gap gives $d_1 \le 0$, false, since $d_1 \ge 1$.)
(3) **The sumset, with repetition.** `kFoldSumset` ranges over an arbitrary index function f with *no* injectivity requirement, so summands may repeat — the intended kA of Definition 1.5. Requiring f injective would model the distinct-summand sumset, a strictly smaller set and a different problem.
(4) **Quantifier order.** `Question` is $\exists r\ \forall B\ \exists A$, with A chosen after B — the game of §1.6, not the easier $\exists A\ \forall B$.
(5) **The ratio over $\mathbb{N}$ rather than $\mathbb{Z}$.** The problem says "an integer r". Passing to $\mathbb{N}$ loses no existence content, and this is now proved rather than argued: the frozen file also carries the integer form `QuestionInt` together with the machine-checked equivalence `question_iff_questionInt`, and the headline theorem in integer form `erdos_1112_int`. So the dichotomy is verified for the problem's literal phrasing.
(6) **A genuine, non-vacuous biconditional.** Both directions are separately witnessed, and both sides are inhabited: none of the predicates is silently trivial. `IsLacunaryWith r` is satisfiable for every r (take $b_i = (r+2)^i$), `HasGapsIn` is satisfiable (take $a_i = 1 + d_1 i$), and `kFoldSumset k a` is nonempty while `Set.range b` is infinite — so disjointness is a real constraint, not a vacuous one.

Scope. Every headline theorem carries the problem's own hypothesis $k \ge 3$. The existence half is proved here in prose for all $k \ge 2$ (§3); the formalization deliberately carries $k \ge 3$ throughout, so the $k = 2$ existence statement stands as an unformalized supplementary observation. Nothing in the mathematical argument of this paper is delegated to the formalization.

Trust base. At the commit recorded in §29, built with the pinned toolchain `leanprover/lean4:v4.27.0` and the Mathlib revision fixed by `lake-manifest.json`,

`#print axioms` reports for each of the three headline theorems exactly

```
[propext, Classical.choice, Quot.sound]
```

— Lean's three standard foundational axioms, and nothing else. In particular there is no `sorryAx` (no gaps) and no `Lean.ofReduceBool` (the axiom `native_decide` would introduce). Every finite check, including the 350 certificate rows and the $a \leq 3000$ scan, is therefore discharged by the trusted kernel rather than by compiled code. The audit is itself the gap check: a `sorry` anywhere would surface in that list.

26.7. **Independent replication.** The proof was posted as a claimed solution on `erdosproblems.com` on 6 July 2026. Two readers have since checked it independently of the author; both reports are public, in the thread for Problem 1112.

Thomas Bloom, who maintains the problem collection, verified on 13 July 2026 that the declaration `erdos_1112` in `Erdos1112Proof/Final.lean` is a correct formalisation of the problem as the collection states it, and that the development compiles with no `sorry`. He was explicit that he had not attempted to follow the mathematics, and that the work of human digestion remains to be done.

That distinction deserves dwelling on, because it lands precisely on the blind spot named in §26.2. The three implementations of the finite layer are independent of each other but not of the specification; what none of them can test is whether the formal statement means what the problem means. Bloom's check is a reading of exactly that translation, by someone who did not write it — a second opinion on §26.6, which is the one part of this verification that no machine can underwrite. It says nothing whatever about whether the proof is right.

On 17 July 2026 Colin Snyder reported a full independent rebuild: cloning the repository, compiling the Lean development from scratch on his own machine, and confirming that the kernel accepts the final theorems on the three standard axioms with no `sorry` and no `native_decide`; running both Python harnesses (all 83,251 hard-core triples passing, and the independent re-check validating all 172 Case-T certificate rows); checking the certificate tables against their published SHA-256 digests; confirming that the declarations of Appendix F are present in the Lean sources; and hand-checking $m(\{3, 4, 5\}) = 4$.

Neither report is a referee's report, and neither claims to be. As of this writing no one has stated publicly that they have read and understood the argument of Parts II–IV. That is the thing this paper is asking for, and it is the reason the paper is written the way it is.

26.8. **What machine-checking still does not settle.** Two honest caveats. "Depends only on three axioms" is a statement about the *logical* axiom report; the operative trusted base is larger, and includes the Lean kernel and elaborator, the Mathlib source at the pinned revision, and the dependency snapshot. And an axiom audit cannot detect a mismatch between a numbered result in this paper and the declaration that purports to formalize it. That correspondence is asserted in §F and must be checked by reading both — which is one more reason for a paper written to be read.

27. Provenance and authorship

This work was produced with substantial assistance from automated systems, and we record the division of labour.

The mathematical constructions, case analyses and proofs, together with the Lean formalization and a first draft of this paper, were produced by Claude (Anthropic; models Fable 5 and Opus 4.8) acting as the reasoning agent, under the author's direction. GPT-5.5 (OpenAI) and Gemini 3.1 (Google DeepMind) were used for advice and for adversarial review of drafts.

The author selected the problem and fixed its formalization; set the intermediate targets and overall strategy; decided which candidate routes to pursue and which to abandon; audited each construction and proof; chose the formal statement; and made the final mathematical judgements. He takes full responsibility for the content, including any error that remains. The automated systems are tools, not authors: they meet no authorship criterion, cannot take responsibility for the work, and are not listed as authors.

Formal verification settles correctness relative to the formal statement only; it does not establish novelty, attribution, or exposition. The one thing no formalization can settle on the reader's behalf — that the formal statement means what the informal problem says — is set out decision by decision in §26.6.

If you are a student reading this: the theorem is true because of the proofs you have just read and can check — every one of them, with enough patience, by hand — and not because any person or machine says so. That is the whole point of mathematics, and it is why a paper like this one can exist at all.

28. Declarations

Funding. None.

Acknowledgements. The author thanks Thomas Bloom for maintaining the problem collection and for checking the formal statement against it, and Colin Snyder for an independent rebuild and re-verification of the whole package. Their reports are described in §26.7; neither implies any endorsement of the mathematics, which remains the author's responsibility.

Competing interests. The author declares no competing interests.

Author contributions. Sole author. The division of labour between the author and the automated systems used is set out in §27.

Use of generative AI. Automated systems played a substantial role in producing this work; the full account — models, tasks, division of labour, and verification — is §27. The author personally checked every proof and every computational specification in this paper, verified each citation used to support an attribution or novelty claim, reviewed every sentence for its intended mathematical meaning, and adopted the final text as his own. Where a result is established only by machine — the finite layer of §26 — the paper says so explicitly and states what the machine did. The author directed the work, audited the output, and takes full responsibility for the content; the systems are tools, not authors. Every mathematical claim used in the proof of Theorem A has been checked by at least one of the routes of §26, and the dichotomy is machine-checked in Lean 4 within the scope stated in §26.6.

A note on repository metadata. The version-control history of the accompanying repository contains `Co-Authored-By` trailers naming the automated systems. These record which tooling produced a commit; they are not a claim of scholarly authorship, which is as stated above.

Data and code availability. See §29.

29. Data availability and reproduction

Everything needed to check this paper is permanently archived at

`doi:10.5281/zenodo.21568276`

— deposited with CERN's Zenodo repository. Code (the Lean development and the Python harnesses) is licensed Apache-2.0; the paper, its figures, and the certificate data are licensed CC BY 4.0. That snapshot carries this LaTeX source, the Lean 4 development [Lean4, Mathlib], the machine-readable certificate exports (`data/table-A.csv`, `data/table-B.csv`, data/SHA256SUMS), the table generator and re-verifier gen-tables.py, the two independent Python harnesses, and the documented prior-art search (paper/novelty-search.md).

The identifier above resolves to that exact snapshot and will not change; the concept identifier doi:10.5281/zenodo.21568275 resolves instead to the most recent version, should later ones exist. Development continues in the open at https://github.com/beetree/math_erdos_1112, but no claim in this paper depends on that repository remaining available.

Pinning. The archived snapshot records, alongside the source, the exact Git commit, the SHA-256 digests of the submitted PDF and of the source archive, the digests of table-A.csv and table-B.csv, the toolchain string leanprover/lean4:v4.27.0, the Mathlib revision fixed by lake-manifest.json, and the identifier of the continuous-integration run that last executed every check below. A referee wishing to confirm that this PDF corresponds to that snapshot can compare the digest printed in the archive against the file in hand.

The machine-checked proof. A clean checkout builds with the pinned toolchain:

```
git clone https://github.com/beetree/math_erdos_1112
cd math_erdos_1112 && git checkout <release tag>
cd lean
lake exe cache get                              # fetch the pinned Mathlib
lake build                                      # compile every file (~1-2 hours)
lake env lean Erdos1112Proof/AxiomsCheck.lean # the 3-axiom report, per theorem
cd ../paper
python3 scripts/sharp6_final.py 120             # constructive harness, fail-closed
python3 scripts/sharp_referee_check.py 120      # independent harness, fail-closed
python3 scripts/check_correspondence.py         # 44/44 declarations resolve
make                                            # re-verifies all 350 rows, builds
```

Continuous integration re-runs the pinned build, the axiom audit, the table re-verification and the correspondence check of §F on every push.

The finite layer. The complete certificate tables are typeset in the Appendix E and exported as CSV with recorded SHA-256 digests. Running make regenerates both from the one canonical source and re-checks every row (budget $\leq M-1$, and coverage of M consecutive integers, by exact bitmask arithmetic, aborting on any failure), so no listing — printed, exported, or kernel-decided — can silently diverge from the others. Any single row can also be checked by hand in a few minutes, by the method of §26.

Appendix A. How to read this proof

Mathematics has a craft as well as a content, and this paper assumes a little of the craft while building all of the content. Here is the whole of what is assumed, in two pages. Skim it now; come back when a phrase in a proof feels like a move you were not taught.

Direct proof, contradiction, contrapositive. To prove "if P then Q", you may assume P and derive Q (direct); or assume P together with "not Q" and derive something absurd (contradiction); or prove "if not Q then not P", which says exactly the same thing (contrapositive). All three appear here, and we name which one we are using whenever it might not be obvious.

"It suffices to show." A proof often replaces its goal with a stronger or more convenient one: *it suffices to show* R, meaning "R implies what we want, so let us prove R instead". This is not a weakening — proving more than you need is always allowed. §4 does this deliberately and says so.

"Fix an arbitrary x." To prove something about *every* x, we take one, refuse to assume anything special about it, and argue. Because nothing special was assumed, the argument

covers all of them. The mirror move, "there exists", is discharged by *exhibiting* one — which is why so many proofs here consist of writing down a construction and then checking it works.

Induction, ordinary and strong. To prove a statement $S(n)$ for every n: prove $S(1)$, then prove that $S(n)$ implies $S(n+1)$. *Strong* induction allows the stronger assumption that $S(1), \dots, S(n)$ *all* hold when proving $S(n+1)$ — used once, in §24, where a statement about a number M is proved using the same statement for all smaller values.

The well-ordering principle. Every nonempty set of positive integers has a smallest element. Obvious-sounding, genuinely load-bearing: it is what licenses phrases like "let n^* be the *first* index such that…", which appear in §15 and §7.

Chains of inequalities. A display like

$$A \;\le\; B \;=\; C \;\le\; D$$

asserts each link separately and concludes $A \le D$. Read it one link at a time, and ask of each: which hypothesis makes *this* step true? Where a link is not immediate we say why, either inline or in Appendix C.

Mechanism versus bookkeeping. This is the distinction most worth internalizing before Part II. Nearly every proof here has two ingredients: a *mechanism* — the construction or idea that makes the thing work — and *bookkeeping* — the inequalities confirming that the particular constants chosen are large enough. The mechanism is what you should remember; the bookkeeping is what you should be able to check but need not carry in your head. We flag the seam explicitly, and passages of pure verification are marked *(first pass: skippable)*.

Passing to a tail. Several arguments discard a finite initial segment of an infinite sequence and work only with what remains — "pass to the tail". This is harmless when the property in question concerns only large numbers, and we always say why it is harmless when we do it.

What is not *assumed.* No calculus, no set theory beyond "$\in$" and "$\subseteq$", no combinatorics, no number theory beyond what §7 builds, no analysis beyond what §11 builds. If you have met proof by induction and can manipulate an inequality without panic, you have enough.

Appendix B. A referee's guide: what to check, and in what order

A hundred pages is a lot to be handed, and not all of it carries weight. This appendix is written for a reader whose job is to decide whether Theorem A is true, and who would like to spend their scepticism where it buys the most. It says which results are load-bearing, what each one rests on, where the proof would most plausibly break, and roughly how long an honest audit takes.

Nothing here is needed to *read* the paper; §1.10 is the guide for that.

B.1. **The audit target.** Theorem A has two halves, and they share almost nothing — no lemma of Part II is used in Parts III–IV, and vice versa. A referee may audit them independently, and may stop after either.

Existence ($d_2 \ge k+1 \Rightarrow r_k \le 192\, d_2$) is Part II: a single explicit construction and the estimates that pay for it. It uses no modular arithmetic, no combinatorics on words, and nothing from Parts III–IV. It is self-contained and, of the two halves, much the shorter check.

Non-existence ($d_2 \le k \Rightarrow$ no r_k) is Parts III–IV. It is the long half, and it is where the finite computation lives.

B.2. **The load-bearing results.** Table 2 lists everything on the critical path. Results not in this table — worked examples, remarks, the three interludes — support the exposition but carry no weight: if one of them were wrong, the theorem would survive.

B.3. **Where we would attack it.** If this proof is wrong, the likeliest places, in order. The ordering below is not our original guess: an earlier draft of this appendix ranked the boundary case first and the induction fourth, and three independent readers reported the reverse — the boundary case survived every attack, while the induction's *statement* did not. We have reordered accordingly, and left the correction visible because it is itself information about where the risk sits: the danger was in the bookkeeping, not in the delicate estimate.

(1) *The induction in Theorem 16.1*, and in particular how Lemma 18.2 is invoked. The recursion is well-founded on $(\max G, |G|)$ ordered lexicographically, and one of the three descents lowers only the second coordinate. An earlier draft stated Lemma 18.2 with an ungraded hypothesis — "if SHARP holds for all irreducible alphabets" — which is not available at the point §24 uses it; the lemma is now graded by the maximum. Check that the grading is honest and that no appeal raises the pair.

(2) *Completeness of the Case-T scan* (**T1** of §26.3). The uniform bound closes $a \geq 3000$ by hand; below that, completeness rests on an exhaustive enumeration. The hand calculation is short and worth checking directly — it is the paragraph establishing $\max K = 435/11$ over the 50 Case-T lines. Note that this calculation is easy to get wrong by inserting a ceiling that the definition of β does not contain; doing so yields 41 and appears to break the bound.

(3) *Hypothesis-matching between prose lemmas and the results that cite them.* This is where the one real defect of the previous draft was found: the even branch of Case L (Lemma 19.3) cited Lemma 17.3(d) on an infinite family with $g = 1$, while that clause was stated for $g \geq 2$. The construction was correct throughout and the formalization accepted it, because the formal staircase statement carries no such restriction — which is exactly why Appendix F matching *names* is not enough. Clause (d) now covers $g \geq 1$. A referee should spot-check other citations the same way.

(4) *The type discipline around rescaling.* Lemma 9.5 divides all gaps by g, destroying the lower bound d_1. Definition 9.2 exists to keep this honest, and every statement in §§12–14 is phrased so that the rescaled object satisfies it. Confirm that no statement applied after rescaling secretly assumes d_1. (Readers who have checked this reported it clean; we leave it on the list because it is cheap to check and fatal if wrong.)

(5) *The boundary case, Proposition 14.6*, and *exhaustiveness of the six-branch routing.* When k is even and $d_2 = k$ the sweep guarantees width $k - 2$ and needs $k - 1$: no slack, and it is closed by a separate argument (border–period duality, Lemma 14.3). We ranked this first for two drafts and it has now been checked closely by several readers without a defect being found. It remains the most delicate estimate in the paper; it is no longer our best guess at where an error would be.

B.4. **Honest time estimates.** For a professional reader checking every line, our estimate of a complete audit is **forty to fifty hours**, distributed roughly as follows.

Statements, architecture, and this appendix	2–3 h
Part II, complete (the whole existence half)	6–8 h
Part III: certificate, reductions, easy cases	3–4 h
Part III: the two-letter campaign, §§12–14	10–12 h
Part IV: machinery and reduction to the hard core	7–8 h
Part IV: the six branches	9–10 h
Finite layer: rerun the harnesses, spot-check rows	1–2 h

Three shorter routes are defensible:

The existence half only (6–8 hours). Self-contained, uses no modular arithmetic and nothing from Parts III–IV, and settles half of Theorem A on its own.

Architecture plus the five attack points above (8–10 hours). Enough to form a confident view on whether the proof is sound in design, without verifying every estimate.

The new tool only (12–15 hours). Theorem 16.1 and Part IV stand alone, independent of the Erdős problem entirely; §1.9 argues they may be of separate interest.

B.5. **What the machine does and does not settle for you.** The Lean development removes one class of doubt completely: no step of the critical path is missing, hand-waved, or circular, and no finite check is wrong. It does not tell you that the formal statement is the problem — §26.6 argues that in prose, and §26.7 records that one reader has now checked it independently. A referee who trusts the Lean build can therefore skip verifying that the estimates close, and concentrate entirely on §26.6 plus the modelling questions: is the game of §1.6 the right formalisation, is "tail-covering" the right intermediate notion, does Definition 9.2 cover the objects the proof actually manipulates. That is a much shorter job — perhaps six hours — and it is the one thing no amount of machine-checking can do.

Appendix C. Every expansion, in full

The main text keeps ideas in the foreground and refers routine expansions here. Each item is pure algebra — no cleverness, only care.

(A1) Lemma 18.3: the run has length $\geq M$. With $x = M - \alpha$, $S = (\alpha - 1)\gamma$ and $\gamma \leq M - \alpha - 1$:

$$\begin{aligned}
\alpha x - S + 1 &= \alpha(M-\alpha) - (\alpha-1)\gamma + 1\\
&\geq \alpha(M-\alpha) - (\alpha-1)(M-\alpha-1) + 1\\
&= \left[\alpha M - \alpha^2\right] - \left[\alpha M - \alpha^2 - M + 1\right] + 1 \quad \text{since } (\alpha-1)(M-\alpha-1) = \alpha M - \alpha^2 - M + 1\\
&= M.
\end{aligned}$$

(The middle expansion: $(\alpha-1)(M-\alpha-1) = \alpha M - \alpha^2 - \alpha - M + \alpha + 1 = \alpha M - \alpha^2 - M + 1$.)

(A2) Lemma 18.4: the covered length. With $x = \beta - 1$, $y = \alpha - 1 + t$, $C = (\alpha-1)(\beta-1) = \alpha\beta - \alpha - \beta + 1$:

$$\begin{aligned}
\alpha x + \beta y - 2C + 1 &= \alpha(\beta-1) + \beta(\alpha - 1 + t) - 2(\alpha\beta - \alpha - \beta + 1) + 1\\
&= \alpha\beta - \alpha + \alpha\beta - \beta + t\beta - 2\alpha\beta + 2\alpha + 2\beta - 2 + 1\\
&= \alpha + \beta - 1 + t\beta.
\end{aligned}$$

(A3) Lemma 18.6: the worst-case rearrangement. At $M = d\beta + 1$: $y = M - d - \beta + 1 = (d-1)(\beta-1) + 1$. Then, with $C = (\alpha-1)(\beta-1)$,

$$\begin{aligned}
L = \alpha x + \beta y - 2C + 1 &= \alpha(\beta-1) + \beta\left[(d-1)(\beta-1)+1\right] - 2(\alpha-1)(\beta-1) + 1\\
&= (\beta-1)\left[\alpha + \beta(d-1) - 2\alpha + 2\right] + \beta + 1 = (\beta-1)\left[\beta(d-1) - \alpha + 2\right] + \beta + 1.
\end{aligned}$$

The claim $L \geq M + 1 = d\beta + 2$ is therefore

$$(\beta-1)\left[\beta(d-1) - \alpha + 2\right] \;\geq\; d\beta + 2 - \beta - 1 \;=\; \beta(d-1) + 1,$$

i.e. $(\beta-1)(u - \alpha + 2) \geq u + 1$ with $u := \beta(d-1)$, as stated. At $d = 2$ ($u = \beta$): expand $(\beta-1)(\beta-\alpha+2) = \beta^2 - \alpha\beta + 2\beta - \beta + \alpha - 2$, so the claim is $\beta^2 - \alpha\beta + \beta + \alpha - 2 \geq \beta + 1$, i.e. $\beta(\beta-\alpha) + \alpha - 3 \geq 0$. Monotonicity in u: the difference $(\beta-1)(u-\alpha+2) - (u+1)$ has

u-coefficient $(\beta-1)-1=\beta-2\geq 0$ for $\beta\geq 2$; and if $\beta=2$ then $\alpha=1$ and the difference is $(u+1)-(u+1)=0$ — equality, which suffices.

(A4) Lemma 18.6: the bridge length.

$$(L-1)d-(d-1)M+1\geq M \iff (L-1)d\geq dM-1 \iff L\geq M+1-\tfrac{1}{d}.$$

Since L is an integer and $0<\frac{1}{d}<1$, the number $M+1-\frac{1}{d}$ is not an integer and lies strictly between M and $M+1$; so the condition is exactly $L\geq M+1$.

(A5) Case P, inequality (E1). Start from the sufficient form

$$2\big[(a-1)(ra-M)+M-1\big] \;\leq\; ar(M-r-1)+2a(r-1).$$

Left side: $2(a-1)ra-2(a-1)M+2M-2=2ra^2-2ra-2aM+4M-2$. Right side: $arM-ar^2-ar+2ar-2a=arM-ar^2+ar-2a$. Moving everything right, the inequality is

$$0 \;\leq\; arM+2aM-4M-ar^2-2ra^2+3ar-2a+2 \;=\; M(ar+2a-4)-ar^2-2ra^2+3ar-2a+2,$$

as displayed in the text. Now insert $2M\geq ra+1$ (valid since $ar+2a-4>0$), multiplying the target by 2:

$$\begin{aligned}2M(ar+2a-4)&-2ar^2-4ra^2+6ar-4a+4\\ &\geq (ra+1)(ar+2a-4)-2ar^2-4ra^2+6ar-4a+4\\ &=\big[a^2r^2+2a^2r-4ar+ar+2a-4\big]-2ar^2-4ra^2+6ar-4a+4\\ &=a^2r^2-2a^2r-2ar^2+3ar-2a \;=\; a\big(ar^2-2ar-2r^2+3r-2\big).\end{aligned}$$

So (E1) holds whenever $ar^2-2ar-2r^2+3r-2\geq 0$, i.e. whenever $a\,r\,(r-2)\geq 2r^2-3r+2$ — the condition $(*)$ of the text.

Appendix D. Symbol glossary

By part, in rough order of first appearance. A few letters are reused across parts (never within one); reuses are flagged.

Global.

k	number of summands	fixed throughout
$d_1\leq d_2$	the gap window	fixed throughout
A, a_i, g_n	the walk, its terms, its gaps	
B, b_i, r	the Adversary's sequence, its ratio	
kA	the k-fold sumset, repetitions allowed	Definition 1.5
$\lfloor\cdot\rfloor$, $\lceil\cdot\rceil$, $\{\cdot\}$	floor, ceiling, fractional part	§3, §5, §13

Part II.

γ	the slope (dial position)	$\in(d_2-1,d_2)$
c	the walk's offset	reused: $y{+}z$ in Part IV, run start in §15
$A_{\gamma,c}$	the Beatty walk	Definition 3.3
t, t_i	shifted target(s) $b-kc$	reused: $\min(\eta,a{-}\eta)$ in Case E
U_t, s_1, J, L, R	unsafe set; gap anatomy	§4.3–5
I_j, γ^*	the nest and its limit point	§6

Part III.

p_n	prefix sums of the gaps	§9
G_∞, T	tail alphabet; tail start	§9
$\delta < \delta'$, e	the two letters; $e = \delta' - \delta$	reused: $e = b - a$ in Part IV
h, h_n	the binary gap word	reused: $h = M - b$ in Part IV
q_n	counting function of h	§11
W_s, $w^{\pm}(s)$, width	the reachable band	§12
$W_2(\sigma)$	two-index width	§12
α, D_n, β	slope, discrepancy, intercept	§13; α, β reused in Lemma 17.1
u/v	α in lowest terms	Step 4
θ, ψ, ζ, κ	depth, rung, margin, padding	§14
$m(G)$, sums	covering cost; subset sums	Definition 15.1

Part IV.

$a < b < M$	the triple	$\gcd(a, b) = 1$ on the hard core
e, h, μ	$b{-}a$, $M{-}b$, $M{-}a = e{+}h$	§17.3
g, e', μ', C', V'	$\gcd(e, \mu)$; divided forms; grid corners	§17.3
(x, y, z), $c = y{+}z$	copies of a, b, M; box total	
$\mathcal{O}_t$	offsets at level t	staircase
(Y, Z), S, $\mathcal{B}$	box; top height; budget	frame lemma
$\bar{e}$, $\bar{\mu}$, η	mod-a data; $\eta = \bar{\mu}\,\bar{e}^{-1}$	§18.5
$t = \min(\eta, a - \eta)$, Z, σ	Case E box data	
n_b, n_M	max unpaired bs, Ms (Case P)	
λ	$\lfloor e/a \rfloor$: rung on a class ladder (Case B)	

Appendix E. The certificate tables

These are the $172 + 178$ finite certificates that Cases T and B consume — the entire finite layer of the proof, printed in full so that this paper is self-contained.

Each entry is a triple (a, b, M) together with a *box certificate* (x, Y, Z), asserting that the multiset S of x copies of a, Y copies of b and Z copies of M satisfies

- **budget:** $x + Y + Z \le M - 1$, and
- **coverage:** sums(S) contains M consecutive integers,

i.e. that S witnesses $m(G) \le M - 1$ for $G = \{a, b, M\}$. The mechanism is the frame lemma (Lemma 17.2): the copies of b and M realize a modulo-a box covering every residue, and the x copies of a lift that residue cover to a solid run.

You can check any row of these tables by hand in a few minutes, and that is the point of printing them. Take the row $(5, 8, 9) \to (5, 2, 1)$, which is Example 22.5's worked instance: the multiset is $\{5, 5, 5, 5, 5,\ 8, 8,\ 9\}$, of size $5 + 2 + 1 = 8 = M - 1$, meeting the budget exactly. Its parts $8j + 9k$ with $j \le 2$, $k \le 1$ take the values $\{0, 8, 16, 9, 17\}$, whose residues modulo 5 are $\{0, 3, 1, 4, 2\}$ — all five. Lifting by the copies of 5 gives the run $[17, 25]$, which is $9 = M$ consecutive integers. That single computation is exactly what the checker performs on each row, and what the Lean kernel re-decides.

These tables are *generated*, not transcribed: `gen-tables.py` parses the canonical source `data/certificate-data.md`, re-verifies every row's budget and coverage by exact arbitrary-precision bitmask, aborts the build on any failure, and only then emits the LaTeX below. The same rows are exported as CSV with recorded digests and transcribed into Lean, where the kernel decides them (§26). No listing — printed, exported, or kernel-checked — can silently diverge from the others.

Table 3: The 172 exceptional Case-T triples (the T-tail lemma of the main paper), all with $a \le 29$; the scan over $a \le 3000$ finds no others. Format $(a,b,M) \to (x,Y,Z)$: the multiset of x copies of a, Y of b, Z of M.

$(4,9,10)\to(7,1,1)$	$(4,13,14)\to(10,1,1)$	$(5,8,9)\to(5,2,1)$	$(5,9,12)\to(7,2,1)$
$(5,11,12)\to(7,1,2)$	$(5,11,13)\to(8,2,1)$	$(5,12,14)\to(9,1,2)$	$(5,13,14)\to(8,2,1)$
$(5,13,16)\to(10,1,2)$	$(6,11,14)\to(9,1,2)$	$(6,11,15)\to(9,2,1)$	$(6,13,14)\to(9,1,2)$
$(6,13,15)\to(10,2,1)$	$(6,13,16)\to(10,1,2)$	$(7,10,12)\to(7,3,1)$	$(7,10,15)\to(8,2,2)$
$(7,11,12)\to(7,2,2)$	$(7,11,13)\to(7,2,2)$	$(7,11,16)\to(8,3,1)$	$(7,12,13)\to(7,3,1)$
$(7,12,15)\to(8,2,2)$	$(7,13,17)\to(9,3,1)$	$(7,13,18)\to(9,2,2)$	$(7,15,16)\to(9,1,3)$
$(7,15,17)\to(9,2,2)$	$(7,15,18)\to(10,3,1)$	$(7,16,17)\to(10,2,2)$	$(7,16,18)\to(11,1,3)$
$(8,11,12)\to(7,3,1)$	$(8,11,15)\to(9,2,2)$	$(8,13,14)\to(7,3,2)$	$(8,13,15)\to(8,2,2)$
$(8,13,17)\to(10,2,2)$	$(8,15,18)\to(9,3,2)$	$(8,15,19)\to(11,2,2)$	$(8,17,18)\to(11,1,3)$
$(8,17,19)\to(10,2,2)$	$(9,13,16)\to(8,3,2)$	$(9,13,20)\to(9,4,1)$	$(9,14,15)\to(8,2,2)$
$(9,14,16)\to(9,4,1)$	$(9,14,17)\to(9,2,3)$	$(9,14,20)\to(9,3,2)$	$(9,16,17)\to(9,4,1)$
$(9,16,19)\to(10,3,2)$	$(9,17,20)\to(11,2,3)$	$(9,19,20)\to(11,1,4)$	$(10,13,15)\to(9,4,1)$
$(10,13,18)\to(10,3,2)$	$(10,13,21)\to(9,3,2)$	$(10,17,18)\to(9,3,2)$	$(10,17,19)\to(9,3,2)$
$(10,17,21)\to(10,2,3)$	$(11,14,15)\to(7,3,4)$	$(11,14,18)\to(9,5,1)$	$(11,14,21)\to(10,3,2)$
$(11,15,16)\to(7,3,2)$	$(11,15,20)\to(9,4,2)$	$(11,16,18)\to(8,4,2)$	$(11,16,19)\to(10,5,1)$
$(11,16,20)\to(9,3,2)$	$(11,17,18)\to(8,2,3)$	$(11,18,19)\to(9,4,3)$	$(11,18,20)\to(11,5,1)$
$(11,18,21)\to(10,2,3)$	$(11,19,20)\to(9,4,2)$	$(11,20,21)\to(11,5,1)$	$(12,17,18)\to(10,5,1)$
$(12,17,20)\to(10,3,2)$	$(12,17,21)\to(10,2,3)$	$(12,17,23)\to(12,4,2)$	$(12,19,20)\to(10,3,2)$
$(12,19,21)\to(11,2,3)$	$(12,19,22)\to(11,3,4)$	$(12,19,23)\to(11,4,2)$	$(13,16,21)\to(10,6,1)$
$(13,16,24)\to(11,2,4)$	$(13,17,20)\to(9,4,2)$	$(13,17,24)\to(10,4,4)$	$(13,18,19)\to(9,4,2)$
$(13,18,20)\to(9,3,3)$	$(13,18,22)\to(11,6,1)$	$(13,18,24)\to(11,3,3)$	$(13,19,24)\to(10,3,3)$
$(13,20,21)\to(10,2,4)$	$(13,20,22)\to(10,4,2)$	$(13,20,23)\to(12,6,1)$	$(13,20,24)\to(11,4,2)$
$(13,21,22)\to(10,4,4)$	$(13,21,24)\to(11,2,4)$	$(13,22,23)\to(11,3,3)$	$(13,22,24)\to(13,6,1)$
$(13,23,24)\to(11,4,2)$	$(14,17,24)\to(12,5,2)$	$(14,19,20)\to(9,3,3)$	$(14,19,21)\to(12,6,1)$
$(14,19,22)\to(9,5,2)$	$(14,19,25)\to(10,4,2)$	$(14,23,24)\to(12,3,5)$	$(14,23,25)\to(11,3,4)$
$(15,19,24)\to(11,2,4)$	$(15,22,24)\to(11,2,4)$	$(15,22,25)\to(11,4,2)$	$(15,22,26)\to(13,7,1)$
$(15,23,24)\to(11,2,4)$	$(15,23,26)\to(12,5,4)$	$(16,19,27)\to(13,6,2)$	$(16,21,23)\to(10,5,2)$
$(16,21,24)\to(13,7,1)$	$(16,21,25)\to(10,4,3)$	$(16,23,24)\to(13,7,1)$	$(16,23,26)\to(11,5,2)$
$(16,23,27)\to(12,3,4)$	$(16,25,26)\to(10,3,4)$	$(16,25,27)\to(12,2,5)$	$(17,21,28)\to(11,3,4)$
$(17,22,23)\to(10,5,6)$	$(17,22,25)\to(10,4,3)$	$(17,22,26)\to(10,6,2)$	$(17,22,28)\to(13,8,1)$
$(17,23,24)\to(10,3,4)$	$(17,24,25)\to(10,5,2)$	$(17,24,26)\to(11,4,4)$	$(17,24,28)\to(11,3,4)$
$(17,25,28)\to(11,6,2)$	$(17,26,27)\to(11,2,5)$	$(17,26,28)\to(11,4,3)$	$(17,27,28)\to(12,4,3)$
$(18,23,27)\to(14,8,1)$	$(18,23,29)\to(12,4,3)$	$(18,25,26)\to(10,5,3)$	$(18,25,27)\to(15,8,1)$
$(18,25,28)\to(11,3,4)$	$(19,24,30)\to(11,5,3)$	$(19,25,30)\to(11,4,3)$	$(19,26,27)\to(10,4,4)$
$(19,26,28)\to(11,3,4)$	$(19,26,29)\to(12,3,4)$	$(19,26,30)\to(12,6,2)$	$(19,27,28)\to(12,6,2)$
$(19,28,30)\to(13,3,5)$	$(19,29,30)\to(13,2,6)$	$(20,27,28)\to(11,3,4)$	$(20,27,29)\to(12,6,2)$
$(20,27,30)\to(16,9,1)$	$(20,27,31)\to(12,5,4)$	$(20,29,30)\to(16,9,1)$	$(21,29,30)\to(10,5,4)$
$(21,29,32)\to(12,3,5)$	$(22,29,31)\to(11,6,3)$	$(22,29,32)\to(11,5,4)$	$(22,29,33)\to(17,10,1)$
$(22,31,32)\to(12,5,3)$	$(22,31,33)\to(18,10,1)$	$(23,30,31)\to(13,7,8)$	$(23,30,33)\to(13,7,2)$
$(23,30,34)\to(13,4,4)$	$(23,31,32)\to(11,3,5)$	$(23,31,34)\to(11,6,4)$	$(23,32,33)\to(13,3,5)$
$(23,32,34)\to(12,4,4)$	$(23,33,34)\to(13,7,2)$	$(24,31,35)\to(12,4,4)$	$(25,33,36)\to(14,8,2)$
$(25,34,35)\to(13,4,4)$	$(25,34,36)\to(13,3,6)$	$(26,35,36)\to(13,3,6)$	$(26,35,37)\to(13,6,3)$
$(27,37,38)\to(14,6,3)$	$(28,37,39)\to(14,4,5)$	$(29,38,39)\to(15,9,10)$	$(29,39,40)\to(14,3,7)$

Table 4: The 178 Case-B class bases ($a \le 11$, $\mu \ge 12$). Format $[a;\bar{e},h]$ followed by the base triple $(a,b,M) \to (x,Y,Z)$. The λ-lift lemma of the main paper extends each certificate to every member of its class with larger e.

$[4;1,1]\,(4,17,18)\to(13,1,1)$	$[5;1,1]\,(5,16,17)\to(10,1,2)$	$[5;1,2]\,(5,16,18)\to(11,2,1)$
$[5;2,2]\,(5,17,19)\to(12,1,2)$	$[5;3,1]\,(5,18,19)\to(11,2,1)$	$[5;3,3]\,(5,18,21)\to(13,1,2)$
$[5;4,3]\,(5,14,17)\to(10,2,1)$	$[6;1,1]\,(6,19,20)\to(13,1,2)$	$[6;1,2]\,(6,19,21)\to(14,2,1)$
$[6;1,3]\,(6,19,22)\to(14,1,2)$	$[6;5,3]\,(6,17,20)\to(13,1,2)$	$[6;5,4]\,(6,17,21)\to(13,2,1)$
$[7;1,1]\,(7,22,23)\to(13,1,3)$	$[7;1,2]\,(7,22,24)\to(13,2,2)$	$[7;1,3]\,(7,22,25)\to(14,3,1)$
$[7;1,4]\,(7,15,19)\to(11,2,2)$	$[7;2,1]\,(7,23,24)\to(14,2,2)$	$[7;2,2]\,(7,23,25)\to(15,1,3)$
$[7;2,4]\,(7,16,20)\to(11,2,2)$	$[7;3,2]\,(7,17,19)\to(11,3,1)$	$[7;3,3]\,(7,17,20)\to(12,1,3)$
$[7;3,5]\,(7,17,22)\to(12,2,2)$	$[7;4,1]\,(7,18,19)\to(11,2,2)$	$[7;4,2]\,(7,18,20)\to(11,2,2)$
$[7;4,4]\,(7,18,22)\to(13,1,3)$	$[7;4,5]\,(7,18,23)\to(12,3,1)$	$[7;5,1]\,(7,19,20)\to(11,3,1)$
$[7;5,3]\,(7,19,22)\to(12,2,2)$	$[7;5,5]\,(7,19,24)\to(14,1,3)$	$[7;6,3]\,(7,20,23)\to(13,2,2)$
$[7;6,4]\,(7,20,24)\to(13,3,1)$	$[7;6,5]\,(7,20,25)\to(13,2,2)$	$[8;1,1]\,(8,25,26)\to(16,1,3)$
$[8;1,2]\,(8,25,27)\to(14,2,2)$	$[8;1,3]\,(8,17,20)\to(12,3,1)$	$[8;1,4]\,(8,17,21)\to(12,2,2)$
$[8;1,5]\,(8,17,22)\to(11,3,2)$	$[8;3,1]\,(8,19,20)\to(12,3,1)$	$[8;3,3]\,(8,19,22)\to(14,1,3)$
$[8;3,4]\,(8,19,23)\to(14,2,2)$	$[8;3,6]\,(8,19,25)\to(12,2,2)$	$[8;5,1]\,(8,21,22)\to(11,3,2)$
$[8;5,2]\,(8,21,23)\to(12,2,2)$	$[8;5,4]\,(8,21,25)\to(15,2,2)$	$[8;5,5]\,(8,21,26)\to(16,1,3)$
$[8;7,3]\,(8,23,26)\to(13,3,2)$	$[8;7,4]\,(8,23,27)\to(16,2,2)$	$[8;7,5]\,(8,15,20)\to(11,3,1)$
$[8;7,6]\,(8,15,21)\to(10,2,2)$	$[9;1,1]\,(9,28,29)\to(16,1,4)$	$[9;1,2]\,(9,19,21)\to(12,2,2)$
$[9;1,3]\,(9,19,22)\to(12,3,2)$	$[9;1,4]\,(9,19,23)\to(12,4,1)$	$[9;1,5]\,(9,19,24)\to(13,2,2)$
$[9;1,6]\,(9,19,25)\to(12,4,2)$	$[9;2,1]\,(9,20,21)\to(12,2,2)$	$[9;2,2]\,(9,20,22)\to(13,1,4)$
$[9;2,3]\,(9,20,23)\to(13,2,3)$	$[9;2,4]\,(9,20,24)\to(13,2,2)$	$[9;2,6]\,(9,20,26)\to(13,3,2)$
$[9;4,2]\,(9,22,24)\to(13,2,2)$	$[9;4,3]\,(9,22,25)\to(13,3,2)$	$[9;4,4]\,(9,22,26)\to(15,1,4)$
$[9;4,6]\,(9,22,28)\to(13,4,2)$	$[9;4,7]\,(9,22,29)\to(14,4,1)$	$[9;5,1]\,(9,23,24)\to(13,2,2)$
$[9;5,2]\,(9,23,25)\to(14,4,1)$	$[9;5,3]\,(9,23,26)\to(14,2,3)$	$[9;5,5]\,(9,23,28)\to(16,1,4)$
$[9;5,6]\,(9,23,29)\to(14,3,2)$	$[9;5,7]\,(9,14,21)\to(10,2,2)$	$[9;7,1]\,(9,25,26)\to(14,4,1)$
$[9;7,3]\,(9,25,28)\to(15,3,2)$	$[9;7,5]\,(9,16,21)\to(11,2,2)$	$[9;7,6]\,(9,16,22)\to(10,4,2)$
$[9;7,7]\,(9,25,32)\to(18,1,4)$	$[9;8,3]\,(9,26,29)\to(16,2,3)$	$[9;8,4]\,(9,17,21)\to(11,2,2)$
$[9;8,5]\,(9,17,22)\to(11,4,1)$	$[9;8,6]\,(9,17,23)\to(11,3,2)$	$[9;8,7]\,(9,17,24)\to(12,2,2)$
$[10;1,1]\,(10,21,22)\to(13,1,4)$	$[10;1,2]\,(10,21,23)\to(11,2,3)$	$[10;1,3]\,(10,21,24)\to(11,3,2)$
$[10;1,4]\,(10,21,25)\to(14,4,1)$	$[10;1,5]\,(10,21,26)\to(14,3,2)$	$[10;1,6]\,(10,21,27)\to(13,3,2)$
$[10;1,7]\,(10,21,28)\to(14,3,3)$	$[10;3,1]\,(10,23,24)\to(12,3,3)$	$[10;3,2]\,(10,23,25)\to(15,4,1)$
$[10;3,3]\,(10,23,26)\to(16,1,4)$	$[10;3,5]\,(10,23,28)\to(16,3,2)$	$[10;3,6]\,(10,23,29)\to(14,2,3)$
$[10;3,8]\,(10,23,31)\to(14,3,2)$	$[10;7,1]\,(10,27,28)\to(14,3,2)$	$[10;7,2]\,(10,27,29)\to(14,3,2)$
$[10;7,4]\,(10,27,31)\to(15,2,3)$	$[10;7,5]\,(10,17,22)\to(12,3,2)$	$[10;7,7]\,(10,27,34)\to(20,1,4)$
$[10;7,8]\,(10,17,25)\to(12,4,1)$	$[10;9,3]\,(10,19,22)\to(11,3,3)$	$[10;9,4]\,(10,19,23)\to(11,3,2)$
$[10;9,5]\,(10,19,24)\to(13,3,2)$	$[10;9,6]\,(10,19,25)\to(13,4,1)$	$[10;9,7]\,(10,19,26)\to(11,3,2)$
$[10;9,8]\,(10,19,27)\to(12,2,3)$	$[11;1,1]\,(11,23,24)\to(13,1,5)$	$[11;1,2]\,(11,23,25)\to(12,2,3)$
$[11;1,3]\,(11,23,26)\to(12,3,2)$	$[11;1,4]\,(11,23,27)\to(13,3,4)$	$[11;1,5]\,(11,23,28)\to(14,5,1)$
$[11;1,6]\,(11,23,29)\to(14,2,4)$	$[11;1,7]\,(11,23,30)\to(13,4,2)$	$[11;1,8]\,(11,23,31)\to(14,4,3)$
$[11;2,1]\,(11,24,25)\to(12,2,4)$	$[11;2,2]\,(11,24,26)\to(15,1,5)$	$[11;2,3]\,(11,24,27)\to(12,4,2)$
$[11;2,4]\,(11,24,28)\to(13,2,3)$	$[11;2,5]\,(11,24,29)\to(13,4,3)$	$[11;2,6]\,(11,24,30)\to(13,3,2)$
$[11;2,8]\,(11,24,32)\to(15,4,2)$	$[11;3,1]\,(11,25,26)\to(12,3,4)$	$[11;3,2]\,(11,25,27)\to(12,4,3)$
$[11;3,3]\,(11,25,28)\to(16,1,5)$	$[11;3,4]\,(11,25,29)\to(15,5,1)$	$[11;3,6]\,(11,25,31)\to(14,2,3)$
$[11;3,7]\,(11,25,32)\to(15,2,4)$	$[11;3,9]\,(11,14,23)\to(9,3,2)$	$[11;4,1]\,(11,26,27)\to(12,3,2)$
$[11;4,2]\,(11,26,28)\to(13,2,4)$	$[11;4,4]\,(11,26,30)\to(17,1,5)$	$[11;4,5]\,(11,26,31)\to(14,3,4)$
$[11;4,6]\,(11,26,32)\to(14,4,2)$	$[11;4,8]\,(11,15,23)\to(10,2,3)$	$[11;4,9]\,(11,15,24)\to(10,5,1)$
$[11;5,2]\,(11,27,29)\to(13,4,2)$	$[11;5,3]\,(11,27,30)\to(16,5,1)$	$[11;5,4]\,(11,27,31)\to(14,3,2)$
$[11;5,5]\,(11,27,32)\to(18,1,5)$	$[11;5,7]\,(11,16,23)\to(10,4,3)$	$[11;5,8]\,(11,16,24)\to(11,3,2)$

Table 4: *(continued)*.

$[11;5,9]\ (11,16,25)\to(11,4,2)$	$[11;6,1]\ (11,28,29)\to(13,2,3)$	$[11;6,2]\ (11,28,30)\to(14,3,4)$
$[11;6,3]\ (11,28,31)\to(14,2,4)$	$[11;6,4]\ (11,28,32)\to(15,4,3)$	$[11;6,6]\ (11,28,34)\to(19,1,5)$
$[11;6,7]\ (11,17,24)\to(10,3,2)$	$[11;6,8]\ (11,17,25)\to(11,5,1)$	$[11;6,9]\ (11,17,26)\to(11,4,2)$
$[11;7,1]\ (11,29,30)\to(14,4,3)$	$[11;7,2]\ (11,29,31)\to(17,5,1)$	$[11;7,3]\ (11,29,32)\to(15,2,3)$
$[11;7,5]\ (11,18,23)\to(10,4,2)$	$[11;7,6]\ (11,18,24)\to(11,4,2)$	$[11;7,7]\ (11,29,36)\to(20,1,5)$
$[11;7,9]\ (11,18,27)\to(13,3,2)$	$[11;8,1]\ (11,30,31)\to(14,4,2)$	$[11;8,2]\ (11,30,32)\to(15,3,2)$
$[11;8,4]\ (11,19,23)\to(11,2,4)$	$[11;8,5]\ (11,19,24)\to(11,2,3)$	$[11;8,7]\ (11,19,26)\to(12,5,1)$
$[11;8,8]\ (11,30,38)\to(21,1,5)$	$[11;8,9]\ (11,19,28)\to(12,4,3)$	$[11;9,1]\ (11,31,32)\to(17,5,1)$
$[11;9,3]\ (11,20,23)\to(11,3,4)$	$[11;9,5]\ (11,20,25)\to(11,3,2)$	$[11;9,6]\ (11,20,26)\to(12,4,3)$
$[11;9,7]\ (11,20,27)\to(12,2,3)$	$[11;9,8]\ (11,20,28)\to(12,4,2)$	$[11;9,9]\ (11,31,40)\to(22,1,5)$
$[11;10,3]\ (11,21,24)\to(11,4,3)$	$[11;10,4]\ (11,21,25)\to(11,4,2)$	$[11;10,5]\ (11,21,26)\to(12,2,4)$
$[11;10,6]\ (11,21,27)\to(13,5,1)$	$[11;10,7]\ (11,21,28)\to(13,4,2)$	$[11;10,8]\ (11,21,29)\to(12,3,2)$
$[11;10,9]\ (11,21,30)\to(13,2,3)$		

Appendix F. Correspondence with the Lean development

Declarations live in the namespace `Erdos1112.Proof`, in files relative to `Erdos1112Proof/`, except the three headline theorems, which are in namespace `Erdos1112`. Every listed result is load-bearing: the development formalizes the full case tree rather than citing anything externally. The name-and-location layer of this table is checked mechanically (`scripts/check_correspondence.py`, re-run in continuous integration), so it cannot silently rot; the *statement-level* match — that each declaration says what the prose result says — remains a human judgement, and is where we invite scrutiny.

One row is marked ALT: for Case P this paper gives a single uniform argument for the whole line (§20) while the formalization instantiates small r separately and routes the rest through an auxiliary lemma. The formal statement proved is the one the assembly consumes; it is not weaker.

Result here	Lean declaration	File
Thm. A (dichotomy)	`erdos_1112`	`Final.lean`
Thm. 6.1	`erdos_1112_existence_bound`, `existence_bound`	`Final.lean`, `Existence/Nested.lean`
Strong non-existence	`erdos_1112_strong_nonexistence`	`Final.lean`
Integer form ($\mathbb{Z} \to \mathbb{N}$ bridge)	`erdos_1112_int`, `question_iff_questionInt`	`Final.lean`, `Erdos1112.lean`
Lem. 5.2 (free gap)	`exists_safe_subinterval`	`Existence/FreeGap.lean`
Lem. 8.6 (certificate)	`cert_hit`	`NonEx/Certificate.lean`
Lems. 9.4, 10.1	`tailCoveringN_of_single_letter`, `..._of_eventually_periodic`	`NonEx/GapWord.lean`
Lem. 12.1 (the band)	`Wset_interval`	`NonEx/TwoLetter/Core.lean`
Lem. 12.2 (sweep)	`sweep`	`NonEx/TwoLetter/Core.lean`
Lem. 12.4 (width dichotomy)	`width_of_unbalanced`	`NonEx/TwoLetter/Core.lean`
Thm. 13.8 (balanced classification)	`balanced_classification` `disc_osc_le_one`, `mechanical_tail`	`NonEx/TwoLetter/Balanced.lean` `NonEx/TwoLetter/MH/`
Lem. 14.5 (boundary width)	`width_even_boundary`	`NonEx/TwoLetter/Core.lean`
Lem. 14.1 (Sturmian)	`tailCovering_of_sturmian`, `walk_enters`	`NonEx/TwoLetter/Sturmian.lean` `NonEx/TwoLetter/MH/Walk.lean`
Lem. 15.3 (slot lemma)	`tailCovering_of_three_letters`	`NonEx/SlotLemma.lean`
Lem. 17.1 (two-generator)	`twoGen_interval`	`Sharp/TwoGen.lean`
Lem. 17.2 (frame)	`frame_lemma`	`Sharp/Frame.lean`
Lem. 17.3 (staircase)	`staircase_phase_base`, `staircase_phase_extended`	`Sharp/Staircase.lean`
Lem. 17.5 (λ-lift)	`FrameCert.lift`	`Sharp/Lift.lean`
Lem. 18.2 (Graham reduction)	`sharpAt_of_hardcore`	`Sharp/Graham.lean`
Lem. 18.8 (minimal alphabets)	`sharp_of_minimal`	`Sharp/Graham.lean`
Lem. 19.1 (Case D)	`caseD`	`Sharp/CaseD.lean`
Lem. 20.1 (Case P) ALT	`caseP_pair`, `caseP_r3`, `caseP_r4`, `caseP_large`	`Sharp/CaseP.lean`
Lem. 19.3 (Case L)	`caseL`	`Sharp/CaseL.lean`
Lem. 21.1 (Case E)	`caseE`	`Sharp/CaseE.lean`
Lem. 22.1, Prop. 22.3 (Case T)	`caseT`, `T_tail_line`, `tSuppT`	`Sharp/CaseT*.lean`
Case B (+ descent)	`caseB`, `rowFor`, `caseBComplete`	`Sharp/CaseB.lean`, `Sharp/CaseBClasses.lean`
Tables A/B (data)	`certTableA_ok`, `certTableB_ok`	`Sharp/TablesData.lean`
Thm. 16.1 (subset-sum lemma)	`sharp`	`Sharp/Main.lean`
Axiom audit	three `#print axioms` commands	`AxiomsCheck.lean`

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Result	**What it gives**	**Why it does not imply Theorem 16.1**
Lev [Lev98, Thm. 1] *(closest antecedent)*	for $A \subseteq [1, l]$ with $l \le \frac{3}{2}\lvert A\rvert - 2$, the subset sums of A contain an interval	same conclusion type (a genuine interval, from genuine $0/1$ subset sums), but a *dense-set* theorem: $\lvert A\rvert \gtrsim \frac{2}{3}M$, vacuous at $\lvert G\rvert = 3$, and no multiset. Ours is its density-free multiset analogue
Lev [Lev97, Thm. 1]	an interval of length exactly $\max(A) + 1$	inside the h-fold sumset $(2k-1)A$ — unbounded multiplicities, many summands — and under a density hypothesis; not $0/1$ sums of a bounded multiset
Lev [Lev99], Sárközy [Sa89, Sa94]	an AP inside the subset sums $S(A)$	the common difference cannot be forced to 1, so not consecutive integers; needs $\lvert A\rvert \sim \sqrt{m \log m}$; vacuous at $\lvert G\rvert = 3$
Szemerédi–Vu [SV06]	subset sums of $A \subseteq [n]$ contain a length-n AP	density threshold $\lvert A\rvert \ge c\sqrt{n}$; an AP, not necessarily consecutive; no uniform bound $\lvert S\rvert \le M - 1$
Conlon–Fox–Pham [CFP21]	a long interval of consecutive integers in subset sums	asymptotic and density-based; yields no uniform small $\lvert S\rvert \le M - 1$
Nathanson [Na72], Granville–Walker [GW21], Lev [Lev22]	the interval structure of mA for large m	the h-fold sumset is a total-degree *simplex* in coefficient space; the subset sums of a bounded multiset are a *box* contained in it, so this is a weaker statement about a larger set
Alon–Freiman [AF88], Lev [Lev03], Chen–Mao–Zhang [CMZ25]	AP or interval in the subset sums of a *dense* set	density $\lvert A\rvert \gtrsim \sqrt{n}$; vacuous at $\lvert G\rvert = 3$
Alon [Al87]	the extremal size of a set *avoiding* a target subset sum	the inverse problem; orthogonal
Folkman–Birch–Graham [Gr64], Erdős–Graham [ErGr80]	subset sums of an *infinite* sequence contain every large integer	infinite sequence, unbounded summands; the filling step is a shared *technique*, not an implication
Frobenius, Ramírez Alfonsín [RA05]	everything past the Frobenius number, as unbounded combinations	Frobenius number $\gg M$; unbounded multiplicity; no length-M window from a bounded multiset
Postage-stamp bases [GS20]	an initial run $[1, N]$ from $\le h$ summands	there the generating set is *chosen* to maximise N; we fix an *arbitrary* G and demand a run of length exactly M

TABLE 1. The nearest results, and the hypothesis that separates each from Theorem 16.1. The literature splits into a *higher-sumset* regime (many summands, unbounded multiplicities) and a *density-threshold* regime (large $\lvert A\rvert$, APs of difference possibly > 1). Theorem 16.1 sits between them: boundedly many summands, drawn with repetition from a fixed and possibly tiny set, covering consecutive integers, uniformly in G down to $\lvert G\rvert = 3$.

Result	What it carries	How checkable
Existence half (Part II)		
Prop. 4.4 (avoidance criterion)	converts "$kA \cap B = \emptyset$" into a condition on one real slope	prose, Lean
Prop. 4.6	shows why the construction cannot work at $d_2 \le k$ — the threshold is not an artefact	prose, Lean
Lem. 5.2 (free gap)	**the steep climb.** Guarantees an interval of safe slopes survives each step	prose, Lean
Lem. 5.3	the geometric-ratio form actually consumed	prose, Lean
Thm. 6.1	nested intervals assemble the above into $r = 192\,d_2$	prose, Lean
Non-existence: reduction (Part III)		
Lem. 8.6 (the certificate)	**the diagonalisation.** One B defeats *every* tail-covering walk	prose, Lean
Lem. 9.3, 9.5	the two reductions. Note the type discipline of Definition 9.2: rescaling takes a walk *out of* the class of (d_1, d_2)-sequences	prose, Lean
Prop. 15.5	every walk with gaps $\le d_2 \le k$ is tail-covering	prose, Lean
Non-existence: the two-letter campaign		
Lem. 12.4 (width dichotomy)	either a reachable band is wide, or the word is balanced	prose, Lean
Lem. 12.2 (sweep)	a wide band covers a residue class	prose, Lean
Thm. 13.8 (balanced classification)	balanced $\Rightarrow$ eventually periodic or Sturmian (Morse–Hedlund; reproved here)	prose, Lean
Lem. 14.1 (Sturmian endgame)	the irrational-slope case	prose, Lean
Prop. 14.6	**the tight corner:** k even, $d_2 = k$, where the sweep is one short	prose, Lean
Thm. 14.7	assembles the campaign	prose, Lean
Non-existence: three or more letters (Parts III–IV)		
Lem. 15.3 (slot lemma)	prices the remaining case at exactly $m(G) \le k - 1$	prose, Lean
Thm. 16.1 (SHARP)	**the new tool.** $m(G) \le \max(G) - 1$	prose, Lean
Lem. 18.2, 18.3–18.8	reduction to the hard core	prose, Lean
Lem. 19.1, 19.3, 20.1, 21.1	four uniform branches, closed-form witnesses	prose, Lean
Lem. 22.1, Prop. 22.3	Case T: uniform tail by hand, 172 rows by table	**partly computational** (**T1**, **T2**)
Lem. 23.1, 17.5	Case B: 178 classes counted by hand, lifted by proof	prose + table (**B1**, **B2**)

TABLE 2. The critical path. "Lean" means the result corresponds to a declaration in the formal development, listed in Appendix F. The bold entries are where we suggest scepticism is best spent; the obligation labels **T1**–**B2** are those of §26.3.